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Data Collection Nonresponse Bias Analysis for Wave 6

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**PATH**Population Assessment
of Tobacco and Health*A collaboration between the NIH and FDA*

PATH Study

Wave 6 Data Collection

Nonresponse Bias Analysis Report

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1. Introduction

Westat prepared this report as a reference document for researchers using the Population Assessment of Tobacco and Health (PATH) Study interview and biomarker data. The PATH Study is a nationally representative, longitudinal cohort study of adults and youth in the United States, collecting data on those ages 12 years and older at Waves 1-4 and Wave 5, on those ages 12 to 17 in the Wave 4.5 special data collection, on those ages 13 to 19 at the Wave 5.5 special data collection, and on those ages 14 and older at Wave 6. A subsample of adults ages 20 and older also was surveyed in the latter part of 2020 via the PATH Adult Telephone Study (PATH-ATS). This report focuses on the Wave 6 response rates and potential nonresponse biases for select demographic, socio-economic, and tobacco-use measures.

Similar reports are also available for previous PATH Study data collections and can be found at the following websites for the National Addiction & HIV Data Archive Program (NAHDAP). The reports for the primary waves (i.e., the waves that included all available sample youth and adults) are located at one website and the reports for the special data collections, where subgroups of participants were surveyed, can be found at a separate website.

- Primary waves: Waves 1-4, 5:
<https://www.icpsr.umich.edu/web/NAHDAP/studies/36231/datadocumentation/>
- Special data collections: Waves 4.5, 5.5, and PATH-ATS:
<https://www.icpsr.umich.edu/web/NAHDAP/studies/37519/datadocumentation/>

In this document, the term “participants” is used to indicate people who agreed to be a part of the PATH Study, whether or not they completed an interview in a particular wave. The term “respondents” is used to indicate the subset of participants who completed an interview or, in the case of shadow youth (i.e., participants younger than 12 years old), verified their information with the study in a particular wave. Where needed for clarity, this term may be qualified to indicate the type of respondent (e.g., shadow youth respondent, adult interview respondent).

The PATH Study data support analyses of two cohorts: the Wave 1 Cohort and the Wave 4 Cohort. The Wave 1 Cohort includes all Wave 1 respondents. These study participants were selected to represent adults ages 18 and older and youth ages 9 to 17 in the U.S. civilian, noninstitutionalized population (CNP) at the time of Wave 1 (September 12, 2013, to December 14, 2014). The Wave 1

Cohort is to be used for longitudinal analysis at Wave 6. All Wave 1 respondents were eligible for Wave 6 data collection unless they were incarcerated, deceased, or resided outside of the United States at the time of Wave 6.

The Wave 4 Cohort consists of two components. The first component is the Wave 4 adult and youth respondents from the Wave 1 Cohort who were in the CNP and ages 12 and older at the time of Wave 4 (December 1, 2016, to January 3, 2018). The second component is the respondents to a replenishment sample designed to supplement the Wave 1 sample. The Wave 4 Cohort serves the purpose of both longitudinal and cross-sectional analyses at Wave 6. All Wave 4 Cohort members were eligible for Wave 6 data collection unless they were incarcerated, deceased, or resided outside of the United States at the time of Wave 6.

This report summarizes the Wave 6 nonresponse bias analyses conducted separately for the two cohorts. Starting with the Wave 5 Nonresponse Bias Analysis Report, analyses for the Wave 1 Cohort use an all-waves perspective, whereas the analyses for Waves 1 through 4 used a single-wave approach (see Section 3.2.1 for an explanation of this change and why it better serves the research community). The analyses for the Wave 4 Cohort use a single-wave perspective and are similar at Wave 6 to those conducted for the Wave 1 Cohort at Wave 3. The potential for nonresponse bias in Wave 6 estimates based on the Wave 1 Cohort is evaluated conditioning on Wave 1 response, whereas potential nonresponse bias in estimates based on the Wave 4 Cohort is assessed conditioning on Wave 4 response. With this in mind, a brief summary of the findings from the Wave 1 nonresponse bias analyses and from the Wave 4 nonresponse bias analyses for the Wave 4 Cohort is provided below.

Wave 1 Cohort Nonresponse Bias Analyses at Wave 1

- For the Wave 1 household screener and adult interview, most demographic and socioeconomic characteristics of the respondents at Wave 1 aligned with estimates from the 2013 American Community Survey 1-year Public Use Microdata Sample (2013 ACS) when using the basic design weight, which is also known as the inverse-of-probability-of-selection (IPS) weight for the PATH Study. Exceptions were found for single-person households, sex, education, and ethnicity when comparing Wave 1 estimates using the IPS weight to the 2013 ACS estimates. Estimates of cigarette-smoking rates among adults at Wave 1 were within the range of estimates found in other national health studies. When the estimates were adjusted for nonresponse using the Wave 1 cross-sectional weight, they more closely approximated the 2013 ACS estimates for

demographic and socio-economic characteristics and adult cigarette-smoking rates remained essentially the same.

- For the Wave 1 youth interview, most demographic characteristics of respondents were consistent with the estimates from the 2013 ACS, with the exception of ethnicity, when using the IPS weight. When the estimates were adjusted for nonresponse among youth, they more closely approximated the 2013 ACS estimates, but the ever-tried cigarette-smoking rates for youth at Wave 1 remained 2 to 12 percentage points lower than those found in other national studies.

Wave 4 Cohort Nonresponse Bias Analyses at Wave 4

- When using the Wave 4 cross-sectional weight, demographic and socio-economic estimates based on Wave 4 adults were nearly identical to corresponding estimates based on the 2016 American Community Survey 1-year Public Use Microdata Sample (2016 ACS). However, adults with health insurance were slightly underrepresented by the Wave 4 Cohort estimate. Estimates of current cigarette-smoking rates among adults at Wave 4 were within the range of estimates found in other national studies.
- When using the Wave 4 cross-sectional weight, demographic estimates based on Wave 4 youth were nearly identical to corresponding estimates based on the 2016 ACS. The ever-tried cigarette-smoking rates for Wave 4 youth were 1 to 6 percentage points lower than those found in other national studies.

This report is organized as follows: Chapter 2 provides an overview of the PATH Study sample design. Chapter 3 describes the methodology used for the analyses in this report. Chapters 4 and 5 present the results of nonresponse bias analyses for the Wave 1 Cohort interview data and Wave 4 Cohort interview data, respectively. Chapter 6 summarizes the findings and discusses their implications.

2. Overview of Sample Design and Data Collection

This chapter provides an overview of the sample design for the PATH Study. The study's design allows for the longitudinal assessment of within-person and between-person change in the use of tobacco products, including initiation, cessation, relapse, and transitions between products, as well as in related health effects associated with use patterns.

At Wave 1, a four-stage, stratified probability sample design was used to select adults ages 18 and older and youth ages 12 to 17 from the CNP; an additional “shadow sample” of youth ages 9 to 11 was selected to be interviewed after reaching age 12 at later waves. All Wave 1 sample respondents together form the Wave 1 Cohort.

At Wave 4, the original set of Wave 1 sample respondents was replenished with a probability sample of adults, youth, and shadow youth ages 10 to 11¹ selected from the CNP at the time of Wave 4. The replenishment effort had two objectives: supplementation of the Wave 1 sample to address attrition and the changing composition of the study's adult respondents as Wave 1 youth (who were not oversampled with respect to tobacco use) turn 18 and age up to the adult interview, and selection of a shadow sample for fielding in future waves. The replenishment effort also gave persons who had been overseas, in the military, or in an institutional setting at the time of Wave 1 (but were no longer at Wave 4) a chance of selection for the study. Because this “replenishment sample” was designed to supplement the Wave 1 sample, it is not intended to be used for estimation purposes on its own; rather, it was intended to be combined for estimation and analysis purposes with Wave 4 adult and youth respondents from the Wave 1 sample who were in the CNP at the time of Wave 4. This combined set of Wave 4 respondents forms the Wave 4 Cohort.

Figure 1 illustrates the components of the Wave 1 and Wave 4 Cohorts. Table 2-1 shows the sample sizes for the two cohorts.

¹ The difference in the shadow youth age range between the Wave 1 sample (ages 9 to 11) and the Wave 4 replenishment sample (ages 10 to 11) reflects the timings of future data collections and periodic efforts to replenish the PATH Study sample that were planned at the time of Wave 4 development.

Figure 1. Illustration of the relationship between the Wave 1 Cohort and the Wave 4 Cohort

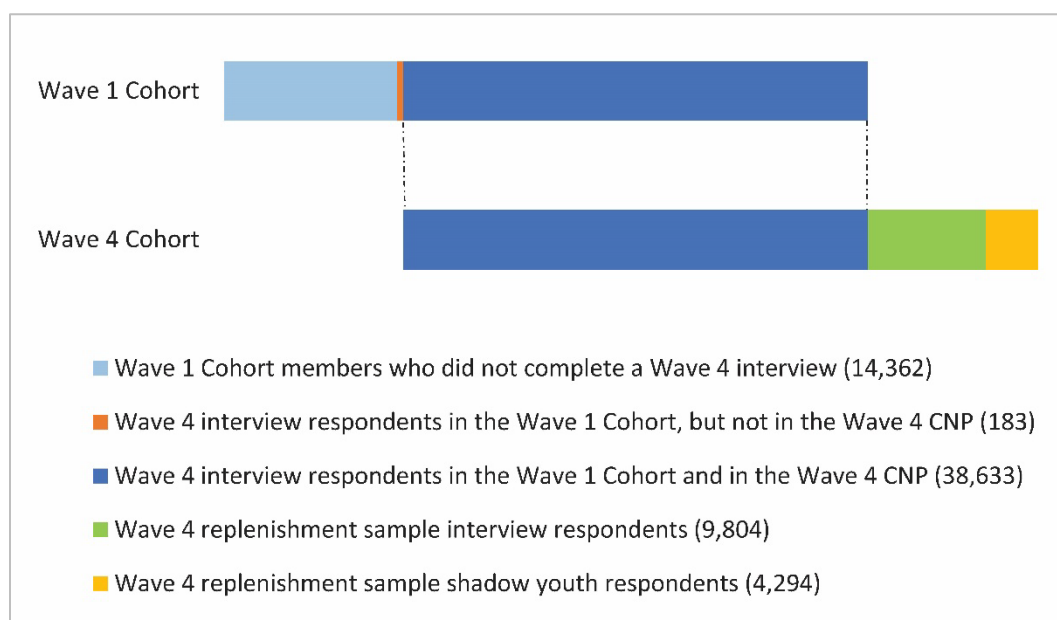


Table 2-1. Wave 1 Cohort and Wave 4 Cohort sample sizes and characteristics at Wave 4

Sample	Wave 1 Cohort	Wave 4 Cohort
Wave 1 Cohort members who did not complete a Wave 4 interview	14,362	
Wave 4 interview respondents in the Wave 1 Cohort, but not in the Wave 4 CNP	183	
Wave 4 interview respondents in the Wave 1 Cohort and in the Wave 4 CNP	38,633	38,633
Wave 4 interview respondents in the Wave 4 replenishment sample		9,804
Wave 4 shadow youth respondents in the Wave 4 replenishment sample		4,294
Total cohort sample size (shadow youth, youth, and adults) ^a	53,178	52,731

^a See Table 2-2 for the counts of interview respondents by cohort for each PATH Study primary and special data collection wave.

Key features of the Wave 1 and Wave 4 sample designs are summarized below. Further details can be found in Chapter 2 of the PATH Study Restricted Use Files (RUFs) User Guide (<https://www.icpsr.umich.edu/web/NAHDAP/studies/36231/datadocumentation/>). At the end of Chapter 2 of this report, Table 2-2 provides data collection start and end dates for each wave of the PATH Study, as well as a summary of the results of the data and biospecimen collection efforts by wave, cohort, and interview type.

2.1 Wave 1

The target population of the PATH Study at Wave 1 is the CNP ages 9 and older at the time of Wave 1 (2013-2014). Active-duty military personnel and those residing in an institutional setting at the time of Wave 1 were excluded. College students living away from home during the school year were identified as members of their permanent residence (e.g., parents' home). For Wave 1, a four-stage, stratified area probability sample design was used with a two-phase design for sampling adults at the final stage. The sampling rates for adults varied by age, race, and tobacco-use status. At the first stage, a stratified sample of geographical primary sampling units (PSUs) was selected, in which a PSU was a county or group of counties. For the second stage, within each selected PSU, smaller geographical segments (consisting of one or more census blocks) were formed and then a systematic sample of these segments was drawn. At the third stage, a systematic sample of addresses within sampled segments was drawn from listings of addresses; the main source of these addresses was the U.S. Postal Service (USPS) Computerized Delivery Sequence Files (CDSFs). The CDSFs provide very high coverage of the residential addresses in the United States. The fourth stage was the random selection of persons within sampled households.

For within-household selection, a roster of all the members of the sampled household was constructed using the household screener. An adult household member, the household screener respondent, was asked to list members of the household and provide demographic, and for adults, tobacco-use information about each person.² This information was used for sampling three main groups of interest:

- Adults ages 18 and older (up to two adults per household);
- Children ages 12 to 17 (i.e., “youth,” generally up to two per household);³ and
- Children ages 9 to 11 (i.e., “shadow youth,” generally up to two per household) to be interviewed at later waves after reaching age 12.

² The household screener collected information on adult household members' use of several different types of tobacco products. For example, it collected information on current use of products with relatively high prevalence or well-established use, such as cigarettes, cigars, and pipes; and on ever use of products with relatively low prevalence or newly emerging use, such as electronic cigarettes or e-cigarettes.

³ Given a special analytic interest in multiple-birth youth (e.g., twins), the shadow youth and youth sampling procedures were modified when households containing multiple-birth youths were encountered so that the multiple-birth youths would have relatively higher probabilities of selection. This resulted in some households with more than two children selected for the youth and/or shadow youth samples.

Two-phase sampling was used for adult selection due to potential misreporting by the household screener respondent of the tobacco-use status of adult household members. Phase 1 sampling depended on the age, race, and tobacco-use information obtained from the household informant during the household screener. Phase 2 sampling was based on self-reported age, race, and tobacco-use status, obtained by interviewing the individuals sampled at Phase 1. Sampling rates for the two phases were designed to oversample adults ages 18 to 24 and adult tobacco users of all ages, thereby increasing precision for estimates relating to these subgroups.⁴

All adult interview respondents were asked to provide urine and blood specimens. A subsample of the adults who provided urine forms the Wave 1 Biomarker Core, and these participants were flagged for ongoing urine collection after Wave 1. Information about the subsequent waves of the PATH Study is provided in the following sections.

2.2 Wave 2 and Wave 3

There was no additional sampling for Wave 2 or Wave 3 of the PATH Study. Wave 1 respondents were eligible for Wave 2 if they were still residents of the United States and not incarcerated.

During the Wave 2 data collection period, attempts were made to contact the Wave 1 adult and youth respondents as well as members of the shadow youth sample established at Wave 1. Shadow youth who reached age 12 by Wave 2 and were permitted by a parent or guardian to participate in the study were asked for assent to be interviewed for the first time at Wave 2. Similarly, Wave 1 youth respondents who reached age 18 by Wave 2 were asked to complete the adult interview.

In addition to the members of the Wave 1 Biomarker Core, the study flagged a subset of other Wave 1 adult respondents who had provided urine at Wave 1 for urine collection at Wave 2. Those who completed the PATH Study adult interview for the first time (referred to as “first-time adult interview respondents”) were asked to provide urine and blood specimens at Wave 2.

Wave 1 respondents were eligible for Wave 3 if they were still residents of the United States and not incarcerated. Wave 2 nonrespondents were fielded for Wave 3 unless they were deceased, had

⁴ In addition, Black or African Americans were slightly oversampled because of their higher rate of use of menthol cigarettes compared to other racial/ethnic groups. For this purpose, Black or African American was defined as Black or African American alone or in combination with other race(s), whether Hispanic or non-Hispanic.

moved out of the United States permanently, were incarcerated long term, had specifically requested withdrawal from the study, were firm or hostile refusers at Wave 2, had a language barrier,⁵ or had a physical or mental disability or chronic illness that prevented participation in the study. Shadow youth from previous waves who reached age 12 by Wave 3 and were permitted by a parent or guardian to participate in the study were asked for assent to be interviewed for the first time at Wave 3. Similarly, youth from previous waves who reached age 18 by Wave 3 were asked to complete the adult interview.

In addition to the members of the Wave 1 Biomarker Core, the study flagged a subset of other adults who had provided urine at a previous wave for urine collection at Wave 3. Those who completed the PATH Study adult interview for the first time at Wave 3 were asked to provide urine and blood specimens.

2.3 Wave 4

Wave 1 respondents were eligible for Wave 4 of the PATH Study if they were still residents of the United States and not incarcerated. Nonrespondents from previous waves were fielded for Wave 4 unless they were deceased, had moved out of the United States permanently, were incarcerated long term, had specifically requested withdrawal from the study, were firm or hostile refusers at a previous wave, had a language barrier, or had a physical or mental disability or chronic illness that prevented participation in the study. In addition, those who had refused participation at both Wave 2 and Wave 3 were not fielded for data collection.⁶ (Note that refusal is a specific type of nonresponse, so some nonrespondents to Wave 2 and Wave 3 who did not refuse at both those waves were fielded for Wave 4.) Shadow youth from previous waves who reached age 12 by Wave 4 and were permitted by a parent or guardian to participate in the study were asked for assent to be interviewed for the first time at Wave 4. Similarly, youth from previous waves who reached age 18 by Wave 4 were asked to complete the adult interview and to provide urine and blood specimens. In addition to the members of the Wave 1 Biomarker Core, the study flagged a subset of other adults

⁵ The PATH Study interviews can be conducted in English or Spanish. At Wave 2, the cases assigned a final nonresponse status due to language problems included two groups: (a) Wave 1 respondents who mentioned difficulty completing the Wave 1 interview in English or Spanish as their main reason for not completing an interview at Wave 2, and (b) Wave 1 shadow youth who did not feel comfortable completing an interview in English or Spanish when asked to do so by the study for the first time.

⁶ For youth, refusal could have come from the youth or their parent/guardian.

who had provided urine at a previous wave for urine collection at Wave 3. Unlike at previous waves, youth interview respondents were asked to provide a urine specimen.

At Wave 4, an additional sample was selected from the CNP to account for the aging of Wave 1 respondents and sample attrition among the Wave 1 Cohort. As part of the Wave 4 replenishment effort, adults, youth, and shadow youth were sampled within the existing PATH Study sample segments from among the addresses not selected for Wave 1.

The Wave 4 within-household sampling procedures mirrored those used at Wave 1. However, the sampling rates were designed to bring the Wave 4 adult and youth sample sizes up to Wave 1 levels by sampling domain. This required consideration for the expected combined effect of the aging of Wave 1 respondents and the loss of sample size due to attrition on the PATH Study sample by the time of Wave 4. In particular, it was necessary to oversample adult tobacco users given that the Wave 1 shadow youth and youth respondents who became adults after Wave 1 (and those who will continue to do so) were not oversampled with respect to tobacco-use status. All adult interview respondents from the replenishment sample were asked to provide urine and blood specimens, and all youth interview respondents from this sample were asked to provide a urine specimen.

2.4 Wave 4.5

Wave 4.5 was a special data collection of the PATH Study; its focus was on continuing the annual collection of data among youth ages 12 to 17 (and their parents). There was no additional sampling for Wave 4.5 and no biospecimens were collected. All study participants were eligible for Wave 4.5 if they were ages 12 to 17, still residents of the United States, and not incarcerated. Nonrespondents from previous waves were asked to participate at Wave 4.5 unless they were deceased, had moved out of the United States permanently, were incarcerated long term, had specifically requested withdrawal from the study, were firm or hostile refusers at a previous wave, had a language barrier, or had a physical or mental disability or chronic illness that prevented participation in the study. In addition, those who had refused participation at any two consecutive waves were not fielded for data collection. Shadow youth from previous waves who reached age 12 by Wave 4.5 and were permitted by a parent or guardian to participate in the study were asked for assent to be interviewed for the first time at Wave 4.5.

2.5 Wave 5

There was no additional sampling for Wave 5, and all study members were eligible if they were still residents of the United States and not incarcerated. Nonrespondents from previous waves were asked to participate at Wave 5 unless they were deceased, had moved out of the United States permanently, were incarcerated long term, had specifically requested withdrawal from the study, were firm or hostile refusers at a previous wave, had a language barrier, or had a physical or mental disability or chronic illness that prevented participation in the study. In addition, those who had refused participation at any two consecutive waves were not fielded for data collection.⁷ Shadow youth from previous waves who reached age 12 by Wave 5 and were permitted by a parent or guardian to participate in the study were asked for assent to be interviewed for the first time at Wave 5 and to provide a urine specimen. Urine specimens were also requested from Wave 5 youth who completed their first youth interview at the Wave 4.5 special data collection (when no biospecimens were requested) or who consented to urine specimen collection at Wave 4.⁸ Similarly, participants who completed their first adult interview at Wave 5 were asked to provide urine and blood specimens. Urine specimens were also requested from all adults belonging to the Wave 1 Biomarker Core.

2.6 Wave 5.5

Wave 5.5 was a special data collection for members of the Wave 4 Cohort, and only youth ages 13 to 17 and young adults ages 18 and 19 at the time of Wave 5.5 were interviewed. By design (see footnote 1), all PATH Study participants were expected to be at least age 13 at Wave 5.5. There was no additional sampling for Wave 5.5 and no biospecimens were collected. All Wave 4 Cohort members ages 13 to 19 at Wave 5.5 were eligible if they were still residents of the United States and not incarcerated. Nonrespondents from previous waves were asked to participate at Wave 5.5 unless they were deceased, had moved out of the United States permanently, were incarcerated long term, had specifically requested withdrawal from the study, were firm or hostile refusers at a previous

⁷ A refusal at Wave 4.5 was counted in the calculation of “two consecutive waves” of refusal for participants who were part of that special data collection.

⁸ In addition to the main groups of Wave 5 youth respondents described above, urine specimens were also requested from approximately 60 youth respondents who did not complete the Wave 4 youth interview or did not consent to urine specimen collection at Wave 4, where the reason for nonresponse or lack of consent was something other than a refusal.

wave, were unable to complete a previous interview in English or Spanish, or had a physical or mental disability or chronic illness that prevented participation in the study. In addition, those who had refused participation at Wave 4.5 and Wave 5 were not fielded for data collection.

Wave 5.5 began on December 1, 2019, with the same in-person data collection methods used at prior waves of the PATH Study, and was originally scheduled to continue until November 30, 2020. However, in-person data collection was suspended on March 17, 2020, due to the Coronavirus Disease (COVID-19) pandemic. Data collection resumed via telephone on July 3, 2020, and continued until December 31, 2020.

If the data collected in person, before COVID-19 and related restrictions were widespread in the United States, had been simply pooled with the data obtained in the latter portion of 2020 via telephone, the resulting Wave 5.5 estimates would have the potential for bias. This is because even though the combined data would represent tobacco-use behavior over most of the year, it would not include the 15-week period during which U.S. COVID-19 cases were rising and most State populations were under stay-at-home orders; such factors may have influenced tobacco-use behaviors. For this reason, it was decided that only data collected via telephone would be used to produce Wave 5.5 estimates, and the target population would be redefined to cover the latter portion of 2020 (specifically, July 3, 2020, to December 31, 2020). However, the PATH Study participants eligible for Wave 5.5 were scheduled for interviews approximately 1 year after their Wave 5 interview, and not in random subsamples. This meant that the participants scheduled for interviews in the latter portion of the year were not a random subsample of those eligible for Wave 5.5, so analyses of data collected from only these participants could also result in biased estimates. To overcome this concern, all participants who completed a Wave 5.5 interview in person before data collection was suspended and were still age eligible for Wave 5.5 (i.e., ages 13 to 19) were re-contacted for an interview by telephone starting August 3, 2020.⁹

⁹ Most participants whose in-person interview status was finalized as “nonrespondent” prior to the resumption of data collection on July 3, 2020, were participants who refused to respond to the interview. Because it was presumed that these participants would also refuse a request to complete a telephone interview, it was decided that no participants finalized as nonrespondent prior to the resumption of data collection would be re-contacted for a Wave 5.5 telephone interview. These cases were accounted for in the weighting process (see Appendix section B.7 of the RUFs User Guide for details).

During the data collection hiatus, some of the 19-year-olds interviewed in person reached age 20, and others would reach age 20 after the resumption of data collection. To ensure that the in-person respondents re-contacted by telephone would still be age eligible for Wave 5.5, August 31, 2020, was used as a date of demarcation to allow some older 19-year-olds (e.g., those with 20th birthdays in September) to be part of Wave 5.5. Those participants ages 13 to 19 on August 31, 2020, who completed the Wave 5.5 interview in person were eligible for the Wave 5.5 telephone re-interview. More generally, participants from the Wave 4 Cohort who were ages 19 and younger¹⁰ on August 31, 2020, were eligible for Wave 5.5 data collection in the latter portion of 2020.

Only data collected via telephone interview are included in the Wave 5.5 data files. The target population for the Wave 4 Cohort at Wave 5.5 is the resident (not incarcerated) U.S. population who were ages 13 to 19 in the latter portion of 2020 and part of the CNP at the time of Wave 4.

2.7 PATH-ATS

The PATH-ATS was a special data collection for which a subsample of adults ages 20 and older were interviewed. No biospecimens were collected for the PATH-ATS. The survey was developed in 2020 in light of new Federal tobacco policies, such as the nationwide Tobacco 21 policy and flavor restrictions, and the COVID-19 pandemic and its potential impact on the way people used and purchased tobacco. The objectives of the PATH-ATS were to (1) provide data that can be used to understand the impact of the Federal policy and regulatory changes in 2020; (2) provide data that can be used to assess the relationship between tobacco use and the COVID-19 pandemic; and (3) fill an unanticipated gap¹¹ in nationally representative data on adult tobacco use, prevalence, cessation, initiation, and relapse.

The PATH-ATS was conducted from September 10, 2020, to December 20, 2020. Because the planned data collection period was shorter than most PATH Study data collection efforts, only a subsample of participants were interviewed. Participants eligible for the PATH-ATS subsample were ages 20 and older on August 31, 2020, part of the Wave 4 Cohort, and respondents to the Wave 5

¹⁰ This includes some participants who were age 12 on August 31, 2020; these participants were contacted for an interview on or after their 13th birthday.

¹¹ Although the PATH Study did not originally plan to collect data on adults ages 20 and older in 2020, the COVID-19 pandemic also interrupted planned data collections for other national surveys used to understand tobacco use.

adult interview. A stratified random sample of 18,601 PATH Study participants was selected for the PATH-ATS from a total of 31,343 eligible participants. Those who at Wave 5 were ages 20 to 24, users of electronic nicotine delivery systems (ENDS), users of cigarettes, and more recent users of tobacco were oversampled. Participants who at Wave 5 were ages 25 and over, users of products other than ENDS or cigarettes, and less recent users of tobacco were similarly undersampled.

The target population for PATH-ATS estimates based on the Wave 4 Cohort is the resident (not incarcerated) U.S. population ages 20 and older in the latter portion of 2020 who were in the CNP at the time of Wave 4 and the resident (not incarcerated) U.S. adult population at Wave 5.

In light of the pandemic, it was of analytic interest to learn how the public health crisis affected the full target population(s) of the PATH Study, adults as well as youth. As noted above, the PATH-ATS was designed to complement the Wave 5.5 sample consisting of those ages 13 to 19 who were interviewed by telephone in the latter portion of 2020. Because the same date of demarcation was applied to determine age eligibility for the Wave 5.5 telephone interview and the PATH-ATS, the two samples are mutually exclusive in terms of their coverage of adults and jointly exhaustive for the adult population. Thus, the Wave 5.5 and the PATH-ATS data may be combined for analysis. The PATH-ATS weights can also be combined with the respective Wave 5.5 weights for longitudinal and cross-sectional analyses for the Wave 4 Cohort.

2.8 Wave 6

There was no additional sampling for Wave 6 and no biospecimens were collected; all study members were eligible if they were still residents of the United States and not incarcerated. Nonrespondents from previous waves were asked to participate at Wave 6 unless they were deceased, had moved out of the United States permanently, were incarcerated long term, had specifically requested withdrawal from the study, were firm or hostile refusers at a previous wave, had a language barrier, or had a physical or mental disability or chronic illness that prevented

participation in the study. In addition, those who had refused participation at any two consecutive waves or otherwise did not respond at three consecutive waves were not fielded for data collection.¹²

Both in-person data collection and telephone data collection were conducted for Wave 6. Due to the COVID-19 pandemic, and for the continued safety of PATH Study participants and field interviewers, Wave 6 data collection began with telephone interviewing. As conditions improved in certain parts of the country, in-person interviews began on May 7, 2021. Where and when in-person visits were permitted by Westat, NIDA and FDA, and local jurisdictions, in-person data collection was given priority, though each participant made the final decision regarding their interview mode. All in-person contacts with participants were conducted in compliance with local and state restrictions for COVID-19 mitigation. Of the 36,168 Wave 6 interviews, 69 percent (24,792) were conducted by telephone. Among all telephone interviews at Wave 6, 54 percent (13,297) were conducted before May 7, 2021.

¹² Refusals and other nonresponse at Wave 4.5 or Wave 5.5 for participants who were part of those special data collections were counted in the calculations of “two consecutive waves” of refusal or “three consecutive waves” of nonresponse. However, for youth and adults ages 18 to 19 approached for re-interview by telephone at Wave 5.5 and for adults selected for the PATH-ATS, refusing or nonresponding to the interview was not counted in the calculation of “two consecutive waves” of refusal nor “three consecutive waves” of nonresponse.

Table 2-2. PATH Study data and biospecimen collection summary by wave, cohort, and interview type

Wave	Data collection start date	Data collection end date	Cohort ^a	Interview type ^b	Interviews conducted	Urine specimens collected ^c	Blood specimens collected ^d
1	September 12, 2013	December 14, 2014	Wave 1 Cohort	Adults	32,320	21,801	14,520
				Youth	13,651	N/A	N/A
2	October 23, 2014	October 30, 2015	Wave 1 Cohort	Adults	28,362	13,696	908
				Youth	12,172	N/A	N/A
3	October 19, 2015	October 23, 2016	Wave 1 Cohort	Adults	28,148	14,979	835
				Youth	11,814	N/A	N/A
4	December 1, 2016	January 3, 2018	Wave 1 Cohort	Adults	27,757	16,131	890
				Youth	11,059	9,892	N/A
			Wave 4 Cohort	Adults	33,644	20,928	3,602
				Youth	14,793	13,095	N/A
4.5	December 1, 2017	December 1, 2018	Wave 1 Cohort	Youth	8,761	N/A	N/A
			Wave 4 Cohort	Youth	12,918	N/A	N/A
5	December 1, 2018	November 30, 2019	Wave 1 Cohort	Adults	28,970	11,270	1,640
				Youth	6,760	5,948	N/A
			Wave 4 Cohort	Adults	32,687	11,429	1,929
				Youth	11,976	10,490	N/A
5.5	July 3, 2020 ^e	December 31, 2020	Wave 4 Cohort	Adults	3,628	N/A	N/A
				Youth	7,129	N/A	N/A
PATH-ATS	September 10, 2020	December 20, 2020	Wave 4 Cohort	Adults	8,874	N/A	N/A
6	March 1, 2021	November 30, 2021	Wave 1 Cohort	Adults	25,604	N/A	N/A
				Youth	2,377	N/A	N/A
			Wave 4 Cohort	Adults	29,516	N/A	N/A
				Youth	5,585	N/A	N/A

^a As illustrated in Figure 1 and Table 2-1, there is substantial overlap between the members of the Wave 1 Cohort and the Wave 4 Cohort, and therefore in the counts of interviews conducted by cohort (for relevant waves).

^b Only Wave 4 Cohort adults ages 18 and 19, and Wave 4 Cohort youth ages 13 to 17 completed interviews at Wave 5.5. Only a subsample of Wave 4 Cohort adults ages 20 and older were interviewed for the PATH-ATS. At Wave 6, all interview respondents were at least age 14. For other waves, there was no cohort restriction or subsampling, and the age range was 18 and older for adults and 12 to 17 for youth.

^c The PATH Study requested urine specimens from: all first-time adult interview respondents at Waves 1 to 4 and Wave 5; subsets of other adult respondents at Waves 2 to 4 and Wave 5; all youth at Wave 4; and a subset of youth respondents at Wave 5.

^d The PATH Study requested blood specimens from first-time adult interview respondents at Waves 1-4 and Wave 5.

^e Data collection for Wave 5.5 originally began in person on December 1, 2019, but was suspended due to the COVID-19 pandemic. The released data for this wave were collected via telephone starting on July 3, 2020.

3. Methodology

This chapter first discusses the factors affecting nonresponse bias, and then describes the methods used for evaluating potential nonresponse bias in this report.

3.1 Factors Affecting Nonresponse Bias

Bias is the difference between a survey estimate and the actual population value. Although nonresponse bias can be a major concern in multi-stage household surveys, nonresponse does not necessarily lead to nonresponse bias in survey estimates. Assuming nonresponse to be a fixed property of an individual in the population, the nonresponse bias of an estimate can be expressed mathematically to show the relationship between the bias and two factors—the amount of nonresponse and the difference between respondents and nonrespondents (Groves, 2006):

$$Bias(\bar{y}_r) = (1 - r)(\bar{Y}_r - \bar{Y}_m)$$

where $\bar{y}_r$ is the estimated population mean based on the sample respondents only, r is the response rate in the target population, $\bar{Y}_r$ is the mean for respondents in the target population, and $\bar{Y}_m$ is the mean for nonrespondents in the target population.

In practice, nonresponse is not necessarily a fixed property of individuals in the population and the magnitude of nonresponse bias depends on the correlation between response propensity and the measure of interest. Within the same survey, different estimates can be subject to different levels of nonresponse bias. Some measures, unrelated to the propensity to respond, can be immune from the biasing effect of nonresponse; others, in the same survey, can be subject to large biases (Groves, 2006). For example, if people who smoke cigarettes were less inclined to respond to the PATH Study than those who do not smoke cigarettes, then the estimated prevalence of cigarette smoking would be subject to bias, prior to any weighting adjustments for nonresponse. However, if users and nonusers of smokeless tobacco were equally likely to respond to the study, then the estimated prevalence of smokeless tobacco use would not be biased due to nonresponse. If propensity to respond is completely random, then nonresponse reduces sample sizes for analysis but does not lead to bias, even if the overall response rate is lower than desired.

Survey practitioners often attempt to decrease potential nonresponse bias by increasing the overall response rate and also improving cooperation from “difficult to reach” subgroups. Effective statistical adjustments can also help reduce potential nonresponse biases in survey estimates.

3.2 Analyses for Evaluating Wave 6 Potential Nonresponse Bias

It is not always possible to measure the actual bias due to nonresponse; however, various approaches can be used to help identify potential sources of nonresponse bias. At Wave 6, the Wave 1 Cohort data serve the purpose of longitudinal analysis, whereas the Wave 4 Cohort data support both cross-sectional and longitudinal analyses. Separate sets of analyses were conducted for the two cohorts to assess potential nonresponse bias in estimates using interview data from Wave 6 of the PATH Study.

3.2.1 Weights and Corresponding Analysis Perspective Used by Cohort

As described in Appendix section B.9 of the PATH Study RUFs User Guide, the study provides three different types of weights for researchers analyzing Wave 6 data: a single-wave weight, an all-waves weight, and a special-collection all-waves weight. Separately for each cohort, a single-wave weight was created for all cohort members who responded at Wave 6. Similarly, an all-waves weight was created for Wave 6 respondents who also responded at each primary wave relevant to the cohort. A special collection all-waves weight was created for each cohort, and applies to Wave 6 respondents who also responded at all previous waves relevant to the cohort and at all previous special data collections as appropriate. Weighting adjustments are often used to account for differential response propensities across population subgroups. (Among numerous sources, the United Nations handbook on household surveys (2005, Chapter 6) and Särndal and Lundström (2005) discuss the methods and theory of using weight adjustments for nonresponse.)

In terms of conducting longitudinal analyses using Wave 1 Cohort data, the single-wave weight has less utility than the all-waves weight. The single-wave weight allows researchers to relate a behavior or outcome at a later wave (Wave *n*) to characteristics of the same person at the first wave of a cohort. An example of an analysis that would use the single-wave weight for Wave 6 data from the Wave 1 Cohort would be estimating the proportion of Wave 6 e-cigarette users who used e-cigarettes at Wave 1. This is a longitudinal analysis using the participants who have data at both

Wave 1 and Wave 6. However, if it is desired to relate Wave *n* outcomes to characteristics from an earlier wave that is not the first wave of a cohort then the all-waves weight is needed. An example of an all-waves analysis would be using logistic regression to estimate a person's e-cigarette use at Wave 6 from the person's e-cigarette use and demographic characteristics at Waves 1 through 5.

As a result, and as waves of data collection accrue, the preponderance of analyses for the Wave 1 Cohort are likely to use all-waves respondents and their corresponding weights. To address the majority of analyses the research community will likely conduct using Wave 6 data from this cohort, the evaluation of potential nonresponse bias in estimates based on the Wave 6 interview data for the Wave 1 Cohort will be from an all-waves perspective.

The Wave 4 Cohort will be evaluated from a single-wave perspective. This is because it is strongly recommended that researchers use the Wave 4 Cohort, along with the Wave 4 Cohort single-wave weight, for any cross-sectional analyses using Wave 6 data (like those presented in Sections 5.1.3, 5.1.4, 5.2.3, and 5.2.4).

3.2.2 Comparison of Response Rates Across Subgroups

Although a response rate does not yield direct estimates of potential nonresponse biases on key measures, examining response rates by subgroups (for example, males versus females) may reveal sources of potential nonresponse bias. Wave 6 interview response rates were calculated for the Wave 1 Cohort and the Wave 4 Cohort, separately, as described below, using the RR3 response rate formula provided by the American Association for Public Opinion Research (AAPOR, 2016).

For the Wave 1 Cohort, the Wave 6 interview all-waves response rates are conditioning on Wave 1 response, so the subgroups are based on Wave 1 information. In contrast, the Wave 6 interview single-wave response rates for the Wave 4 Cohort are conditioning on Wave 4 response, so most of the subgroups are based on Wave 4 information—the exceptions are the recruitment wave subgroups (which distinguish cohort members recruited at Wave 1 from those recruited at Wave 4) and the response history subgroups (which distinguish participants who responded at Wave 5 from those who did not).

At Wave 6, participants ages 18 and older were asked to complete an adult interview and those younger than age 18 were asked to complete a youth interview. Wave 6 nonrespondents do not have

a Wave 6 interview date, so their Wave 6 age and interview eligibility (i.e., adult interview versus youth interview) were determined using the latest available date of birth or age information. Each study participant was assigned an “anniversary month” for Wave 6, based primarily on the most recent interview date of the study member(s) in their household from prior waves. The age classification date for Wave 6 nonrespondents was 1 month after the last day of their anniversary month or the final date of the Wave 6 data collection, whichever was earlier. Nonrespondents were included in the Wave 6 youth analyses if their age was determined to be younger than 18 on the age classification date, or included in the Wave 6 adult analyses if their age was determined to be 18 or older on the age classification date.

Both weighted and unweighted response rates are reported for the adult and youth interviews. The unweighted response rate measures the success of field operations in obtaining responses from the sampled persons. The weighted response rate estimates the proportion of the population represented by the sampled persons that would have responded if they all had been asked to participate in the study, and thus provides a measure of the potential impact of nonresponse on the study estimates.

The basic design weight associated with the Wave 1 sample selection, also known as the Wave 1 IPS weight, was used for calculating the Wave 1 Cohort weighted all-waves response rates. A Wave 4 IPS weight does not exist for the Wave 4 Cohort. As discussed in Chapter 2, the Wave 4 Cohort consists of the Wave 4 adult and youth respondents from the Wave 1 sample (except those not in the CNP at Wave 4) and the respondents from the replenishment sample. The Wave 1 sample and the Wave 4 replenishment sample were selected at different times from two different sampling frames, so the probabilities of selection (and IPS weights) were computed for the two samples separately. An adjustment to the IPS weights computed for the separate samples is needed for persons with multiple chances of selection for the Wave 4 Cohort. The information needed to make this adjustment (e.g., Wave 1 age, race, and tobacco-use status from which the probability of an adult being selected at Wave 1 could be determined) is known only for some replenishment sample respondents and is unknown for persons from the replenishment sample who did not respond at Wave 4. Because the Wave 4 cross-sectional weight is the closest available approximation to an IPS

weight for the Wave 4 Cohort,¹³ the Wave 4 cross-sectional weight was used for calculating the Wave 4 Cohort weighted single-wave response rates.

3.2.3 Comparison Using Data from Cohort's First Wave

In a longitudinal study, the information collected for respondents at the cohort's first wave is available regardless of whether they are respondents or nonrespondents for future waves, and thus can be used for evaluating potential nonresponse bias (Bose & West, 2002; Javitz & Wagner, 2005; Brownstein et al., 2009). For the PATH Study Wave 1 Cohort, estimates of Wave 1 demographic, socio-economic, and tobacco-use characteristics based on Wave 6 all-waves respondents were compared to estimates of the same Wave 1 characteristics based on cohort members eligible to be Wave 6 all-waves respondents. For the Wave 4 Cohort, estimates of Wave 4 demographic, socio-economic, and tobacco-use characteristics based on Wave 6 respondents were compared to estimates of the same Wave 4 characteristics based on cohort members eligible to be Wave 6 respondents.

For the Wave 1 Cohort, this comparison helps identify characteristics that might be associated with bias due to nonresponse at any primary wave since Wave 1, after compensating for possible undercoverage of the target population at Wave 1 (due to sampling frame deficiencies) and Wave 1 nonresponse. For the Wave 4 Cohort, the comparison helps identify characteristics that might be associated with bias due to nonresponse at Wave 6, after compensating for possible undercoverage and Wave 4 nonresponse. Participants known to have been ineligible at any wave were excluded from the Wave 1 Cohort all-waves response rates and nonresponse bias analyses in this report. This is analogous to the single-wave approach; participants known to be ineligible at Wave 6 were excluded from the Wave 4 Cohort single-wave analyses. In this sense, the focus of both analyses is on assessing bias due to nonresponse by participants known to be eligible or whose eligibility status was unknown.

¹³ The sum of the Wave 4 cross-sectional weights provides the closest approximation to the Wave 4 Cohort target population at Wave 4, i.e., the CNP ages 10 and older as of Wave 4. Details on the creation of the Wave 4 cross-sectional weight can be found in Appendix sections B.4.3 and B.4.4 of the PATH Study RUFs User Guide.

Wave 1 Cohort

One set of Wave 1 Cohort estimates was computed based on the cohort members who were eligible to be Wave 6 all-waves respondents. These estimates were computed using the Wave 1 cross-sectional weight, which was designed to reduce potential nonresponse bias at Wave 1.

Two sets of Wave 1 Cohort estimates were computed based on the Wave 6 all-waves respondents. The first set of estimates used the Wave 1 cross-sectional weight. Comparing this set of estimates to the estimates based on cohort members eligible to be Wave 6 all-waves respondents (which also used the Wave 1 cross-sectional weight) shows the representativeness of the Wave 6 all-waves respondents *before* statistical adjustment for nonresponse at any primary wave since Wave 1.

However, the study provides a Wave 6 all-waves weight for the Wave 1 Cohort that was specifically designed to reduce potential bias due to nonresponse at any primary wave since Wave 1. The Wave 5 nonresponse-adjusted all-waves weights for the Wave 1 Cohort (prior to raking and trimming) served as the initial weights for developing the Wave 6 all-waves weights. The initial weights were adjusted to account for nonresponse to the Wave 6 interview, and the resulting weights were raked to control totals. Some of the control totals came from the 2013 ACS; others involving tobacco use were sample-based rather than population-based and reflected estimated Wave 1 population characteristics. Raking to sample-based control totals, often employed in longitudinal studies (see, for example, Brick, L  , & West (2003)), can limit “drifting” in some important baseline characteristics that might arise through the applications of nonresponse adjustments over time. Lundstr  m and S  r  ndal (1999) provide theoretical discussions about sample-based calibration together with empirical evidence that such calibration can help reduce both variance and nonresponse bias. General discussions about the calibration method can be found in S  r  ndal and Lundstr  m (2005) and S  r  ndal (2007). Details about the construction of the Wave 6 all-waves weight for the Wave 1 Cohort appear in Appendix section B.9.1 of the PATH Study RUFs User Guide.

Therefore, the second set of estimates based on the Wave 6 all-waves respondents used the Wave 6 all-waves weight for the Wave 1 Cohort. Comparing this set of estimates to the estimates based on Wave 1 Cohort members eligible to be Wave 6 all-waves respondents shows the representativeness of the Wave 6 all-waves respondents *after* statistical adjustment for nonresponse at any primary wave. That is, the results of this comparison show the extent to which the statistical weighting

adjustments might have reduced potential nonresponse bias in Wave 1 Cohort estimates based on the Wave 6 all-waves respondents.

For ease of reference, the terms “before Wave 6 weighting adjustment” and “after Wave 6 weighting adjustment” are used to refer to the two sets of estimates and comparisons in Sections 4.1.2 and 4.2.2.

Wave 4 Cohort

One set of Wave 4 Cohort estimates was computed based on the cohort members who were eligible for the Wave 6 interview. These estimates were computed using the Wave 4 cross-sectional weight, which was designed to reduce potential nonresponse bias at Wave 4.

Two sets of Wave 4 Cohort estimates were computed based on the Wave 6 respondents. The first set of estimates used the Wave 4 cross-sectional weight. Comparing this set of estimates to the estimates based on the Wave 4 cohort members eligible to be Wave 6 respondents (which also used the Wave 4 cross-sectional weight) shows the representativeness of the Wave 6 respondents *before* statistical adjustment for nonresponse at Wave 6.¹⁴

However, the PATH Study also provides a Wave 6 single-wave weight for the Wave 4 Cohort and this weight was devised to reduce potential bias due to nonresponse at Wave 6. The Wave 4 cross-sectional weights served as the initial weights for developing the Wave 6 single-wave weights for the Wave 4 Cohort. The initial weights were adjusted to account for nonresponse to the Wave 6 interview and the resulting weights were raked to control totals. Some of the control totals came from the 2016 ACS; others involving tobacco use were sample-based rather than population-based and reflected estimated Wave 4 population characteristics. Details about the construction of the Wave 6 single-wave weight for the Wave 4 Cohort appear in Appendix section B.9.6 of the PATH Study RUFs User Guide.

Hence, the second set of estimates based on the Wave 6 respondents used the Wave 6 single-wave weight for the Wave 4 Cohort. Comparing this set of estimates to the estimates based on Wave 4 Cohort members eligible to be Wave 6 respondents shows the representativeness of the Wave 6

¹⁴ As noted in Section 3.2.1, the special data collections were ignored for this report.

respondents *after* statistical adjustment for nonresponse. The results of this comparison show the extent to which the statistical weighting adjustments might have reduced potential nonresponse bias in Wave 4 Cohort estimates based on the Wave 6 respondents.

Similar to the Wave 1 Cohort analyses, the terms “before Wave 6 weighting adjustment” and “after Wave 6 weighting adjustment” are used to refer to the two sets of estimates and comparisons in Sections 5.1.2 and 5.2.2.

3.2.4 Wave 4 Cohort: Comparison to External Data Sources

For the first three waves of the PATH Study, the Wave 1 Cohort can be used to produce cross-sectional estimates (which are approximate in nature for Waves 2 and 3). For Wave 4 and subsequent waves up to and including Wave 6, the Wave 4 Cohort serves the purpose of cross-sectional analysis (which is approximate in nature after Wave 4). A widely used approach for assessing potential nonresponse bias in cross-sectional estimates is benchmarking to external data sources. A strength of this method is that estimates from external sources are independent of the PATH Study. On the other hand, key outcomes from the PATH Study are not necessarily available from external sources; the measurements may not be exactly the same across studies, and the external sources may be subject to coverage and nonresponse errors as well (Groves, 2006).

However, responses to the Wave 6 questions on cigarette smoking and electronic nicotine delivery system (ENDS)¹⁵ use can be compared with estimates from other national studies of tobacco and health (for the most similar time frames for which data were available). Wave 6 estimates for adults and for youth ages 14 and older were obtained using the Wave 6 single-wave weight for the Wave 4 Cohort.

The external data sources used for analyses of cigarette smoking were the Tobacco Use Supplement to the Current Population Survey 2018-2019 (TUS-CPS 2018-2019), the National Youth Risk Behavior Survey 2019 (NYRBS 2019), the National Health and Nutrition Examination Survey 2017-prepandemic 2020 (NHANES 2017-2020), the National Survey on Drug Use and Health 2020 (NSDUH 2020), Monitoring the Future 2021 (MTF 2021), the National Health Interview Survey

¹⁵ Although the term “electronic nicotine products” is used in the PATH-ATS instrument to represent all electronic nicotine devices such as e-cigarettes, e-cigars, and e-hookahs, this report uses the more technical term “ENDS” because it is more commonly used in the tobacco-use literature.

2021 (NHIS 2021), and the National Youth Tobacco Survey 2021 (NYTS 2021). The external data sources used for comparison of ENDS use were TUS-CPS 2018-2019, NYRBS 2019, NSDUH 2020, MTF 2021, and NYTS 2021. Estimates from these surveys were obtained using public use files, with the exception of MTF 2021. The variance stratum and cluster variables, which are needed to account appropriately for MTF’s complex sample design when estimating the variance of estimates and corresponding confidence intervals, were available only on restricted use files. When computing estimates of youth cigarette-smoking or youth ENDS use, the data from the other national studies (NYRBS 2019, NHANES 2017-2020, NSDUH 2020, MTF 2021, and NYTS 2021) were restricted to ages 14 to 17 to match the age range for PATH Study youth at Wave 6.

With data collection for Wave 6 of the PATH Study occurring in 2021, and given the possible effects of the COVID-19 pandemic on tobacco-use behaviors, each of the external data sources ideally would include data from after the start of the pandemic in mid-March 2020. In addition, over the past 5 years, estimates of ENDS use have steadily increased and estimates of cigarette smoking have slightly decreased. However, the TUS-CPS, NHANES, and NYRBS are not fielded every year and new data are not scheduled to be released until 2023.

Appendixes A and B describe the questions used to define cigarette smoking and ENDS use on these surveys as well as the PATH Study, and outline differences in their target populations. The PATH Study’s reports on nonresponse bias assessments for waves prior to Wave 5.5 included two estimates of adult current cigarette smoking based on NSDUH data—one using NSDUH’s “original definition” and one using a “modified definition.” (See Ryan, Trosclair, & Gfroerer, 2012, for the reasoning for creating a modified definition.) The latter adds the restriction that the adult has smoked at least 100 cigarettes in their lifetime, and is the only definition considered in this report. Although not labeled as such, the NSDUH 2020 estimates of adult current cigarette smoking therefore correspond to NSDUH estimates using the “modified definition” in PATH Study reports about nonresponse bias prior to Wave 5.5.

For both the PATH Study and the external data sources for cigarette smoking and ENDS use, item nonresponse was handled by excluding respondents with missing values for an item from the counts and estimates regarding that item. The proportions of item-missing values were generally very low: no greater than 3 percent for all adult estimates at the overall level, and below 3 percent for youth

estimates at the overall level for all studies except NHANES 2017-2020 and NYRBS 2019.¹⁶ The missing data rates for youth estimates for NHANES 2017-2020 were 4.8 percent for ever tried cigarette smoking and 11.5 percent missing for past 30-day cigarette smoking. For NYRBS 2019, the missing data rates were 27.7 percent for ever tried cigarette smoking, 9.8 percent for past 30-day cigarette smoking, 3.2 percent for ever used ENDS, and 6.3 percent for past 30-day ENDS use among high school students.

3.3 Estimation Method and Software Package

SAS/STAT software version 15.2 was used to calculate all the estimates presented in this report. Fay's balanced repeated replication (BRR) method with a factor of 0.3 was used for variance estimation with the Wave 6 data. This method accounts for the impact of the stratification and clustering involved in the study's sample design and the weighting adjustments made to the IPS weights. For estimates involving data from one of the external sources, different variance estimation methods were used according to each study's sample design. Confidence intervals were estimated using the modified Wilson approach (Wilson, 1927; SAS Institute, 2020). No adjustments were made for multiple comparisons; however, all the statistical tests were based on pre-planned (i.e., not post-hoc) comparisons and are presented in this report.

The preferred approach for testing whether two point estimates differ is to examine the confidence interval for the difference between the two point estimates. If the 95 percent confidence interval does not include zero (i.e., the lower and upper confidence interval endpoints are both negative or both positive), it can be concluded that the difference between the two estimates is statistically significant at the 0.05 significance level.¹⁷ This approach was used for comparisons of the type described in Section 3.2.2, which involve only PATH Study estimates. Due to the nature of these analyses, some of the point estimates and confidence interval endpoints for the difference between two estimates are very close to zero. Although it is usually of no practical importance, a negative sign

¹⁶ However, note that the NSDUH 2020 public use file included only “usable” cases that met specified minimum item response requirements.

¹⁷ Testing whether or not a $100(1-\alpha)$ percent confidence interval for a difference between two population proportions includes zero is equivalent to a two-sided test of the null hypothesis that the difference is zero at the α significance level. See, for example, Hanushek and Jackson (1977) and Heeringa, West, and Berglund (2017, Section 5.6.1).

appears in front of a point estimate or confidence interval endpoint when the exact value is less than zero but the displayed value rounds to zero (e.g., “-0.0%”).

Another approach is to examine whether the confidence intervals for the two point estimates overlap; if the confidence intervals for two proportions do not overlap, then the difference between the two proportions is considered statistically significant. This approach was used to compare estimates between the PATH Study and the external data sources (as described in Section 3.2.3) due to the different complex sample designs used by each study. However, Schenker and Gentleman (2001) show that using the second approach results in a conservative test.

4. Results of Wave 1 Cohort Nonresponse Bias Analyses at Wave 6

This chapter presents results of nonresponse bias analyses for the Wave 1 Cohort interview data at Wave 6. The nonresponse bias analyses for the Wave 1 Cohort at Wave 6 are all conditioning on Wave 1 response. Sections 4.1.1 and 4.2.1 describe how the all-waves response rates vary across select subgroups. Sections 4.1.2 and 4.2.2 compare demographic and socio-economic characteristics as well as tobacco-use status between those eligible to be Wave 6 all-waves respondents and Wave 6 all-waves respondents. The results are presented separately for adults and youth.

4.1 Adult Interview

4.1.1 All-Waves Response Rates Conditioning on Wave 1 Response

Among those in the Wave 1 Cohort, unweighted and weighted all-waves response rates were calculated for the Wave 6 adult interview using the following formulas:

$$RR_{A_AW} = C_{A_AW} / (C_{A_AW} + N_{A_AW} + e_{A_AW} \times U_{A_AW})$$

$$e_{A_AW} = (C_{A_AW} + N_{A_AW}) / (C_{A_AW} + N_{A_AW} + I_{A_AW})$$

where

RR_{A_AW} = Wave 6 adult interview all-waves response rate;

C_{A_AW} = number of Wave 6 adults who were a respondent at each of primary Waves 2 through 6;

N_{A_AW} = number of Wave 6 adults known to be eligible at each wave, but a nonrespondent at least once for primary Waves 2 through 6;

U_{A_AW} = number of Wave 6 adults with unknown eligibility status at least once and never ineligible for primary Waves 2 through 6;

e_{A_AW} = estimated proportion of eligible adults among the Wave 6 adults with unknown eligibility status (from an all-waves perspective); and

I_{A_AW} = number of Wave 6 adults who were ineligible at least once for primary Waves 2 through 6.

Unweighted counts and weighted counts based on the Wave 1 IPS weight were obtained for the all-waves response status categories C_{A_AW} , N_{A_AW} , U_{A_AW} , and I_{A_AW} for the unweighted all-waves response rates (RR_{A_AW}) and weighted all-waves response rates ($RR_{A_W_AW}$), respectively.

Table 4-1 provides all-waves response rates for the Wave 6 adult interview. In addition to the overall row, the table includes rows for subgroups based on Wave 1 sex, age, race/ethnicity, census region, highest education level attained, health insurance status, and tobacco-use status. Participants for whom the Wave 1 interview (or Wave 1 household screener in the case of shadow youth) had missing information for the demographic, socio-economic, or tobacco-use characteristic were excluded from the response rate calculation for that subgroup. In addition, the education level and health insurance status subgroups exclude the Wave 6 adults who were youth or shadow youth at Wave 1 because they were not asked about their highest level of education attained or health insurance status at that time.

The unweighted and weighted all-waves response rates for the Wave 6 adult interview were 51.3 percent and 54.4 percent, respectively. The unweighted and weighted single-wave response rates for the Wave 6 adult interview were higher than the all-waves response rates, at 55.2 percent and 57.5 percent, respectively.

For many subgroups (e.g., males) the weighted all-waves response rate was at least one percentage point higher than the unweighted all-waves response rate, with Non-Hispanic White alone adults having the greatest difference between weighted (54.8 percent) and unweighted (50.2 percent) response rates. Table 4-1 shows some differences in the all-waves response rates across the various subgroups (e.g., male versus female). The largest variations in weighted response rates for Wave 6 were associated with age group and education level. It is generally considered harder to reach younger adults, and the Wave 6 weighted all-waves response rates were lower for participants younger than age 18 at Wave 1 (50.5 percent) and ages 18 to 24 at Wave 1 (47.0 percent) than for those in the older Wave 1 age groups (who had response rates ranging from 52.9 percent to 60.6 percent). Moreover, adults with a bachelor's degree or advanced degree at Wave 1 had a higher propensity to respond at every primary wave (61.9 percent) than adults with less education (who had response rates ranging from 50.6 percent to 54.3 percent). Notable variations in response rate occurred for the other Wave 1 characteristics as well. The weighted all-waves response rate was lower among males (51.4 percent) than among females (57.0 percent); this pattern is consistent with

most household surveys (Groves & Couper, 1998; Stoop, 2005). Non-Hispanic Black alone adults had a moderately higher weighted all-waves response rate (56.8 percent) than the other Wave 1 race/ethnicity groups (53.1 percent for Hispanic, 54.8 percent for Non-Hispanic White alone, and 52.1 percent for Non-Hispanic other race or multiple races). Across the census regions, those residing in the Midwest at Wave 1 had the highest weighted all-waves response rate (56.5 percent) whereas those in the Northeast had the lowest (52.7 percent). Adults with health insurance at Wave 1 were more likely to respond at every primary wave (56.4 percent) than adults without health insurance (48.8 percent). Finally, the weighted all-waves response rates were 8.2 percentage points lower for Wave 1 adult current established tobacco users than for Wave 1 adults who were not current established tobacco users, and 9.7 percentage points lower for Wave 1 youth who had ever used tobacco than for Wave 1 youth who were never users.

Variation in response rates by subgroups is to be expected in large-scale data collection efforts, especially at later waves of a longitudinal study. In particular, Cunradi, Moore, Killoran, and Ames (2005) and Young, Powers, and Bell (2006) have found that people who smoke were less likely to be retained in subsequent waves of surveys than those who do not smoke. Therefore, given the differences in response propensity shown in Table 4-1, it is pertinent to examine the degree to which the Wave 6 all-waves weighting adjustments addressed differences in response propensity. The results of these analyses are discussed in the next section.

Table 4-1. Wave 6 adult interview all-waves response rates conditioning on Wave 1 response

Wave 1 characteristic ^a	<i>C_{A,AW}</i> : Completed Wave 6 adult interview and responded at each primary wave (n)	<i>I_{A,AW}</i> : Ineligible for at least one primary wave (n)	<i>N_{A,AW}</i> : Nonresponse for at least one primary wave and known to be eligible for each primary wave (n)	<i>U_{A,AW}</i> : Nonresponse with unknown eligibility status for at least one primary wave and never ineligible for a primary wave (n)	<i>RR_{A,AW}</i> : Unweighted all-waves response rate ^b (%)	<i>RR_{A,W,AW}</i> : Weighted all-waves response rate ^b (%)
Overall	23,078	2,526	5,844	17,505	51.3	54.4
Sex						
Male	10,878	1,532	3,116	9,287	48.6	51.4
Female	12,174	988	2,718	8,186	53.9	57.0
Age						
Under 18	8,150	338	2,322	5,823	50.6	50.5
18-24	3,873	374	996	3,866	45.8	47.0
25-44	5,315	464	1,244	4,244	50.5	52.9
45-64	4,546	716	928	2,627	58.3	60.6
65 or older	1,191	634	353	932	54.0	54.5
Race/ethnicity ^c						
Hispanic	4,956	459	1,199	3,665	51.8	53.1
Non-Hispanic White alone	12,579	1,381	3,396	9,852	50.2	54.8
Non-Hispanic Black alone	3,392	392	683	2,237	55.5	56.8
Non-Hispanic other race or multiple races	1,808	215	477	1,433	50.3	52.1
Census region						
Northeast	3,408	335	962	2,829	48.7	52.7
Midwest	5,560	619	1,371	3,730	53.7	56.5
South	8,428	1,030	2,156	6,907	49.9	53.0
West	5,682	542	1,355	4,039	52.7	55.6
Education (Wave 1 adults only)						
Less than high school or GED	2,666	741	696	2,347	50.4	51.1
High school graduate	3,210	506	889	2,943	47.8	50.6
Some college but no degree, or associate's degree	5,280	582	1,276	4,161	50.9	54.3
Bachelor's degree or advanced degree	3,713	335	641	2,120	58.7	61.9

Table 4-1. Wave 6 adult interview all-waves response rates conditioning on Wave 1 response (continued)

Wave 1 characteristic ^a	<i>C_{A,AW}</i> : Completed Wave 6 adult interview and responded at each primary wave (n)	<i>I_{A,AW}</i> : Ineligible for at least one primary wave (n)	<i>N_{A,AW}</i> : Nonresponse for at least one primary wave and known to be eligible for each primary wave (n)	<i>U_{A,AW}</i> : Nonresponse with unknown eligibility status for at least one primary wave and never ineligible for a primary wave (n)	<i>RR_{A,AW}</i> : Unweighted all-waves response rate ^b (%)	<i>RR_{A,W,AW}</i> : Weighted all-waves response rate ^b (%)
Health insurance (Wave 1 adults only)						
Yes	12,211	1,742	2,845	8,956	52.9	56.4
No	2,591	410	637	2,518	47.4	48.8
Wave 1 current established tobacco use ^d (Wave 1 adults only)						
Current established tobacco user	6,002	1,219	1,649	5,499	48.4	48.9
Not current established tobacco user	8,535	861	1,783	5,830	54.4	57.1
Wave 1 ever tobacco use ^e (Wave 1 youth only)						
Ever user	1,160	115	446	1,146	43.4	43.4
Never user	5,312	174	1,415	3,340	53.2	53.1

^a For each characteristic, the sum of the counts across the categories may not be equal to the count in the overall row due to missing values.

^b The unweighted adult interview all-waves response rate formula is provided earlier in this section. The weighted adult interview all-waves response rates were calculated using a similar approach except that the Wave 1 IPS weight was used to obtain weighted counts.

^c Hispanic refers to Hispanic or Latino; Black refers to Black or African American. Alone means that the participant reported only one race.

^d A current established user of a tobacco product is someone who currently uses the product every day or some days and: for cigarettes, has smoked at least 100 cigarettes in their lifetime and, for any other tobacco product, has reported they ever used that product fairly regularly. The products covered by the Wave 1 adult interview are cigarettes, traditional cigars, cigarillos, filtered cigars, pipes, smokeless tobacco, snus, hookah, and dissolvable tobacco.

^e An ever user of a tobacco product is defined as someone who has ever used the product, even one or two times. The products covered by the Wave 1 youth interview are cigarettes, traditional cigars, cigarillos, filtered cigars, pipes, smokeless tobacco, snus, hookah, dissolvable tobacco, e-cigarettes, bidis, and kreteks.

4.1.2 Comparison Between Those Eligible to be Wave 6 All-Waves Respondents and Wave 6 All-Waves Respondents (Among the Wave 1 Cohort)

This section compares weighted estimates between the Wave 1 respondents who were eligible to be Wave 6 adult all-waves respondents and the Wave 6 adult all-waves respondents in the Wave 1 Cohort. Section 3.2.3 describes the methods used to obtain these estimates. The weighted estimates cover Wave 1 demographic and socio-economic characteristics (shown in Table 4-2) as well as Wave 1 tobacco-use status (shown in Tables 4-3 and 4-4). Each table shows two sets of comparisons. The first set is for the “before Wave 6 weighting adjustment” comparison, and the second set is for the “after Wave 6 weighting adjustment” comparison.

In Table 4-2, the percentages sum to 100 percent¹⁸ over the categories associated with each characteristic. For example, for both the Wave 6 all-waves respondents and those eligible to be Wave 6 all-waves respondents, the estimated percentage of males and the estimated percentage of females add to 100 percent. Table 4-2 has two columns labeled “Unweighted count” and Tables 4-3 and 4-4 each have two columns labeled “Sample size.” These columns provide the unweighted number of adults in the subgroup (e.g., males) associated with the table row. They are labeled differently in each table because the (weighted) number constitutes the numerator of the corresponding percentage in Table 4-2 (e.g., the estimated percentage of adults who were male), whereas it is the denominator of the corresponding percentage in Tables 4-3 and 4-4 (e.g., for Table 4-3, the estimated percentage of adult males who were current established users of tobacco at Wave 1).¹⁹

¹⁸ The sum may be not exactly 100 percent due to rounding.

¹⁹ These definitions generalize to all tables in this report that include a column heading of “Unweighted count” or “Sample size.”

Table 4-2. Comparison of Wave 1 demographic and socio-economic characteristics between Wave 1 respondents eligible to be Wave 6 adult all-waves respondents and Wave 1 respondents who were Wave 6 adult all-waves respondents

Wave 1 characteristic	Wave 1 respondents who were eligible to be Wave 6 adult all-waves respondents		Wave 1 respondents who were Wave 6 adult all-waves respondents				
	Unweighted count	A: Weighted percentage, using Wave 1 cross-sectional weight [95% confidence interval]	Unweighted count	Before Wave 6 weighting adjustment		After Wave 6 weighting adjustment	
				B: Weighted percentage, using Wave 1 cross-sectional weight [95% confidence interval]	Difference in weighted percentages [B - A] [95% confidence interval]	C: Weighted percentage, using Wave 6 all-waves weight [95% confidence interval]	Difference in weighted percentages [C - A] [95% confidence interval]
Sex							
Male	23,281	48.0% [47.6%, 48.5%]	10,878	45.2% [44.5%, 45.9%]	-2.8% [-3.5%, -2.1%]	48.0% [47.4%, 48.7%]	0.0% [-0.1%, 0.1%]
Female	23,078	52.0% [51.5%, 52.4%]	12,174	54.8% [54.1%, 55.5%]	2.8% [2.1%, 3.5%]	52.0% [51.3%, 52.6%]	-0.0% [-0.1%, 0.1%]
Age							
Under 18	16,295	11.7% [11.4%, 12.0%]	8,150	11.3% [10.9%, 11.7%]	-0.5% [-0.7%, -0.2%]	11.8% [11.4%, 12.2%]	0.0% [-0.0%, 0.1%]
18-24	8,735	11.8% [11.6%, 12.1%]	3,873	10.5% [10.1%, 10.9%]	-1.4% [-1.7%, -1.1%]	12.0% [11.5%, 12.4%]	0.1% [0.0%, 0.2%]
25-44	10,803	31.4% [31.0%, 31.8%]	5,315	30.7% [30.0%, 31.4%]	-0.7% [-1.4%, -0.1%]	31.6% [31.0%, 32.2%]	0.2% [0.1%, 0.4%]
45-64	8,101	30.8% [30.4%, 31.3%]	4,546	34.6% [33.8%, 35.4%]	3.7% [3.0%, 4.4%]	31.0% [30.4%, 31.6%]	0.1% [0.0%, 0.3%]
65 or older	2,476	14.2% [13.8%, 14.5%]	1,191	13.0% [12.4%, 13.6%]	-1.1% [-1.7%, -0.6%]	13.6% [13.2%, 14.1%]	-0.5% [-0.8%, -0.3%]
Race/ethnicity ^a							
Hispanic	9,820	16.2% [15.9%, 16.6%]	4,956	15.8% [15.2%, 16.4%]	-0.5% [-1.1%, 0.1%]	16.3% [15.8%, 16.8%]	0.0% [-0.1%, 0.1%]
Non-Hispanic White alone	25,827	64.6% [64.1%, 65.0%]	12,579	64.9% [64.1%, 65.6%]	0.3% [-0.4%, 1.1%]	64.4% [63.8%, 65.1%]	-0.1% [-0.3%, 0.1%]
Non-Hispanic Black alone	6,312	11.4% [11.2%, 11.7%]	3,392	11.9% [11.4%, 12.3%]	0.4% [-0.0%, 0.8%]	11.4% [11.0%, 11.8%]	-0.0% [-0.1%, 0.1%]
Non-Hispanic other race or multiple races	3,718	7.8% [7.5%, 8.0%]	1,808	7.5% [7.1%, 8.0%]	-0.3% [-0.6%, 0.1%]	7.9% [7.5%, 8.2%]	0.1% [-0.0%, 0.3%]

Table 4-2. Comparison of Wave 1 demographic and socio-economic characteristics between Wave 1 respondents eligible to be Wave 6 adult all-waves respondents and Wave 1 respondents who were Wave 6 adult all-waves respondents (continued)

Wave 1 characteristic	Wave 1 respondents who were eligible to be Wave 6 adult all-waves respondents		Wave 1 respondents who were Wave 6 adult all-waves respondents				
	Unweighted count	A: Weighted percentage, using Wave 1 cross-sectional weight [95% confidence interval]	Unweighted count	Before Wave 6 weighting adjustment		After Wave 6 weighting adjustment	
				B: Weighted percentage, using Wave 1 cross-sectional weight [95% confidence interval]	Difference in weighted percentages [B – A] [95% confidence interval]	C: Weighted percentage, using Wave 6 all-waves weight [95% confidence interval]	Difference in weighted percentages [C – A] [95% confidence interval]
Census region							
Northeast	7,199	18.1% [17.8%, 18.5%]	3,408	17.8% [17.2%, 18.4%]	-0.3% [-0.9%, 0.2%]	18.2% [17.7%, 18.7%]	0.0% [-0.1%, 0.2%]
Midwest	10,661	21.4% [21.0%, 21.8%]	5,560	22.3% [21.6%, 23.1%]	0.9% [0.2%, 1.6%]	21.3% [20.8%, 21.9%]	-0.1% [-0.2%, 0.0%]
South	17,491	37.0% [36.6%, 37.5%]	8,428	35.8% [34.8%, 36.8%]	-1.2% [-2.2%, -0.3%]	37.0% [36.4%, 37.6%]	-0.0% [-0.2%, 0.1%]
West	11,076	23.5% [23.1%, 23.9%]	5,682	24.1% [23.3%, 24.9%]	0.7% [-0.1%, 1.4%]	23.5% [23.0%, 24.1%]	0.1% [-0.1%, 0.2%]
Education (Wave 1 adults only)							
Less than high school or GED	5,709	15.8% [15.4%, 16.2%]	2,666	14.0% [13.4%, 14.6%]	-1.8% [-2.4%, -1.3%]	15.6% [15.0%, 16.2%]	-0.3% [-0.5%, -0.0%]
High school graduate	7,042	24.0% [23.5%, 24.5%]	3,210	21.8% [21.1%, 22.6%]	-2.2% [-2.9%, -1.4%]	23.9% [23.2%, 24.6%]	-0.1% [-0.3%, 0.1%]
Some college but no degree, or associate's degree	10,717	31.6% [31.0%, 32.1%]	5,280	31.4% [30.7%, 32.2%]	-0.2% [-0.8%, 0.5%]	31.6% [30.8%, 32.3%]	-0.0% [-0.2%, 0.2%]
Bachelor's degree or advanced degree	6,474	28.6% [28.1%, 29.2%]	3,713	32.8% [32.0%, 33.6%]	4.2% [3.4%, 4.9%]	29.0% [28.3%, 29.7%]	0.4% [0.2%, 0.5%]
Health insurance (Wave 1 adults only)							
Yes	24,012	85.3% [84.7%, 85.9%]	12,211	87.1% [86.3%, 87.8%]	1.8% [1.3%, 2.2%]	85.6% [84.8%, 86.3%]	0.2% [-0.2%, 0.7%]
No	5,746	14.7% [14.1%, 15.3%]	2,591	12.9% [12.2%, 13.7%]	-1.8% [-2.2%, -1.3%]	14.4% [13.7%, 15.2%]	-0.2% [-0.7%, 0.2%]

^a Hispanic refers to Hispanic or Latino; Black refers to Black or African American. Alone means that the participant reported only one race.

Table 4-3. Comparison of Wave 1 current established tobacco-use estimates between Wave 1 adults eligible to be Wave 6 all-waves respondents and Wave 1 adults who were Wave 6 all-waves respondents*

Wave 1 characteristic ^a	Wave 1 adults who were eligible to be Wave 6 all-waves respondents		Wave 1 adults who were Wave 6 all-waves respondents				
	Sample size	A: Weighted estimate, using Wave 1 cross-sectional weight [95% confidence interval]	Sample size	Before Wave 6 weighting adjustment		After Wave 6 weighting adjustment	
				B: Weighted estimate, using Wave 1 cross-sectional weight [95% confidence interval]	Difference in weighted estimates [B – A] [95% confidence interval]	C: Weighted estimate, using Wave 6 all-waves weight [95% confidence interval]	Difference in weighted estimates [C – A] [95% confidence interval]
Overall	29,298	23.4% [22.8%, 24.0%]	14,537	20.2% [19.5%, 20.9%]	-3.3% [-3.7%, -2.8%]	23.2% [22.5%, 23.9%]	-0.3% [-0.4%, -0.1%]
Sex							
Male	14,612	29.2% [28.3%, 30.0%]	6,695	24.5% [23.3%, 25.6%]	-4.7% [-5.5%, -4.0%]	28.7% [27.6%, 29.8%]	-0.4% [-0.7%, -0.2%]
Female	14,662	18.2% [17.5%, 18.9%]	7,830	16.7% [15.9%, 17.5%]	-1.5% [-2.0%, -1.1%]	18.1% [17.3%, 19.0%]	-0.1% [-0.2%, 0.0%]
Age							
18-24	8,601	28.8% [27.4%, 30.2%]	3,821	24.6% [22.9%, 26.5%]	-4.1% [-5.2%, -3.1%]	28.6% [27.1%, 30.1%]	-0.2% [-0.4%, 0.0%]
25-44	10,591	28.5% [27.6%, 29.5%]	5,220	24.4% [23.2%, 25.6%]	-4.2% [-5.0%, -3.3%]	28.2% [27.0%, 29.4%]	-0.4% [-1.0%, 0.3%]
45-64	7,798	22.1% [21.1%, 23.0%]	4,377	19.3% [18.2%, 20.5%]	-2.7% [-3.4%, -2.1%]	21.6% [20.4%, 22.9%]	-0.4% [-1.1%, 0.2%]
65 or older	2,297	9.7% [8.6%, 11.0%]	1,116	8.3% [6.8%, 10.1%]	-1.4% [-2.4%, -0.4%]	9.5% [7.9%, 11.4%]	-0.2% [-1.4%, 1.0%]
Race/ethnicity ^b							
Hispanic	5,017	17.1% [16.1%, 18.2%]	2,483	15.8% [14.4%, 17.3%]	-1.3% [-2.2%, -0.4%]	17.0% [15.6%, 18.5%]	-0.1% [-0.4%, 0.2%]
Non-Hispanic White alone	17,626	24.9% [24.1%, 25.8%]	8,685	20.6% [19.7%, 21.5%]	-4.4% [-4.9%, -3.8%]	24.6% [23.7%, 25.6%]	-0.3% [-0.5%, -0.1%]
Non-Hispanic Black alone	4,046	26.2% [24.8%, 27.6%]	2,158	26.5% [24.4%, 28.7%]	0.3% [-1.1%, 1.7%]	26.6% [24.8%, 28.5%]	0.5% [0.0%, 0.9%]
Non-Hispanic other race or multiple races	2,188	19.2% [17.5%, 20.9%]	1,023	15.5% [13.3%, 17.9%]	-3.7% [-5.6%, -1.8%]	18.4% [16.1%, 20.9%]	-0.8% [-1.5%, -0.1%]

Table 4-3. Comparison of Wave 1 current established tobacco-use estimates between Wave 1 adults eligible to be Wave 6 all-waves respondents and Wave 1 adults who were Wave 6 all-waves respondents* (continued)

Wave 1 characteristic ^a	Wave 1 adults who were eligible to be Wave 6 all-waves respondents		Wave 1 adults who were Wave 6 all-waves respondents				
	Sample size	A: Weighted estimate, using Wave 1 cross-sectional weight [95% confidence interval]	Sample size	Before Wave 6 weighting adjustment		After Wave 6 weighting adjustment	
				B: Weighted estimate, using Wave 1 cross-sectional weight [95% confidence interval]	Difference In weighted estimates [B – A] [95% confidence interval]	C: Weighted estimate, using Wave 6 all-waves weight [95% confidence interval]	Difference In weighted estimates [C – A] [95% confidence interval]
Census region							
Northeast	4,615	21.8% [20.3%, 23.3%]	2,207	18.3% [16.5%, 20.2%]	-3.5% [-4.4%, -2.6%]	21.5% [19.9%, 23.3%]	-0.3% [-0.5%, -0.0%]
Midwest	6,971	26.2% [24.5%, 27.9%]	3,682	23.1% [21.3%, 25.0%]	-3.1% [-4.0%, -2.1%]	25.8% [24.2%, 27.5%]	-0.3% [-0.7%, 0.0%]
South	11,017	25.2% [23.9%, 26.6%]	5,263	21.8% [20.3%, 23.4%]	-3.4% [-4.3%, -2.5%]	24.9% [23.5%, 26.3%]	-0.3% [-0.6%, -0.1%]
West	6,695	19.4% [17.8%, 21.0%]	3,385	16.4% [14.9%, 18.0%]	-3.0% [-3.9%, -2.1%]	19.3% [17.7%, 21.0%]	-0.1% [-0.3%, 0.2%]
Education (Wave 1 adults only)							
Less than high school or GED	5,524	35.0% [33.3%, 36.8%]	2,586	35.0% [32.6%, 37.4%]	-0.1% [-1.8%, 1.6%]	35.1% [33.3%, 37.0%]	0.1% [-0.6%, 0.8%]
High school graduate	6,812	27.6% [26.2%, 29.1%]	3,110	24.8% [23.2%, 26.5%]	-2.8% [-4.0%, -1.7%]	27.4% [25.8%, 29.0%]	-0.3% [-0.7%, 0.2%]
Some college but no degree, or associate's degree	10,473	26.1% [25.0%, 27.2%]	5,156	22.6% [21.4%, 23.8%]	-3.5% [-4.3%, -2.7%]	25.8% [24.6%, 27.0%]	-0.3% [-0.6%, -0.1%]
Bachelor's degree or advanced degree	6,320	10.7% [9.9%, 11.4%]	3,633	8.6% [7.8%, 9.6%]	-2.0% [-2.5%, -1.5%]	10.6% [9.6%, 11.6%]	-0.1% [-0.3%, 0.1%]
Health insurance (Wave 1 adults only)							
Yes	23,368	20.9% [20.3%, 21.5%]	11,897	17.7% [17.1%, 18.4%]	-3.2% [-3.6%, -2.7%]	20.3% [19.6%, 21.1%]	-0.6% [-0.8%, -0.3%]
No	5,609	38.0% [36.4%, 39.7%]	2,527	36.5% [34.1%, 38.9%]	-1.6% [-3.3%, 0.1%]	39.7% [37.4%, 42.0%]	1.6% [0.1%, 3.2%]

*A current established user of a tobacco product is someone who currently uses the product every day or some days and: for cigarettes, has smoked at least 100 cigarettes in their lifetime and, for any other tobacco product, has reported they ever used that product fairly regularly. The products covered by the Wave 1 adult Interview are cigarettes, traditional cigars, cigarillos, filtered cigars, pipes, smokeless tobacco, snus, hookah, dissolvable tobacco, and e-cigarettes.

^a For each characteristic, the sum of the counts across the categories may not be equal to the count in the overall row due to missing values.

^b Hispanic refers to Hispanic or Latino; Black refers to Black or African American. Alone means that the participant reported only one race.

Table 4-4. Comparison of Wave 1 ever tobacco-use estimates between Wave 1 youth eligible to be Wave 6 all-waves respondents and Wave 1 youth who were Wave 6 all-waves respondents (among Wave 6 adults)*

Wave 1 characteristic ^a	Wave 1 youth who were eligible to be Wave 6 adult all-waves respondents		Wave 1 youth who were Wave 6 adult all-waves respondents				
	Sample size	A: Weighted estimate, using Wave 1 cross-sectional weight [95% confidence interval]	Sample size	Before Wave 6 weighting adjustment		After Wave 6 weighting adjustment	
				B: Weighted estimate, using Wave 1 cross-sectional weight [95% confidence interval]	Difference in weighted estimates [B – A] [95% confidence interval]	C: Weighted estimate, using Wave 6 all-waves weight [95% confidence interval]	Difference in weighted estimates [C – A] [95% confidence interval]
Overall	12,819	21.4% [20.4%, 22.4%]	6,472	17.9% [16.8%, 19.0%]	-3.5% [-4.3%, -2.7%]	20.5% [19.5%, 21.5%]	-0.9% [-1.4%, -0.4%]
Sex							
Male	6,450	22.7% [21.5%, 23.9%]	3,183	19.4% [18.0%, 20.9%]	-3.3% [-4.5%, -2.1%]	21.7% [20.3%, 23.2%]	-1.0% [-1.8%, -0.2%]
Female	6,339	20.1% [19.0%, 21.3%]	3,278	16.5% [15.0%, 18.0%]	-3.7% [-4.7%, -2.6%]	19.3% [17.9%, 20.7%]	-0.8% [-1.6%, -0.1%]
Race/ethnicity ^b							
Hispanic	3,619	21.0% [19.4%, 22.8%]	1,898	17.6% [15.5%, 19.9%]	-3.4% [-5.0%, -1.8%]	20.0% [18.0%, 22.1%]	-1.1% [-2.4%, 0.2%]
Non-Hispanic White alone	6,145	23.1% [21.6%, 24.6%]	2,928	19.0% [17.4%, 20.6%]	-4.1% [-5.3%, -3.0%]	21.7% [20.2%, 23.4%]	-1.3% [-2.1%, -0.5%]
Non-Hispanic Black alone	1,682	18.3% [16.3%, 20.4%]	924	18.1% [15.3%, 21.2%]	-0.2% [-2.0%, 1.6%]	20.0% [17.2%, 23.0%]	1.7% [-0.0%, 3.4%]
Non-Hispanic other race or multiple races	1,150	18.5% [16.4%, 20.8%]	611	13.7% [11.2%, 16.7%]	-4.8% [-7.1%, -2.5%]	16.2% [13.5%, 19.3%]	-2.3% [-4.6%, -0.1%]
Census region							
Northeast	1,937	20.8% [18.6%, 23.2%]	907	17.5% [15.2%, 20.1%]	-3.3% [-5.3%, -1.3%]	19.6% [16.9%, 22.6%]	-1.2% [-2.9%, 0.6%]
Midwest	2,795	21.8% [19.8%, 24.1%]	1,447	19.5% [17.2%, 22.0%]	-2.4% [-3.8%, -0.9%]	22.4% [19.8%, 25.2%]	0.6% [-0.9%, 2.0%]
South	4,826	21.9% [20.4%, 23.6%]	2,395	17.4% [15.7%, 19.3%]	-4.5% [-6.0%, -3.1%]	20.1% [18.3%, 22.0%]	-1.9% [-3.2%, -0.6%]
West	3,261	20.7% [18.5%, 23.0%]	1,723	17.4% [15.3%, 19.7%]	-3.3% [-4.8%, -1.7%]	20.0% [17.7%, 22.5%]	-0.7% [-2.3%, 1.0%]

*An ever user of a tobacco product is someone who has ever used the product, even one or two times. The products covered by the Wave 1 youth interview are cigarettes, traditional cigars, cigarillos, filtered cigars, pipes, smokeless tobacco, snus, hookah, e-cigarettes, dissolvable tobacco, bidis, and kreteks.

^a For each characteristic, the sum of the counts across the categories may not be equal to the count in the overall row due to missing values.

^b Hispanic refers to Hispanic or Latino; Black refers to Black or African American. Alone means that the participant reported only one race.

The majority of the 95 percent confidence intervals for the estimated differences in the first set of comparisons in Table 4-2 did not include zero. Prior to Wave 6 all-waves weighting adjustments, the most noticeable underrepresentation among the Wave 6 adult all-waves respondents was for males, who tend to have a lower response propensity than females in most household surveys (Groves & Couper, 1998; Stoop, 2005). Adults younger than age 45 and older than age 64 at Wave 1, those who resided in the South census region at Wave 1, and adults who had never attended college at Wave 1 were also underrepresented among the Wave 6 adult all-waves respondents; whereas adults ages 45 to 64 at Wave 1, those residing in the Midwest census region at Wave 1, adults with a bachelor's degree or advanced degree at Wave 1, and adults with health insurance at Wave 1 were overrepresented.

As shown in the last column of Table 4-2, the Wave 6 all-waves weighting adjustments for the Wave 1 Cohort reduced all the differences between the Wave 1 respondents eligible to be Wave 6 adult all-waves respondents and the Wave 1 respondents who were Wave 6 all-waves respondents. The 95 percent confidence intervals for the estimated differences either included or were in the proximity of zero, with three exceptions. Among the Wave 6 adult all-waves respondents, adults ages 65 and older at Wave 1 were slightly underrepresented, whereas adults ages 25 to 44 at Wave 1 and adults with a bachelor's degree or advanced degree at Wave 1 were slightly overrepresented. However, the magnitude of these differences is unlikely to be of practical importance.

Tables 4-3 and 4-4 show the comparisons for the Wave 1 tobacco-use measures between those eligible to be Wave 6 adult all-waves respondents and the Wave 6 adult all-waves respondents, among the Wave 1 Cohort. The Wave 1 respondents were divided into two groups based on their Wave 1 interview type (i.e., adult versus youth) and analyzed separately. Table 4-3 covers current established tobacco use for the Wave 1 adult respondents and Table 4-4 covers ever use of tobacco for the Wave 1 youth respondents.

For the “before Wave 6 weighting adjustment” estimates, most of the 95 percent confidence intervals did not include zero, with the differences consistently indicating an underrepresentation of Wave 1 tobacco users among the Wave 6 adult all-waves respondents. The Wave 6 all-waves weighting adjustments for the Wave 1 Cohort greatly reduced almost all of the differences and the majority of the corresponding confidence intervals either included or were in the proximity of zero. However, using the Wave 6 all-waves weight, the Wave 6 adult all-waves respondents tended to

underrepresent Wave 1 current established users of tobacco overall, and among males, non-Hispanic White alone adults, non-Hispanic adults of other or multiple races, adults who resided in the South census region at Wave 1, adults with some college but no degree or with an associate's degree at Wave 1, and adults with health insurance at Wave 1; whereas Wave 1 current established users were slightly overrepresented among adults with no health insurance at Wave 1 (see Table 4-3). After Wave 6 weighting adjustments, the Wave 6 adult all-waves respondents also tended to underrepresent Wave 1 ever users of tobacco overall, and among males, females, non-Hispanic White alone adults, non-Hispanic adults of other or multiple races, and adults in the South region who were youth at Wave 1 (see Table 4-4).

Overall, these results reflect positively on the use of demographic, socio-economic, and tobacco-use variables from Wave 1 for calibrating the Wave 6 all-waves weight for the Wave 1 Cohort. However, they reflect the challenges faced by panel studies as waves of nonresponse accrue. Assuming that the Wave 1 characteristics in Tables 4-2 through 4-4 are correlated with key tobacco- and health-related outcome measures at Wave 6 of the PATH Study, the results suggest a small amount of potential bias in the adult estimates for the Wave 1 Cohort due to nonresponse at any primary wave since Wave 1. Researchers are encouraged to think carefully about the choice of covariates in their analyses and to consider conducting sensitivity analyses to support their findings.

4.2 Youth Interview

4.2.1 All-Waves Response Rates Conditioning on Wave 1 Response

Among those in the Wave 1 Cohort, unweighted and weighted all-waves response rates were calculated for the Wave 6 youth interview using the following formulas:

$$RR_{Y_AW} = C_{Y_AW} / (C_{Y_AW} + N_{Y_AW} + e_{Y_AW} \times U_{Y_AW})$$

$$e_{Y_AW} = (C_{Y_AW} + N_{Y_AW}) / (C_{Y_AW} + N_{Y_AW} + I_{Y_AW})$$

where

RR_{Y_AW} = Wave 6 youth interview all-waves response rate;

C_{Y_AW} = number of Wave 6 youth who were respondents at each of primary Waves 2 through 6;

- N_{Y_AW} = number of Wave 6 youth known to be eligible at each wave, but a nonrespondent at least once for primary Waves 2 through 6;
- U_{Y_AW} = number of Wave 6 youth with unknown eligibility status at least once and never ineligible for primary Waves 2 through 6;
- e_{Y_AW} = estimated proportion of eligible youth among the Wave 6 youth with unknown eligibility status (from an all-waves perspective); and
- I_{Y_AW} = number of Wave 6 youth who were ineligible at least once for primary Waves 2 through 6.

Unweighted counts and weighted counts based on the Wave 1 IPS weight were obtained for the response status categories C_{Y_AW} , N_{Y_AW} , U_{Y_AW} , and I_{Y_AW} for the unweighted all-waves response rates (RR_{Y_AW}) and weighted all-waves response rates ($RR_{Y_W_AW}$), respectively.

Table 4-5 provides all-waves response rates for the Wave 6 youth interview. In addition to the overall row, the table includes rows for subgroups based on Wave 1 sex, race/ethnicity, and census region.²⁰ Participants for whom the Wave 1 interview (or Wave 1 household screener in the case of shadow youth) had missing information for the demographic characteristic were excluded from the response rate calculation for that subgroup.

²⁰ Response rates by Wave 1 tobacco-use status are not included in Table 4-5 because almost all of the Wave 6 youth in the Wave 1 Cohort were shadow youth at Wave 1 and so were not asked about their tobacco use at that time. These participants were ages 9 and 10 at Wave 1, and responses rates are not provided by Wave 1 single year of age.

Table 4-5. Wave 6 youth interview all-waves response rates conditioning on Wave 1 response

Wave 1 characteristic ^a	<i>C_{Y,AW}</i> : Completed Wave 6 youth interview and responded at each primary wave (n)	<i>I_{Y,AW}</i> : Ineligible for at least one primary wave (n)	<i>N_{Y,AW}</i> : Nonresponse for at least one primary wave and known to be eligible for each primary wave (n)	<i>U_{Y,AW}</i> : Nonresponse with unknown eligibility status for at least one primary wave and never ineligible for a primary wave (n)	<i>RR_{Y,AW}</i> : Unweighted all- waves response rate ^b (%)	<i>RR_{Y,W,AW}</i> : Weighted all- waves response rate ^b (%)
Overall	2,242	33	477	1,473	53.7	53.7
Sex						
Male	1,163	21	274	752	53.4	53.5
Female	1,079	12	202	719	54.1	54.1
Race/ethnicity ^c						
Hispanic	638	15	143	403	54.2	54.2
Non-Hispanic White alone	1,094	4	228	730	53.4	53.4
Non-Hispanic Black alone	311	7	59	203	54.6	54.2
Non-Hispanic other race or multiple races	197	7	46	135	52.6	52.7
Census region						
Northeast	305	7	69	226	51.2	51.5
Midwest	532	9	107	329	55.2	55.2
South	807	7	194	550	52.2	51.8
West	598	10	107	368	56.0	56.2

^a For each characteristic, the sum of the counts across the categories may not be equal to the count in the overall row due to missing values.

^b The unweighted youth interview all-waves response rate formula is provided earlier in this section. The weighted youth interview all-waves response rates were calculated using a similar approach except that the Wave 1 IPS weight was used to obtain weighted counts.

^c Hispanic refers to Hispanic or Latino; Black refers to Black or African American. Alone means that the participant reported only one race.

The unweighted and weighted all-waves response rates for the Wave 6 youth interview were both 53.7 percent; the unweighted and weighted all-waves response rates were also very similar to each other for most subgroups. The unweighted and weighted single-wave response rates for the Wave 6 youth interview were higher than the all-waves response rates, at 56.7 percent and 56.6 percent, respectively.

The Wave 6 weighted all-waves response rates varied most by Wave 1 census region, to a lesser extent by race/ethnicity, and only slightly by sex. Similar to the response rates observed by Wave 1 census region for adults, youth in the Northeast and in the South at Wave 1 had the lowest weighted Wave 6 all-waves youth response rates (51.5 percent and 51.8 percent, respectively). Weighted all-waves response rates also differed somewhat across Wave 1 race/ethnicity groups, with the highest among non-Hispanic Black alone youth and Hispanic youth (each 54.2 percent) and lowest among non-Hispanic youth of other or multiple races (52.7 percent).

As mentioned in Section 4.1.1, variation in response rates by subgroups is to be expected in large-scale data collection efforts. None of the differences among the demographic subgroups in Table 4-5 causes serious concern about potential nonresponse bias in the Wave 6 youth interview estimates for all-waves respondents from the Wave 1 Cohort. However, it is pertinent to examine the degree to which the Wave 6 all-waves weighting adjustments addressed differences in response propensity. The results of these analyses are discussed in the next section.

4.2.2 Comparison Between Those Eligible to be Wave 6 All-Waves Respondents and Wave 6 All-Waves Respondents (Among the Wave 1 Cohort)

This section compares weighted estimates between the Wave 1 respondents who were eligible to be Wave 6 youth all-waves respondents and the Wave 6 youth all-waves respondents in the Wave 1 Cohort. The weighted estimates in Table 4-6 cover Wave 1 demographic characteristics. The structure of this table is similar to Table 4-2 in Section 4.1.2 for the adult interview. For example, the percentages sum to 100 percent²¹ over the categories associated with each characteristic.

²¹ The sum may be not exactly 100 percent due to rounding.

The estimated 95 percent confidence intervals for both the “before Wave 6 weighting adjustment” differences and the “after Wave 6 weighting adjustment” differences included zero for all of the Wave 1 demographics examined. These results reflect positively on the use of demographic variables from Wave 1 for calibrating the Wave 6 all-waves weight for the Wave 1 Cohort. Assuming that the Wave 1 demographic characteristics in Table 4-6 are correlated with key tobacco- and health-related outcome measures at Wave 6 of the PATH Study, the results do not raise concern about bias in the youth interview estimates for the Wave 1 Cohort due to nonresponse at any primary wave since Wave 1. However, almost all of the Wave 6 youth in the Wave 1 Cohort were shadow youth at Wave 1, and limited data are available for assessing nonresponse bias for this group.

Table 4-6. Comparison of Wave 1 demographic characteristics between Wave 1 respondents eligible to be Wave 6 youth all-waves respondents and Wave 1 respondents who were Wave 6 youth all-waves respondents

Wave 1 characteristic	Wave 1 respondents who were eligible to be Wave 6 youth all-waves respondents		Wave 1 respondents who were Wave 6 youth all-waves respondents				
	Unweighted count	A: Weighted percentage, using Wave 1 cross-sectional weight [95% confidence interval]	Unweighted count	Before Wave 6 weighting adjustment		After Wave 6 weighting adjustment	
				B: Weighted percentage, using Wave 1 cross-sectional weight [95% confidence interval]	Difference in weighted percentages [B – A] [95% confidence interval]	C: Weighted percentage, using Wave 6 all-waves weight [95% confidence interval]	Difference in weighted percentages [C – A] [95% confidence interval]
Sex							
Male	2,189	51.3% [49.8%, 52.8%]	1,163	50.7% [48.7%, 52.8%]	-0.5% [-1.9%, 0.8%]	51.3% [49.2%, 53.4%]	0.0% [-0.6%, 0.7%]
Female	2,000	48.7% [47.2%, 50.2%]	1,079	49.3% [47.2%, 51.3%]	0.5% [-0.8%, 1.9%]	48.7% [46.6%, 50.8%]	-0.0% [-0.7%, 0.6%]
Race/ethnicity ^a							
Hispanic	1,184	23.7% [22.4%, 25.0%]	638	23.6% [21.9%, 25.4%]	-0.1% [-1.2%, 1.1%]	23.8% [22.1%, 25.6%]	0.1% [-0.4%, 0.6%]
Non-Hispanic White alone	2,052	52.9% [51.4%, 54.4%]	1,094	52.8% [50.7%, 54.8%]	-0.1% [-1.6%, 1.3%]	53.0% [50.9%, 55.0%]	0.0% [-0.6%, 0.7%]
Non-Hispanic Black alone	573	13.2% [12.2%, 14.3%]	311	13.2% [11.8%, 14.6%]	-0.0% [-1.1%, 1.0%]	13.2% [11.8%, 14.6%]	-0.0% [-0.5%, 0.4%]
Non-Hispanic other race or multiple races	378	10.2% [9.3%, 11.1%]	197	10.4% [9.2%, 11.7%]	0.2% [-0.7%, 1.2%]	10.1% [8.9%, 11.4%]	-0.1% [-0.4%, 0.3%]
Census region							
Northeast	600	16.0% [14.9%, 17.1%]	305	15.6% [14.2%, 17.2%]	-0.4% [-1.4%, 0.6%]	16.1% [14.6%, 17.6%]	0.1% [-0.4%, 0.6%]
Midwest	968	21.9% [20.7%, 23.2%]	532	22.7% [21.0%, 24.5%]	0.8% [-0.7%, 2.2%]	22.1% [20.5%, 23.9%]	0.2% [-0.3%, 0.7%]
South	1,551	38.2% [36.7%, 39.6%]	807	36.5% [34.5%, 38.5%]	-1.6% [-3.4%, 0.1%]	38.1% [36.1%, 40.1%]	-0.1% [-0.6%, 0.5%]
West	1,073	23.9% [22.7%, 25.3%]	598	25.2% [23.5%, 27.1%]	1.3% [-0.1%, 2.6%]	23.7% [22.0%, 25.5%]	-0.2% [-0.8%, 0.3%]

^a Hispanic refers to Hispanic or Latino; Black refers to Black or African-American. Alone means that the participant reported only one race.

5. Results of Wave 4 Cohort Nonresponse Bias Analyses at Wave 6

This chapter presents results of nonresponse bias analyses for the Wave 4 Cohort interview data at Wave 6. The nonresponse bias analyses for the Wave 4 Cohort at Wave 6 are all conditioning on Wave 4 response. Sections 5.1.1 and 5.2.1 describe how the single-wave response rates vary across select subgroups, including recruitment wave. Sections 5.1.2 and 5.2.2 compare demographic and socio-economic characteristics as well as tobacco-use status between those eligible for the Wave 6 interview and the Wave 6 respondents. Four sections compare cross-sectional estimates of cigarette smoking (Sections 5.1.3 and 5.2.3) and ENDS use (Sections 5.1.4 and 5.2.4) between the PATH Study Wave 4 Cohort at Wave 6 and other national studies. All results are presented separately for adults and youth.

5.1 Adult Interview

5.1.1 Single Wave Response Rates Conditioning on Wave 4 Response

This section summarizes the Wave 4 Cohort single-wave response rates for the Wave 6 adult interview, conditioning on Wave 4 response. The unweighted and weighted single-wave response rates were calculated for the Wave 6 adult interview among those in the Wave 4 Cohort using the following formulas:

$$RR_{A_SW} = C_{A_SW} / (C_{A_SW} + N_{A_SW} + e_{A_SW} \times U_{A_SW})$$

$$e_{A_SW} = (C_{A_SW} + N_{A_SW}) / (C_{A_SW} + N_{A_SW} + I_{A_SW})$$

where

RR_{A_SW} = Wave 6 adult interview single-wave response rate;

C_{A_SW} = number of completed Wave 6 adult interviews;

N_{A_SW} = number of Wave 6 adult nonrespondents known to be eligible at Wave 6;

U_{A_SW} = number of Wave 6 adult nonrespondents whose eligibility status could not be determined;

e_{A_SW} = estimated proportion of eligible adults among the Wave 6 adult nonrespondents with unknown eligibility status; and

I_{A_SW} = number of adults determined to be ineligible at Wave 6.²²

Unweighted counts and weighted counts based on the Wave 4 cross-sectional weight were obtained for the response status categories C_{A_SW} , N_{A_SW} , U_{A_SW} , and I_{A_SW} for the unweighted single-wave response rates (RR_{A_SW}) and weighted single-wave response rates ($RR_{A_W_SW}$), respectively.

Table 5-1 provides single-wave response rates for the Wave 6 adult interview. In addition to the overall row, the table includes rows for subgroups based on response status at Wave 5 and recruitment wave as well as the following Wave 4 characteristics: sex, age, race/ethnicity, census region, highest education level attained, health insurance status, and tobacco-use status. Participants for whom the Wave 4 interview had missing information for the demographic, socio-economic, or tobacco-use characteristic were excluded from the response rate calculation for that subgroup. In addition, the education level and health insurance status subgroups exclude the Wave 6 adults who were youth at Wave 4 because they were not asked about their highest level of education attained or health insurance status at that time.

The unweighted and weighted single-wave response rates for the Wave 6 adult interview were 69.4 percent and 73.5 percent, respectively. The difference between the unweighted and weighted adult response rates was larger than for Wave 5, both overall and for most subgroups. The largest difference was for Wave 6 adults residing in the Northeast census region at Wave 4, with a weighted response rate 5.2 percentage points higher than the unweighted response rate.

The Wave 6 adult single-wave response rates were noticeably lower than the Wave 5 adult single-wave response rates. Wave 6 was the first wave of the PATH Study for which participants were not fielded for data collection if previously they had not responded at three consecutive waves.²³ Also, as noted in Section 2.8, telephone interviewing was the only option available at some times and locations during Wave 6. Telephone surveys generally achieve lower response rates than in-person surveys (Groves & Couper, 1998); however, the COVID-19 pandemic may have affected survey

²² Excludes participants determined to be permanently ineligible (i.e., deceased, moved out of the United States permanently, or incarcerated long term) prior to Wave 6.

²³ See Section 2.8 for details and note that the decision to exclude participants from data collection who have previously *refused* participation at two consecutive waves has been in place since Wave 4.

participants' preferences with respect to mode. These factors will have had some effect on the Wave 6 response rates.

The weighted response rates varied moderately by Wave 4 sex, race/ethnicity, census region, health insurance status, current established tobacco use (among those who were adults at Wave 4), and ever tobacco use (among those who were youth at Wave 4). The response rates varied to a greater extent by Wave 4 age and educational level, and recruitment wave; and varied substantially by Wave 5 response status. Adults ages 45 to 64 at Wave 4 had the highest weighted single-wave response rate (78.5 percent) among the age groups, whereas those ages 18 to 24 at Wave 4 had the lowest (65.6 percent). Adults with a bachelor's degree or advanced degree at Wave 4 had a much higher propensity to respond (81.6 percent) than adults who did not graduate from high school or had a GED (General Education Development Test) at that time (67.2 percent). The weighted single-wave response rate for Wave 4 Cohort adults recruited at Wave 1 was 13.3 percentage points higher than for those recruited as part of the Wave 4 replenishment sample. Finally, the weighted single-wave response rate for the Wave 6 adult interview was 81.0 percent for respondents to Wave 5 and 17.4 percent for Wave 5 nonrespondents. A possible explanation for the large differences in subgroup response rates by recruitment wave and Wave 5 response status is that Wave 4 Cohort members who have responded to multiple previous waves may be more cooperative than other cohort members. The Wave 1 sample members of the Wave 4 Cohort have persisted with the PATH Study over a number of years and all completed an interview at Wave 4.

Variation in response rates by subgroups is to be expected in large-scale data collection efforts. However, it is pertinent to examine the effects of the Wave 6 single-wave weighting adjustments for the Wave 4 Cohort, given the differences in response propensity noted above. The results of these analyses are discussed in the next section.

Table 5-1. Wave 6 adult interview single-wave response rates conditioning on Wave 4 response (Wave 4 Cohort)

Characteristic ^a	<i>C_{A,SW}</i> : Completed Wave 6 adult interview (n)	<i>I_{A,SW}</i> : Ineligible at Wave 6 ^b (n)	<i>N_{A,SW}</i> : Nonresponse known to be eligible at Wave 6 (n)	<i>U_{A,SW}</i> : Nonresponse with unknown eligibility status at Wave 6 (n)	<i>RR_{A,SW}</i> : Unweighted single-wave response rate ^c (%)	<i>RR_{A,W,SW}</i> : Weighted single-wave response rate ^c (%)
Overall	29,516	881	6,822	6,367	69.4	73.5
Response status at Wave 5						
Responded at Wave 5	28,531	627	4,800	3,624	77.3	81.0
Did not respond at Wave 5	985	254	2,022	2,743	17.8	17.4
Wave 4 sex						
Male	14,028	547	3,742	3,289	66.9	71.3
Female	15,451	332	3,066	3,065	71.8	75.5
Wave 4 age						
Under 18	6,802	95	1,620	1,870	66.2	65.9
18-24	7,145	138	1,937	1,952	64.9	65.6
25-44	7,818	172	1,649	1,632	70.6	73.2
45-64	5,793	234	1,130	672	76.5	78.5
65 or older	1,957	242	486	239	73.6	72.8
Wave 4 race/ethnicity ^d						
Hispanic	6,414	170	1,466	1,614	67.8	70.6
Non-Hispanic White alone	15,856	452	3,875	3,125	69.6	74.5
Non-Hispanic Black alone	4,320	151	805	973	71.2	73.3
Non-Hispanic other race or multiple races	2,393	73	546	530	69.2	72.9
Wave 4 census region						
Northeast	4,296	108	1,151	777	69.2	74.4
Midwest	6,988	231	1,594	1,368	70.5	74.5
South	10,951	366	2,518	2,697	68.0	71.6
West	7,281	176	1,559	1,524	70.5	74.8
Wave 4 education (Wave 4 adults only)						
Less than high school or GED	3,890	264	1,001	1,045	66.1	67.2
High school graduate	5,168	195	1,371	1,267	66.5	69.7
Some college but no degree, or associate's degree	8,119	194	1,920	1,541	70.3	73.5
Bachelor's degree or advanced degree	5,463	126	878	610	78.7	81.6

Table 5-1. Wave 6 adult interview single-wave response rates conditioning on Wave 4 response (Wave 4 Cohort) (continued)

Characteristic ^a	<i>C_{A,SW}</i> Completed Wave 6 adult interview (n)	<i>I_{A,SW}</i> Ineligible at Wave 6 ^b (n)	<i>N_{A,SW}</i> Nonresponse known to be eligible at Wave 6 (n)	<i>U_{A,SW}</i> Nonresponse with unknown eligibility status at Wave 6 (n)	<i>RR_{A,SW}</i> Unweighted single-wave response rate ^c (%)	<i>RR_{A,W,SW}</i> Weighted single-wave response rate ^c (%)
Wave 4 health insurance (Wave 4 adults only)						
Yes	19,072	640	4,262	3,332	71.8	75.1
No	3,422	132	878	1,105	63.7	67.3
Wave 4 current established tobacco use ^e (Wave 4 adults only)						
Current established tobacco user	7,966	406	2,013	1,968	67.1	67.5
Not current established tobacco user	14,678	374	3,171	2,510	72.3	75.9
Wave 4 ever tobacco use ^f (Wave 4 youth only)						
Ever user	1,639	39	483	583	60.8	60.4
Never user	4,756	47	1,009	1,147	68.9	68.6
Recruitment wave						
Wave 1	24,604	703	5,160	4,628	71.8	75.4
Wave 4	4,912	178	1,662	1,739	59.4	62.1

^a For each characteristic, the sum of the counts across the categories may not be equal to the count in the overall row due to missing values.

^b Excludes participants determined to be permanently ineligible (i.e., deceased, moved out of the United States permanently, or incarcerated long term) prior to Wave 6.

^c The unweighted adult interview single-wave response rate formula is provided earlier in this section. The weighted adult interview single-wave response rates were calculated using a similar approach except that the Wave 4 cross-sectional weight was used to obtain weighted counts.

^d Hispanic refers to Hispanic or Latino; Black refers to Black or African American. Alone means that the participant reported only one race.

^e A current established user of a tobacco product is someone who currently uses the product every day or some days and: for cigarettes, has smoked at least 100 cigarettes in their lifetime and, for any other tobacco product, has reported they ever used that product fairly regularly. The products covered by the Wave 4 adult interview are cigarettes, traditional cigars, cigarillos, filtered cigars, pipes, smokeless tobacco, snus, hookah, and electronic nicotine delivery systems or ENDS (such as e-cigarettes, vape pens, personal vaporizers and mods, e-cigars, e-pipes, e-hookahs, and hookah pens).

^f An ever user of a tobacco product is someone who has ever used the product, even one or two times. The products covered by the Wave 4 youth interview are cigarettes, traditional cigars, cigarillos, filtered cigars, pipes, smokeless tobacco, snus, hookah, electronic nicotine delivery systems or ENDS (such as e-cigarettes, vape pens, personal vaporizers and mods, e-cigars, e-pipes, e-hookahs, and hookah pens), dissolvable tobacco, bidis, and kreteks.

5.1.2 Comparison Between Those Eligible for Wave 6 Interview and Wave 6 Respondents (Among the Wave 4 Cohort)

This section compares weighted estimates between the members of the Wave 4 Cohort who were eligible for the Wave 6 adult interview and the Wave 6 adult respondents in the Wave 4 Cohort. The weighted estimates cover Wave 4 demographic and socio-economic characteristics (shown in Table 5-2) as well as Wave 4 tobacco-use status (shown in Tables 5-3 and 5-4). The structure of these tables is similar to the tables in Section 4.1.2.

For the “before Wave 6 weighting adjustments” in Table 5-2, the largest overrepresentation among Wave 6 adult respondents was of those with a bachelor’s degree or advanced degree at Wave 4. Females, adults ages 45 to 64 at Wave 4, non-Hispanic White alone adults, and adults with health insurance at Wave 4 were also overrepresented among the Wave 6 adult respondents, whereas adults younger than age 25 at Wave 4, Hispanic adults, adults residing in the South census region at Wave 4, and adults with at most a high school diploma at Wave 4 were underrepresented.

For the “after Wave 6 weighting adjustment” differences, the estimated 95 percent confidence intervals (in the last column of Table 5-2) either included zero or were in the proximity of zero for most of the Wave 4 characteristics examined. However, among the Wave 6 responding adults from the Wave 4 Cohort, those with a bachelor’s degree or advanced degree at Wave 4 were slightly overrepresented and adults ages 65 and older were slightly underrepresented.

Table 5-2. Comparison of Wave 4 demographic and socio-economic characteristics between Wave 4 respondents who were eligible for Wave 6 adult interview and Wave 6 adult interview respondents (Wave 4 Cohort)

Wave 4 characteristic	Wave 4 respondents who were eligible for Wave 6 adult interview		Wave 4 respondents who completed Wave 6 adult interview				
	Unweighted count	A: Weighted percentage, using Wave 4 cross-sectional weight [95% confidence interval]	Unweighted count	Before Wave 6 weighting adjustment		After Wave 6 weighting adjustment	
				B: Weighted percentage, using Wave 4 cross-sectional weight [95% confidence interval]	Difference in weighted percentages [B – A] [95% confidence interval]	C: Weighted percentage, using Wave 6 single-wave weight [95% confidence interval]	Difference in weighted percentages [C – A] [95% confidence interval]
Sex							
Male	21,059	48.1% [47.6%,48.5%]	14,028	46.6% [46.0%, 47.1%]	-1.5% [-1.9%, -1.1%]	48.0% [47.4%, 48.6%]	-0.1% [-0.1%, -0.0%]
Female	21,582	51.9% [51.5%, 52.4%]	15,451	53.4% [52.9%, 54.0%]	1.5% [1.1%, 1.9%]	52.0% [51.4%, 52.6%]	0.1% [0.0%, 0.1%]
Age							
Under 18	10,292	6.8% [6.6%, 7.1%]	6,802	6.1% [5.9%, 6.4%]	-0.7% [-0.8%, -0.6%]	6.8% [6.5%, 7.1%]	-0.0% [-0.0%, 0.0%]
18-24	11,034	11.9% [11.6%, 12.2%]	7,145	10.6% [10.2%, 10.9%]	-1.3% [-1.4%, -1.1%]	11.9% [11.5%, 12.2%]	0.0% [0.0%, 0.0%]
25-44	11,099	32.6% [32.1%, 33.0%]	7,818	32.5% [32.0%, 33.1%]	-0.1% [-0.5%, 0.3%]	32.6% [32.1%, 33.2%]	0.1% [0.0%, 0.1%]
45-64	7,595	32.0% [31.6%, 32.5%]	5,793	34.3% [33.7%, 34.8%]	2.2% [1.8%, 2.7%]	32.1% [31.6%, 32.6%]	0.1% [0.0%, 0.1%]
65 or older	2,682	16.7% [16.4%, 17.1%]	1,957	16.5% [16.0%, 17.0%]	-0.2% [-0.7%, 0.2%]	16.6% [16.1%, 17.0%]	-0.2% [-0.2%, -0.1%]
Race/ethnicity ^a							
Hispanic	9,494	16.5% [16.2%, 16.9%]	6,414	15.8% [15.3%, 16.3%]	-0.7% [-1.1%, -0.2%]	16.5% [16.1%, 16.9%]	-0.0% [-0.1%, 0.0%]
Non-Hispanic White alone	22,856	63.8% [63.3%, 64.3%]	15,856	64.6% [64.0%, 65.2%]	0.8% [0.3%, 1.3%]	63.9% [63.4%, 64.5%]	0.1% [0.0%, 0.2%]
Non-Hispanic Black alone	6,098	11.4% [11.1%, 11.7%]	4,320	11.3% [11.0%, 11.7%]	-0.1% [-0.3%, 0.2%]	11.4% [11.0%, 11.7%]	-0.1% [-0.1%, 0.0%]
Non-Hispanic other race or multiple races	3,469	8.3% [8.0%, 8.6%]	2,393	8.2% [7.9%, 8.6%]	-0.1% [-0.4%, 0.2%]	8.2% [7.9%, 8.6%]	-0.1% [-0.1%, 0.0%]

Table 5-2. Comparison of Wave 4 demographic and socio-economic characteristics between Wave 4 respondents who were eligible for Wave 6 adult interview and Wave 6 adult interview respondents (Wave 4 Cohort) (continued)

Wave 4 characteristic	Wave 4 respondents who were eligible for Wave 6 adult interview		Wave 4 respondents who completed Wave 6 adult interview				
	Unweighted count	A: Weighted percentage, using Wave 4 cross-sectional weight [95% confidence interval]	Unweighted count	Before Wave 6 weighting adjustment		After Wave 6 weighting adjustment	
				B: Weighted percentage, using Wave 4 cross-sectional weight [95% confidence interval]	Difference in weighted percentages [B – A] [95% confidence interval]	C: Weighted percentage, using Wave 6 single-wave weight [95% confidence interval]	Difference in weighted percentages [C – A] [95% confidence interval]
Census region							
Northeast	6,224	17.8% [17.5%, 18.2%]	4,296	18.1% [17.6%, 18.5%]	0.2% [-0.2%, 0.6%]	17.8% [17.4%, 18.3%]	0.0% [-0.0%, 0.1%]
Midwest	9,950	21.0% [20.6%, 21.4%]	6,988	21.3% [20.9%, 21.8%]	0.3% [-0.1%, 0.8%]	21.0% [20.6%, 21.5%]	0.0% [-0.0%, 0.0%]
South	16,166	37.4% [37.0%, 37.9%]	10,951	36.4% [35.9%, 37.0%]	-1.0% [-1.5%, -0.4%]	37.4% [36.8%, 37.9%]	-0.1% [-0.1%, -0.0%]
West	10,364	23.8% [23.3%, 24.2%]	7,281	24.2% [23.7%, 24.7%]	0.4% [-0.0%, 0.9%]	23.8% [23.3%, 24.3%]	0.0% [0.0%, 0.1%]
Education (Wave 4 adults only)							
Less than high school or GED	5,936	15.3% [14.9%, 15.7%]	3,890	13.8% [13.3%, 14.2%]	-1.5% [-1.9%, -1.1%]	15.1% [14.7%, 15.6%]	-0.1% [-0.2%, -0.0%]
High school graduate	7,806	23.4% [23.0%, 23.9%]	5,168	22.0% [21.5%, 22.5%]	-1.4% [-1.8%, -1.0%]	23.3% [22.8%, 23.9%]	-0.1% [-0.2%, 0.0%]
Some college but no degree, or associate's degree	11,580	31.5% [31.0%, 32.0%]	8,119	31.3% [30.7%, 31.9%]	-0.2% [-0.6%, 0.2%]	31.6% [31.0%, 32.2%]	0.0% [-0.1%, 0.1%]
Bachelor's degree or advanced degree	6,951	29.8% [29.3%, 30.3%]	5,463	32.9% [32.3%, 33.5%]	3.1% [2.7%, 3.5%]	30.0% [29.4%, 30.6%]	0.2% [0.1%, 0.3%]

Table 5-2. Comparison of Wave 4 demographic and socio-economic characteristics between Wave 4 respondents who were eligible for Wave 6 adult interview and Wave 6 adult interview respondents (Wave 4 Cohort) (continued)

Wave 4 characteristic	Wave 4 respondents who were eligible for Wave 6 adult interview		Wave 4 respondents who completed Wave 6 adult interview				
	Unweighted count	A: Weighted percentage, using Wave 4 cross-sectional weight [95% confidence interval]	Unweighted count	Before Wave 6 weighting adjustment		After Wave 6 weighting adjustment	
				B: Weighted percentage, using Wave 4 cross-sectional weight [95% confidence interval]	Difference in weighted percentages [B – A] [95% confidence interval]	C: Weighted percentage, using Wave 6 single-wave weight [95% confidence interval]	Difference in weighted percentages [C – A] [95% confidence interval]
Health insurance (Wave 4 adults only)							
Yes	26,666	88.0% [87.5%, 88.5%]	19,072	89.1% [88.5%, 89.7%]	1.1% [0.8%, 1.5%]	88.0% [87.3%, 88.6%]	-0.0% [-0.4%, 0.3%]
No	5,405	12.0% [11.5%, 12.5%]	3,422	10.9% [10.3%, 11.5%]	-1.1% [-1.5%, -0.8%]	12.0% [11.4%, 12.7%]	0.0% [-0.3%, 0.4%]

^a Hispanic refers to Hispanic or Latino; Black refers to Black or African-American. Alone means that the participant reported only one race.

Table 5-3. Comparison of Wave 4 current established tobacco-use estimates between Wave 4 adults who were eligible for Wave 6 adult interview and Wave 4 adults who completed Wave 6 adult interview (Wave 4 Cohort)*

Wave 4 characteristic ^a	Wave 4 adults who were eligible for Wave 6 adult interview		Wave 4 adults who completed Wave 6 adult interview				
	Sample size	A: Weighted estimate, using Wave 4 cross-sectional weight [95% confidence interval]	Sample size	Before Wave 6 weighting adjustment		After Wave 6 weighting adjustment	
				B: Weighted estimate, using Wave 4 cross-sectional weight [95% confidence interval]	Difference in weighted estimates [B – A] [95% confidence interval]	C: Weighted estimate, using Wave 6 single-wave weight [95% confidence interval]	Difference in weighted estimates [C – A] [95% confidence interval]
Overall	32,306	22.6% [22.0%, 23.1%]	22,644	20.5% [19.9%, 21.1%]	-2.0% [-2.4%, -1.7%]	22.6% [22.0%, 23.1%]	0.0% [-0.2%, 0.2%]
Sex							
Male	15,734	28.0% [27.1%, 28.9%]	10,640	25.3% [24.2%, 26.4%]	-2.7% [-3.3%, -2.1%]	27.9% [27.0%, 28.8%]	-0.1% [-0.4%, 0.3%]
Female	16,539	17.6% [17.0%, 18.2%]	11,980	16.4% [15.8%, 17.1%]	-1.2% [-1.5%, -0.8%]	17.7% [17.0%, 18.4%]	0.1% [-0.2%, 0.3%]
Age							
18-24	10,994	22.4% [21.2%, 23.6%]	7,121	20.3% [19.0%, 21.6%]	-2.1% [-2.8%, -1.4%]	22.4% [21.2%, 23.7%]	0.1% [-0.5%, 0.6%]
25-44	11,068	28.3% [27.3%, 29.4%]	7,795	25.2% [24.1%, 26.4%]	-3.1% [-3.7%, -2.5%]	27.9% [26.8%, 29.1%]	-0.4% [-1.0%, 0.1%]
45-64	7,578	23.1% [22.2%, 24.1%]	5,782	21.3% [20.3%, 22.4%]	-1.8% [-2.4%, -1.2%]	23.3% [22.2%, 24.5%]	0.2% [-0.3%, 0.7%]
65 or older	2,664	10.3% [9.2%, 11.5%]	1,945	9.7% [8.5%, 11.1%]	-0.6% [-1.1%, -0.0%]	10.5% [9.3%, 12.0%]	0.2% [-0.3%, 0.8%]
Race/ethnicity ^b							
Hispanic	6,412	17.1% [16.0%, 18.3%]	4,335	15.9% [14.7%, 17.1%]	-1.2% [-2.0%, -0.4%]	17.1% [15.9%, 18.4%]	0.0% [-0.5%, 0.6%]
Non-Hispanic White alone	18,191	24.3% [23.5%, 25.1%]	12,875	21.6% [20.8%, 22.5%]	-2.7% [-3.1%, -2.3%]	24.2% [23.3%, 25.0%]	-0.1% [-0.4%, 0.1%]
Non-Hispanic Black alone	4,724	25.4% [24.1%, 26.8%]	3,370	25.2% [23.5%, 27.0%]	-0.2% [-1.2%, 0.8%]	26% [24.5%, 27.6%]	0.6% [-0.0%, 1.3%]
Non-Hispanic other race or multiple races	2,508	15.7% [14.2%, 17.4%]	1,743	14.1% [12.5%, 15.9%]	-1.6% [-2.6%, -0.6%]	15.8% [14.1%, 17.6%]	0.1% [-0.9%, 1.1%]

Table 5-3. Comparison of Wave 4 current established tobacco-use estimates between Wave 4 adults who were eligible for Wave 6 adult interview and Wave 4 adults who completed Wave 6 adult interview (Wave 4 Cohort)* (continued)

Wave 4 characteristic ^a	Wave 4 adults who were eligible for Wave 6 adult interview		Wave 4 adults who completed Wave 6 adult interview				
	Sample size	A: Weighted estimate, using Wave 4 cross-sectional weight [95% confidence interval]	Sample size	Before Wave 6 weighting adjustment		After Wave 6 weighting adjustment	
				B: Weighted estimate, using Wave 4 cross-sectional weight [95% confidence interval]	Difference in weighted estimates [B – A] [95% confidence interval]	C: Weighted estimate, using Wave 6 single-wave weight [95% confidence interval]	Difference in weighted estimates [C – A] [95% confidence interval]
Census region							
Northeast	4,784	20.6% [19.0%, 22.4%]	3,400	19.1% [17.3%, 20.9%]	-1.6% [-2.4%, -0.8%]	20.7% [19.0%, 22.6%]	0.1% [-0.4%, 0.7%]
Midwest	7,700	25.7% [24.1%, 27.3%]	5,510	23.3% [21.8%, 24.9%]	-2.4% [-3.1%, -1.7%]	25.6% [24.1%, 27.2%]	-0.1% [-0.5%, 0.3%]
South	12,197	24.8% [23.5%, 26.0%]	8,332	22.7% [21.4%, 24.0%]	-2.1% [-2.6%, -1.6%]	24.7% [23.4%, 26.0%]	-0.1% [-0.5%, 0.3%]
West	7,625	17.8% [16.5%, 19.2%]	5,402	16% [14.6%, 17.5%]	-1.8% [-2.5%, -1.2%]	17.9% [16.6%, 19.3%]	0.1% [-0.3%, 0.5%]
Education (Wave 4 adults only)							
Less than high school or GED	5,900	36.2% [34.3%, 38.2%]	3,862	35.7% [33.3%, 38.1%]	-0.6% [-1.8%, 0.7%]	36.6% [34.5%, 38.7%]	0.4% [-0.3%, 1.0%]
High school graduate	7,773	28.1% [26.7%, 29.5%]	5,150	26.5% [24.9%, 28.1%]	-1.6% [-2.5%, -0.7%]	28.4% [27.0%, 30.0%]	0.4% [-0.1%, 0.9%]
Some college but no degree, or associate’s degree	11,558	23.8% [22.9%, 24.8%]	8,104	21.8% [20.7%, 22.9%]	-2.1% [-2.6%, -1.5%]	23.7% [22.7%, 24.8%]	-0.1% [-0.5%, 0.3%]
Bachelor’s degree or advanced degree	6,942	9.7% [9.0%, 10.5%]	5,457	9.0% [8.2%, 9.8%]	-0.7% [-1.0%, -0.4%]	9.6% [8.8%, 10.5%]	-0.1% [-0.4%, 0.2%]
Health insurance (Wave 4 adults only)							
Yes	26,599	20.7% [20.1%, 21.2%]	19,029	18.8% [18.2%, 19.4%]	-1.8% [-2.2%, -1.5%]	20.6% [20.0%, 21.2%]	-0.0% [-0.3%, 0.2%]
No	5,371	36.3% [34.6%, 38.0%]	3,399	34.2% [32.2%, 36.2%]	-2.1% [-3.4%, -0.8%]	36.4% [34.5%, 38.4%]	0.1% [-1.1%, 1.4%]

* A 'current established user' of a tobacco product is someone who currently uses the product every day or some days and: for cigarettes, has smoked at least 100 cigarettes in their lifetime and, for any other tobacco product, has reported they ever used that product fairly regularly. The products covered by the Wave 4 adult interview are cigarettes, traditional cigars, cigarillos, filtered cigars, pipes, smokeless tobacco, snus, hookah, dissolvable tobacco, and electronic nicotine delivery systems or ENDS (such as e-cigarettes, vape pens, personal vaporizers and mods, e-cigars, e-pipes, e-hookahs, and hookah pens).

^a For each characteristic, the sum of the counts across the categories may not be equal to the count in the overall row due to missing values.

^b Hispanic refers to Hispanic or Latino; Black refers to Black or African-American. Alone means that the participant reported only one race.

Table 5-4. Comparison of Wave 4 ever tobacco-use estimates between Wave 4 youth who were eligible for Wave 6 adult interview and Wave 4 youth who completed Wave 6 adult interview (Wave 4 Cohort)*

Wave 4 characteristic ^a	Wave 4 youth who were eligible for Wave 6 adult interview		Wave 4 youth who completed Wave 6 adult interview				
	Sample size	A: Weighted estimate, using Wave 4 cross-sectional weight [95% confidence interval]	Sample size	Before Wave 6 weighting adjustment		After Wave 6 weighting adjustment	
				B: Weighted estimate, using Wave 4 cross-sectional weight [95% confidence interval]	Difference in weighted estimates [B – A] [95% confidence interval]	C: Weighted estimate, using Wave 6 single-wave weight [95% confidence interval]	Difference in weighted estimates [C – A] [95% confidence interval]
Overall	9,617	28.2% [27.0%, 29.4%]	6,395	25.6% [24.3%, 27.0%]	-2.6% [-3.3%, -1.8%]	27.5% [26.3%, 28.8%]	-0.7% [-1.1%, -0.2%]
Sex							
Male	4,885	30.0% [28.6%, 31.5%]	3,136	26.4% [24.7%, 28.2%]	-3.6% [-4.8%, -2.5%]	28.9% [27.3%, 30.5%]	-1.1% [-2.0%, -0.3%]
Female	4,708	26.3% [24.7%, 27.9%]	3,250	24.9% [23.2%, 26.8%]	-1.3% [-2.3%, -0.3%]	26.1% [24.5%, 27.8%]	-0.1% [-0.9%, 0.6%]
Race/ethnicity ^b							
Hispanic	2,827	28.2% [26.1%, 30.4%]	1,931	26.7% [24.3%, 29.2%]	-1.5% [-3.0%, -0.1%]	28.2% [25.6%, 30.9%]	-0.1% [-1.4%, 1.3%]
Non-Hispanic White alone	4,383	30.2% [28.3%, 32.1%]	2,821	27.1% [25.1%, 29.2%]	-3.0% [-4.1%, -2.0%]	29.4% [27.5%, 31.4%]	-0.7% [-1.5%, -0.0%]
Non-Hispanic Black alone	1,250	23.4% [20.4%, 26.7%]	865	22.4% [18.9%, 26.2%]	-1.0% [-3.0%, 0.9%]	23.3% [19.8%, 27.2%]	-0.1% [-2.0%, 1.8%]
Non-Hispanic other race or multiple races	884	24.2% [21.0%, 27.7%]	596	21.6% [18.5%, 25.1%]	-2.6% [-5.4%, 0.2%]	22.8% [19.6%, 26.3%]	-1.4% [-4.0%, 1.1%]
Census region							
Northeast	1,329	30.1% [27.7%, 32.6%]	841	27.7% [24.8%, 30.8%]	-2.4% [-3.9%, -0.8%]	30.1% [27.1%, 33.3%]	0.0% [-1.8%, 1.9%]
Midwest	2,102	29.6% [27.2%, 32.1%]	1,384	27.1% [24.7%, 29.7%]	-2.5% [-3.6%, -1.4%]	29.6% [27.2%, 32.0%]	-0.0% [-1.3%, 1.2%]
South	3,659	27.0% [25.4%, 28.6%]	2,427	24.6% [22.5%, 26.8%]	-2.4% [-3.7%, -1.1%]	26.1% [24.1%, 28.2%]	-0.9% [-2.0%, 0.2%]
West	2,527	27.4% [24.0%, 31.1%]	1,743	24.6% [21.2%, 28.4%]	-2.8% [-4.4%, -1.2%]	26.0% [22.3%, 30.0%]	-1.4% [-2.8%, -0.1%]

* An 'ever user' of a tobacco product is someone who has ever used the product, even one or two times. The products covered by the Wave 4 youth interview are cigarettes, traditional cigars, cigarillos, filtered cigars, pipes, smokeless tobacco, snus, hookah, dissolvable tobacco, electronic nicotine delivery systems or ENDS (such as e-cigarettes, vape pens, personal vaporizers and mods, e-cigars, e-pipes, e-hookahs, and hookah pens), bidis, and kreteks.

^a For each characteristic, the sum of the counts across the categories may not be equal to the count in the overall row due to missing values.

^b Hispanic refers to Hispanic or Latino; Black refers to Black or African-American. Alone means that the participant reported only one race.

Tables 5-3 and 5-4 show the comparisons for the Wave 4 tobacco-use measures between those eligible for the Wave 6 adult interview and the Wave 6 adult respondents, among the Wave 4 Cohort. (As in Section 4.1.2, separate analyses were conducted based on the participant’s interview type at Wave 4.)

For the “before Wave 6 weighting adjustment” estimates in Table 5-3, the largest difference between those eligible for the Wave 6 adult interview and the Wave 6 adult respondents was 3.1 percentage points, for adults ages 25 to 44 at Wave 4. Nearly all of the 95 percent confidence intervals for the estimated differences were below zero, indicating an underrepresentation of Wave 4 adult current established users of tobacco among the Wave 6 respondents. Similarly, most of the confidence intervals for the estimated differences in Table 5-4 were below zero, indicating an underrepresentation of Wave 4 youth ever users of tobacco among the Wave 6 adult respondents—the largest difference was 3.6 percentage points for males.

For the “after Wave 6 weighting adjustment” estimates in Table 5-3, the differences were mostly negligible and all of the 95 percent confidence intervals included zero, indicating a relatively unbiased estimation of Wave 4 current established tobacco use for the Wave 6 respondents who were adults at Wave 4. In Table 5-4, most of the confidence intervals for the differences in ever use of tobacco at Wave 4 between the Wave 4 youth who were eligible for the Wave 6 adult interview and the Wave 4 youth who were Wave 6 adult respondents included zero. However, the confidence intervals were below zero overall, for males, and for youth residing in the West census region at Wave 4, indicating an underrepresentation of Wave 4 youth ever users of tobacco among the Wave 6 adult respondents. The largest underrepresentation was again for males, but the magnitude of the difference (1.1 percentage points) was notably reduced after applying the Wave 6 weighting adjustments.

Overall, these results reflect positively on the use of demographic, socio-economic, and tobacco-use variables from Wave 4 for calibrating the Wave 6 single-wave weight for the Wave 4 Cohort. Assuming that the Wave 4 demographic, socio-economic, and tobacco-use characteristics in Tables 5-2 through 5-4 are correlated with key tobacco- and health-related outcome measures at Wave 6 of the PATH Study, the results suggest a very small amount of potential bias in the adult interview estimates for the Wave 4 Cohort due to nonresponse at Wave 6.

5.1.3 Comparison of Cigarette-Smoking Estimates for Adults to Other National Studies

This section compares adult cigarette-smoking estimates based on the Wave 4 Cohort respondents at Wave 6 of the PATH Study to similar estimates based on data from TUS-CPS 2018-2019, NHANES 2017-2020, NSDUH 2020, and NHIS 2021.

Table 5-5 presents estimates of the prevalence of current cigarette smoking²⁴ for the adult population as a whole and for subgroups. These estimates are accompanied by 95 percent confidence intervals. The point estimates from the PATH Study were calculated using the Wave 6 single-wave weight for the Wave 4 Cohort. The corresponding replicate weights were used to calculate variances and confidence intervals. Point estimates and 95 percent confidence intervals are reported for the other national studies as well.

Table 5-5 shows that the Wave 6 estimates of current cigarette smoking among adults were similar to the estimates from NHANES 2017-2020 and NSDUH 2020; these three studies had overlapping 95 percent confidence intervals for almost all estimates in the table. The estimates from Wave 6 of the PATH Study tended to be higher than the estimates from TUS-CPS 2018-2019 and NHIS 2021, with mostly non-overlapping confidence intervals. Table 5-5 shows no evidence of nonresponse bias in the Wave 6 estimates with respect to current cigarette smoking among adults, in the sense that the Wave 4 Cohort estimates were within the range of estimates from comparable surveys.

²⁴ For the PATH Study, following common practice for tobacco surveys, a person who currently smokes cigarettes is someone who (1) has smoked at least 100 cigarettes in their lifetime and (2) currently smokes cigarettes every day or some days. The questions used to define current cigarette smoking for each survey are provided in Appendix A.

Table 5-5. Comparison of cigarette-smoking estimates from Wave 6 adult interview (Wave 4 Cohort) and other national studies

Cigarette-smoking measure and subgroup ^a	PATH Study Wave 6 (2021) sample size	Percentage from PATH Study Wave 6 (2021) [95% confidence interval]	Percentage from TUS-CPS 2018-2019 [95% confidence interval]	Percentage from NHANES 2017-2020 [95% confidence interval]	Percentage from NSDUH 2020 ^b [95% confidence interval]	Percentage from NHIS 2021 [95% confidence interval]
Current, overall	29,499	14.4% [13.9%, 14.9%]	11.4% [11.2%, 11.6%]	16.4% [14.4%, 18.5%]	14.9% [13.9%, 15.9%]	11.5% [11.1%, 12.0%]
Current, male	14,018	16.2% [15.3%, 17.1%]	12.9% [12.6%, 13.3%]	18.3% [16.0%, 20.8%]	16.4% [15.0%, 18.0%]	13.1% [12.4%, 13.9%]
Current, female	15,443	12.8% [12.2%, 13.3%]	10.0% [9.7%, 10.3%]	14.6% [12.6%, 16.8%]	13.4% [12.2%, 14.8%]	10.1% [9.5%, 10.7%]
Current, age 18-24	10,628	6.1% [5.4%, 6.8%]	7.4% [6.7%, 8.2%]	11.7% [9.0%, 15.1%]	N/A	5.3% [4.3%, 6.6%]
Current, age 25-44	9,978	18.2% [17.4%, 19.1%]	12.5% [12.2%, 12.9%]	21.6% [18.0%, 25.8%]	N/A	12.6% [11.8%, 13.5%]
Current, age 45-64	5,940	17.7% [16.7%, 18.7%]	14.3% [13.9%, 14.7%]	17.8% [15.8%, 20.0%]	N/A	14.9% [14.0%, 15.9%]
Current, age 65 or older	2,953	8.4% [7.4%, 9.4%]	7.2% [6.9%, 7.6%]	8.0% [6.6%, 9.6%]	N/A	8.3% [7.6%, 9.0%]
Current, Hispanic	6,409	11.2% [10.4%, 12.1%]	7.5% [7.0%, 8.0%]	13.0% [11.1%, 15.2%]	10.8% [9.1%, 12.7%]	7.7% [6.8%, 8.7%]
Current, non-Hispanic White alone	15,852	15.2% [14.5%, 15.9%]	12.6% [12.3%, 12.9%]	16.7% [14.2%, 19.5%]	15.9% [14.7%, 17.2%]	12.9% [12.3%, 13.6%]
Current, non-Hispanic Black alone	4,316	18.0% [16.8%, 19.3%]	12.6% [11.9%, 13.3%]	20.6% [17.3%, 24.3%]	16.5% [13.9%, 19.4%]	11.7% [10.4%, 13.2%]
Current, non-Hispanic other race or multiple races	2,392	8.9% [7.7%, 10.1%]	8.7% [8.1%, 9.4%]	15.0% [11.7%, 19.0%]	13.2% [10.9%, 15.9%]	8.7% [7.4%, 10.2%]
Current every-day, overall	29,502	11.3% [10.9%, 11.7%]	8.7% [8.5%, 8.9%]	12.7% [10.9%, 14.8%]	N/A	8.8% [8.4%, 9.3%]
Current some-days, overall	29,502	3.1% [2.9%, 3.3%]	2.7% [2.6%, 2.8%]	3.6% [3.1%, 4.3%]	N/A	2.7% [2.5%, 3.0%]

^a For the race/ethnicity subgroups, Hispanic refers to Hispanic or Latino, Black refers to Black or African American, and alone means that the respondent reported only one race.

^b NSDUH's definition of a person who currently smokes cigarettes is someone who has smoked part or all of a cigarette in the past 30 days, which is more expansive than the definition used in the other surveys. However, NSDUH contains questions on lifetime smoking and current smoking. These questions were used to construct a modified measure of "current smoking" that is comparable to that of the other surveys (Ryan, Trosclair, & Gfroerer, 2012). The construction of this variable is described in Appendix A.

Aside from the possibility of nonresponse bias in one or more of the studies compared, the differences between estimates in Table 5-5 can be explained by a number of potential reasons. In addition to the varying degrees of sampling and measurement errors, the surveys differ in timing of data collection, survey mode, question order, and context.

The differences between the timing of data collection across the survey estimates compared are important to note, particularly given the COVID-19 pandemic. Data collection for Wave 6 of the PATH Study began on March 1, 2021, and ended on November 30, 2021—whereas data collection for TUS-CPS 2018-2019 and NHANES 2017-2020 occurred prior to the pandemic. The differences between tobacco-use estimates from the PATH Study and other national studies could be in part due to a material change in behavior in the population. In addition, legislation that raised the Federal minimum age for sale of tobacco products from 18 to 21 years went into effect on December 20, 2019, near or after the close of data collection for TUS-CPS 2018-2019 and NHANES 2017-2020. In Table 5-5, the pattern of estimates for adults ages 18 to 24 might reflect lower tobacco use among the young adult population due to this regulation.

Due to the COVID-19 pandemic, NHANES, which had originally planned to release a 2019-2020 public use file, suspended 2020 data collection efforts in mid-March and did not resume. To provide a public use file that would generate estimates with an acceptable level of precision, the National Center for Health Statistics (NCHS) decided to combine the 2017-2018 and 2019-2020 data into one file along with an appropriate set of weights. NHIS and NSDUH adjusted their data collection protocols in response to the pandemic. Like the PATH Study, NHIS and NSDUH adopted multi-mode data collection approaches. NHIS began data collection in 2021 primarily via telephone, with in-person follow-up allowed. In May, interviewers returned to regular survey interviewing procedures, whereby first contact with household members was attempted in person, and follow-up was allowed by telephone. However, interviewers were given flexibility to continue using telephone first contact attempts based on local COVID-19 conditions. The majority of NHIS interviews were conducted at least partially by telephone in 2021. NSDUH conducted interviews in person until mid-March 2020 and then suspended data collection through the end of September. From October through December, NSDUH collected data in person in areas with low risk of COVID-19 transmission and via the Web in other areas.

For in-person data collection, Wave 6 of the PATH Study and NSDUH 2020 used audio computer-assisted self-interview (ACASI) administration for the tobacco-use questions, so that the interviewer did not see or hear responses to the questions. In contrast, TUS-CPS 2018-2019, NHANES 2017-2020, NHIS 2021, and telephone data collection for Wave 6 of the PATH Study used direct questioning by an interviewer: NHANES was conducted in person, and TUS-CPS, NHIS, and Wave 6 of the PATH Study were conducted in person and by telephone. Direct questions by an interviewer can produce underreports of socially undesirable behavior compared to self-administration (Tourangeau & Smith, 1996), and this may explain some of the differences between the cigarette-smoking estimates shown in Table 5-5.

The contexts and purposes of these surveys also differ: CPS is a general survey on unemployment, NHANES and NHIS are general health surveys, and NSDUH is a cross-sectional survey on substance use (including tobacco use) and health, including mental health. Unlike the cross-sectional prevalence surveys, the PATH Study uses a longitudinal cohort design to assess within-person changes and between-person differences in tobacco-use behaviors and health over time. Differences among the questions used in the instruments of these different studies are outlined in Appendix A.

5.1.4 Comparison of ENDS-Use Estimates for Adults to Other National Studies

This section compares adult ENDS-use estimates based on the Wave 4 Cohort respondents at Wave 6 of the PATH Study to similar estimates based on data from TUS-CPS 2018-2019 and NHIS 2021. As with adult cigarette smoking, the primary measure of ENDS use is whether the adult currently uses ENDS.²⁵

Table 5-6 presents estimates of the prevalence of current ENDS use for the adult population as a whole, and for subgroups. These estimates are accompanied by 95 percent confidence intervals. The point estimates from the PATH Study were calculated using the Wave 6 single-wave weight for the Wave 4 Cohort. The corresponding replicate weights were used to calculate variances and

²⁵ For the PATH Study, a current ENDS user is someone who (1) at any time has used ENDS fairly regularly and (2) currently uses ENDS every day or some days. The questions used to define ENDS use for each survey are provided in Appendix B.

confidence intervals. Point estimates and 95 percent confidence intervals are reported for the other national studies as well.

Table 5-6. Comparison of ENDS-use estimates from Wave 6 adult interview (Wave 4 Cohort) and other national studies

ENDS-use measure and subgroup^a	PATH Study Wave 6 (2021) sample size	Percentage from PATH Study Wave 6 (2021) [95% confidence interval]	Percentage from TUS-CPS 2018-2019 [95% confidence interval]	Percentage from NHIS 2021 [95% confidence interval]
Current, overall	29,496	5.1% [4.8%, 5.3%]	2.3% [2.2%, 2.4%]	4.5% [4.2%, 4.9%]
Current, male	14,018	5.6% [5.2%, 6.0%]	2.8% [2.7%, 3.0%]	5.1% [4.7%, 5.6%]
Current, female	15,440	4.6% [4.2%, 4.9%]	1.8% [1.7%, 1.9%]	4.0% [3.6%, 4.4%]
Current, age 18-24	10,629	14.5% [13.7%, 15.3%]	5.3% [4.8%, 6.0%]	11.0% [9.4%, 12.7%]
Current, age 25-44	9,978	7.2% [6.7%, 7.7%]	3.0% [2.8%, 3.2%]	6.5% [5.9%, 7.1%]
Current, age 45-64	5,937	2.5% [2.1%, 2.9%]	1.6% [1.5%, 1.8%]	2.7% [2.3%, 3.1%]
Current, age 65 or older	2,952	0.7% [0.5%, 1.1%]	0.6% [0.5%, 0.7%]	0.9% [0.7%, 1.1%]
Current, Hispanic	6,413	4.1% [3.6%, 4.6%]	1.5% [1.3%, 1.8%]	3.3% [2.8%, 4.0%]
Current, non-Hispanic White alone	15,850	5.7% [5.4%, 6.1%]	2.8% [2.6%, 2.9%]	5.2% [4.8%, 5.7%]
Current, non-Hispanic Black alone	4,309	3.3% [2.8%, 4.0%]	1.1% [0.9%, 1.4%]	2.4% [1.8%, 3.2%]
Current, non-Hispanic other race or multiple races	2,391	4.6% [3.8%, 5.5%]	1.9% [1.6%, 2.3%]	4.5% [3.5%, 5.7%]
Current every-day, overall	29,497	2.9% [2.8%, 3.1%]	1.1% [1.0%, 1.2%]	2.2% [2.0%, 2.4%]
Current some-days, overall	29,496	2.1% [2.0%, 2.3%]	1.2% [1.1%, 1.3%]	2.3% [2.1%, 2.6%]

^a For the race/ethnicity subgroups, Hispanic refers to Hispanic or Latino, Black refers to Black or African American, and alone means that the respondent reported only one race.

The Wave 6 estimates of current ENDS use among adults were generally higher than the estimates from TUS-CPS 2018-2019 and NHIS 2021. In Table 5-6, only one out of the thirteen PATH Study 95 percent confidence intervals overlapped with that for TUS-CPS. The estimates from NHIS were more similar to the PATH Study estimates, with eleven of the thirteen confidence intervals overlapping.

As discussed in Section 5.1.3, some differences in the estimates may be attributable to differences in timing of data collection, survey mode, context, question order and wording, and measurement error. The most pronounced difference was between the PATH Study and TUS-CPS. May 2019 was the last month covered by the TUS-CPS 2018-2019 data collection, almost 2 years before the

beginning of data collection for Wave 6 of the PATH Study. Therefore, some of the differences in ENDS-use estimates between these two studies might reflect an increase in the prevalence of ENDS use over time.

5.2 Youth Interview

5.2.1 Single Wave Response Rates Conditioning on Wave 4 Response

This section summarizes the Wave 4 Cohort single-wave response rates for the Wave 6 youth interview, conditioning on Wave 4 response. Unweighted and weighted single-wave response rates were calculated for the Wave 6 youth interview among those in the Wave 4 Cohort using the following formulas:

$$RR_{Y_SW} = C_{Y_SW} / (C_{Y_SW} + N_{Y_SW} + e_{Y_SW} \times U_{Y_SW})$$

$$e_{Y_SW} = (C_{Y_SW} + N_{Y_SW}) / (C_{Y_SW} + N_{Y_SW} + I_{Y_SW})$$

where

- RR_{Y_SW} = Wave 6 youth interview single-wave response rate;
- C_{Y_SW} = number of completed Wave 6 youth interviews;
- N_{Y_SW} = number of Wave 6 youth nonrespondents known to be eligible at Wave 6;
- U_{Y_SW} = number of Wave 6 youth nonrespondents whose eligibility status could not be determined;
- e_{Y_SW} = estimated proportion of eligible youth among the Wave 6 youth nonrespondents with unknown eligibility status; and
- I_{Y_SW} = number of youth determined to be ineligible at Wave 6.²⁶

Unweighted counts and weighted counts based on the Wave 4 cross-sectional weight were obtained for the response status categories C_{Y_SW} , N_{Y_SW} , U_{Y_SW} , and I_{Y_SW} for the unweighted single-wave response rates (RR_{Y_SW}) and weighted single-wave response rates ($RR_{Y_W_SW}$), respectively.

²⁶ Excludes participants who were permanently ineligible (i.e., deceased, moved out of the United States permanently, or incarcerated long term) prior to Wave 6.

Table 5-7 provides single-wave response rates for the Wave 6 youth interview. In addition to the overall row, the table includes rows for subgroups based on response status at Wave 5 and recruitment wave as well as the following Wave 4 characteristics: sex, age, race/ethnicity, census region, and tobacco-use status. Participants for whom the Wave 4 interview (or Wave 4 household screener in the case of shadow youth) had missing information for the demographic or tobacco-use characteristic were excluded from the response rate calculation for that subgroup.

The unweighted and weighted single-wave response rates for the Wave 6 youth interview were 64.8 percent and 63.6 percent, respectively; the unweighted and weighted response rates for each subgroup were also similar to each other. Like the adult response rates, the Wave 6 single-wave youth response rates were noticeably lower than for Wave 5, and the possible reasons discussed in Section 5.1.1 apply here as well. The weighted response rates varied moderately by Wave 4 sex, race/ethnicity, census region, and ever use of tobacco.

However, the response rates by Wave 4 age, Wave 5 response status, and recruitment wave showed more variation. The weighted response rate for youth younger than age 12 at Wave 4 (56.7 percent) was noticeably lower than for youth ages 12 to 17 at that time (71.1 percent).²⁷ (In previous waves, a similar pattern has been observed for the youth interview response rates by age group among the Wave 1 and Wave 4 Cohorts.) For the Wave 4 Cohort, this is likely because the participants younger than age 12 were shadow youth at Wave 4, and became cohort members without having completed a Wave 4 interview. About half of the Wave 4 shadow youth were first asked to complete a PATH Study interview at the Wave 4.5 special data collection; the other half were first asked to complete an interview at Wave 5.

In terms of the Wave 5 response status subgroups, the Wave 6 weighted youth interview response rate was much higher for respondents to Wave 5 (77.1 percent) than for Wave 5 nonrespondents (8.2 percent). The Wave 6 weighted youth response rate was 73.2 percent for those recruited at Wave 1 and 58.5 percent for those recruited at Wave 4. A possible explanation for the large differences in subgroup response rates for both of these characteristics is that Wave 4 Cohort members who have responded to multiple previous waves, particularly those recruited at Wave 1, are

²⁷ In theory, all participants who were ages 14 to 17 at Wave 4 should have reached adulthood by Wave 6. However, some were still youth at Wave 6 as the result of age corrections since Wave 4. Due to the small sample size, these participants were combined with the youth ages 12 to 13 at Wave 4 for the purposes of this report.

likely to be more invested in the PATH Study and may be more cooperative than other cohort members. The Wave 1 sample members of the Wave 4 Cohort have persisted with the PATH Study over a number of years and all completed an interview at Wave 4, including the Wave 1 shadow youth. The Wave 4 replenishment sample members of the Wave 4 Cohort do not have the same length of history with the study and include Wave 4 shadow youth, who did not have to complete an interview at Wave 4 in order to become a study participant.

Variation in response rates by subgroups is to be expected in large-scale data collection efforts. However, it is pertinent to examine the effect of the Wave 6 single-wave weighting adjustments for the Wave 4 Cohort, given the differences in response propensity noted above. The results of these analyses are discussed in the next section.

Table 5-7. Wave 6 youth interview single-wave response rates conditioning on Wave 4 response (Wave 4 Cohort)

Characteristic ^a	<i>C_{Y,SW}</i> : Completed Wave 6 youth interview (n)	<i>I_{Y,SW}</i> : Ineligible at Wave 6 ^b (n)	<i>N_{Y,SW}</i> : Nonresponse known to be eligible at Wave 6 (n)	<i>U_{Y,SW}</i> : Nonresponse with unknown eligibility status at Wave 6 (n)	<i>RR_{Y,SW}</i> : Unweighted single-wave response rate ^c (%)	<i>RR_{Y,W,SW}</i> : Weighted single-wave response rate ^c (%)
Overall	5,585	34	1,242	1,800	64.8	63.6
Response status at Wave 5						
Responded at Wave 5	5,464	23	1,041	563	77.3	77.1
Did not respond at Wave 5	121	11	201	1,237	8.0	8.2
Wave 4 sex						
Male	2,925	17	657	960	64.5	63.5
Female	2,647	17	582	835	65.2	63.7
Wave 4 age						
Under 12	2,483	18	640	1,140	58.3	56.7
12-17	3,102	16	601	659	71.2	71.1
Wave 4 race/ethnicity ^d						
Hispanic	1,628	14	344	513	65.6	64.1
Non-Hispanic White alone	2,553	9	606	841	63.9	62.9
Non-Hispanic Black alone	649	4	143	221	64.1	62.2
Non-Hispanic other race or multiple races	557	5	112	185	65.3	65.5
Wave 4 census region						
Northeast	776	1	203	254	62.9	62.1
Midwest	1,249	5	317	433	62.5	61.5
South	2,006	19	483	671	63.6	62.4
West	1,554	9	238	442	69.6	68.4
Wave 4 ever tobacco use ^e (Wave 4 youth only)						
Ever user	184	3	45	43	67.8	66.6
Never user	2,845	13	543	601	71.4	71.5

Table 5-7. Wave 6 youth interview single-wave response rates conditioning on Wave 4 response (Wave 4 Cohort) (continued)

Characteristic ^a	<i>C_{Y,SW}</i> : Completed Wave 6 youth interview (n)	<i>I_{Y,SW}</i> : Ineligible at Wave 6 ^b (n)	<i>N_{Y,SW}</i> : Nonresponse known to be eligible at Wave 6 (n)	<i>U_{Y,SW}</i> : Nonresponse with unknown eligibility status at Wave 6 (n)	<i>RR_{Y,SW}</i> : Unweighted single-wave response rate ^c (%)	<i>RR_{Y,W,SW}</i> : Weighted single- wave response rate ^c (%)
Recruitment wave						
Wave 1	2,310	7	423	417	73.4	73.2
Wave 4	3,275	27	819	1,383	59.9	58.5

^a For each characteristic, the sum of the counts across the categories may not be equal to the count in the overall row due to missing values.

^b Excludes participants determined to be permanently ineligible (i.e., deceased, moved out of the United States permanently, or incarcerated long term) prior to Wave 6.

^c The unweighted youth interview single-wave response rate formula is provided earlier in this section. The weighted youth interview single-wave response rates were calculated using a similar approach except that the Wave 4 cross-sectional weight was used to obtain weighted counts.

^d Hispanic refers to Hispanic or Latino; Black refers to Black or African American. Alone means that the participant reported only one race.

^e An ever user of a tobacco product is someone who has ever used the product, even one or two times. The products covered by the Wave 4 youth interview are cigarettes, traditional cigars, cigarillos, filtered cigars, pipes, smokeless tobacco, snus, hookah, electronic nicotine delivery systems or ENDS (such as e-cigarettes, vape pens, personal vaporizers and mods, e-cigars, e-pipes, e-hookahs, and hookah pens), dissolvable tobacco, bidis, and kreteks.

5.2.2 Comparison Between Those Eligible for Wave 6 Interview and Wave 6 Respondents (Among the Wave 4 Cohort)

This section compares weighted estimates between the members of the Wave 4 Cohort who were eligible for the Wave 6 youth interview and the Wave 6 youth respondents in the Wave 4 Cohort. The weighted estimates cover Wave 4 demographic characteristics (shown in Table 5-8) and Wave 4 tobacco-use status (shown in Table 5-9). The structure of Table 5-8 is similar to Table 4-6 in Section 4.2.2, and the structure of Table 5-9 is similar to Table 5-4 in Section 5.1.2.

Before application of Wave 6 weighting adjustments, the distributions among those eligible for the Wave 6 youth interview were very similar to the distributions among the Wave 6 youth respondents for Wave 4 sex and race/ethnicity (see Table 5-8). The most noticeable underrepresentation among the Wave 6 youth respondents from the Wave 4 Cohort was for those younger than age 12 at Wave 4, with a corresponding overrepresentation of youth ages 12 to 17 at that time. Possible reasons for this result were discussed in Section 5.2.1. In addition, the Wave 6 youth respondents overrepresented those residing in the West census region at Wave 4 while slightly underrepresenting those residing in the Midwest census region. These results align with the Wave 6 youth interview response rates presented in Table 5-7.

As shown in the last column of Table 5-8, the Wave 6 single-wave weighting adjustments reduced the differences between the Wave 4 Cohort members eligible for the Wave 6 youth interview and those who were Wave 6 youth respondents. Furthermore, the estimated 95 percent confidence intervals for the “after Wave 6 weighting adjustment” differences included zero for all of the Wave 4 demographics examined.

Table 5-8. Comparison of Wave 4 demographic characteristics between Wave 4 respondents who were eligible for Wave 6 youth interview and Wave 6 youth interview respondents (Wave 4 Cohort)

Wave 4 characteristic	Wave 4 respondents who were eligible for Wave 6 youth interview		Wave 4 respondents who completed Wave 6 youth interview				
	Unweighted count	A: Weighted percentage, using Wave 4 cross-sectional weight [95% confidence interval]	Unweighted count	Before Wave 6 weighting adjustment		After Wave 6 weighting adjustment	
				B: Weighted percentage, using Wave 4 cross-sectional weight [95% confidence interval]	Difference in weighted percentages [B – A] [95% confidence interval]	C: Weighted percentage, using Wave 6 single-wave weight [95% confidence interval]	Difference in weighted percentages [C – A] [95% confidence interval]
Sex							
Male	4,542	50.9% [49.9%, 52.0%]	2,925	50.8% [49.5%, 52.2%]	-0.1% [-1.0%, 0.8%]	51.1% [49.7%, 52.4%]	0.1% [-0.0%, 0.3%]
Female	4,064	49.1% [48.0%, 50.1%]	2,647	49.2% [47.8%, 50.5%]	0.1% [-0.8%, 1.0%]	48.9% [47.6%, 50.3%]	-0.1% [-0.3%, 0.0%]
Age							
Under 12	4,263	52.2% [51.1%, 53.3%]	2,483	46.5% [45.2%, 47.8%]	-5.7% [-6.5%, -4.8%]	52.0% [50.7%, 53.3%]	-0.2% [-0.4%, 0.0%]
12-17	4,362	47.8% [46.7%, 48.9%]	3,102	53.5% [52.2%, 54.8%]	5.7% [4.8%, 6.5%]	48.0% [46.7%, 49.3%]	0.2% [-0.0%, 0.4%]
Race/ethnicity ^a							
Hispanic	2,485	25.2% [24.3%, 26.2%]	1,628	25.5% [24.4%, 26.7%]	0.3% [-0.5%, 1.1%]	25.2% [24.1%, 26.4%]	-0.0% [-0.2%, 0.2%]
Non-Hispanic White alone	4,000	51.3% [50.2%, 52.4%]	2,553	50.9% [49.6%, 52.2%]	-0.4% [-1.3%, 0.6%]	51.3% [49.9%, 52.6%]	-0.0% [-0.2%, 0.2%]
Non-Hispanic Black alone	1,013	13.2% [12.5%, 13.9%]	649	12.9% [12.1%, 13.9%]	-0.2% [-0.8%, 0.3%]	13.2% [12.3%, 14.1%]	-0.0% [-0.2%, 0.1%]
Non-Hispanic other race or multiple races	854	10.3% [9.7%, 11.0%]	557	10.6% [9.8%, 11.5%]	0.3% [-0.2%, 0.8%]	10.3% [9.5%, 11.2%]	0.0% [-0.1%, 0.1%]

Table 5-8. Comparison of Wave 4 demographic characteristics between Wave 4 respondents who were eligible for Wave 6 youth interview and Wave 6 youth interview respondents (Wave 4 Cohort) (continued)

Wave 4 characteristic	Wave 4 respondents who were eligible for Wave 6 youth interview		Wave 4 respondents who completed Wave 6 youth interview				
	Unweighted count	A: Weighted percentage, using Wave 4 cross-sectional weight [95% confidence interval]	Unweighted count	Before Wave 6 weighting adjustment		After Wave 6 weighting adjustment	
				B: Weighted percentage, using Wave 4 cross-sectional weight [95% confidence interval]	Difference in weighted percentages [B – A] [95% confidence interval]	C: Weighted percentage, using Wave 6 single-wave weight [95% confidence interval]	Difference in weighted percentages [C – A] [95% confidence interval]
Census region							
Northeast	1,233	16.1% [15.3%, 16.9%]	776	15.7% [14.8%, 16.7%]	-0.4% [-1.1%, 0.3%]	16.1% [15.1%, 17.0%]	-0.0% [-0.2%, 0.2%]
Midwest	1,999	21.2% [20.3%, 22.0%]	1,249	20.5% [19.4%, 21.5%]	-0.7% [-1.4%, -0.1%]	21.2% [20.1%, 22.3%]	0.0% [-0.1%, 0.2%]
South	3,160	38.4% [37.3%, 39.4%]	2,006	37.6% [36.3%, 38.9%]	-0.8% [-1.7%, 0.2%]	38.3% [37.0%, 39.6%]	-0.1% [-0.3%, 0.1%]
West	2,234	24.4% [23.5%, 25.3%]	1,554	26.2% [25.1%, 27.4%]	1.8% [1.0%, 2.7%]	24.4% [23.3%, 25.6%]	0.0% [-0.1%, 0.2%]

^a Hispanic refers to Hispanic or Latino; Black refers to Black or African-American. Alone means that the participant reported only one race.

Table 5-9. Comparison of Wave 4 ever tobacco-use estimates between Wave 4 youth who were eligible for Wave 6 youth interview and Wave 4 youth who completed Wave 6 youth interview (Wave 4 Cohort)*

Wave 4 characteristic ^a	Wave 4 youth who were eligible for Wave 6 youth interview		Wave 4 youth who completed Wave 6 youth interview				
	Sample size	A: Weighted estimate, using Wave 4 cross-sectional weight [95% confidence interval]	Sample size	Before Wave 6 weighting adjustment		After Wave 6 weighting adjustment	
				B: Weighted estimate, using Wave 4 cross-sectional weight [95% confidence interval]	Difference in weighted estimates [B – A] [95% confidence interval]	C: Weighted estimate, using Wave 6 single-wave weight [95% confidence interval]	Difference in weighted estimates [C – A] [95% confidence interval]
Overall ^b	4,261	6.2% [5.4%, 7.1%]	3,029	5.8% [4.9%, 6.7%]	-0.4% [-0.9%, 0.1%]	5.9% [5.0%, 7.0%]	-0.2% [-0.7%, 0.2%]
Sex							
Male	2,243	8.0% [6.9%, 9.3%]	1,576	7.5% [6.3%, 8.9%]	-0.5% [-1.3%, 0.2%]	7.7% [6.4%, 9.1%]	-0.3% [-1.1%, 0.5%]
Female	2,002	4.2% [3.4%, 5.4%]	1,442	4.1% [3.1%, 5.4%]	-0.1% [-0.7%, 0.4%]	4.2% [3.1%, 5.6%]	-0.1% [-0.7%, 0.5%]
Race/ethnicity ^c							
Hispanic	1,237	7.2% [5.6%, 9.1%]	889	6.7% [5.0%, 8.9%]	-0.5% [-1.6%, 0.6%]	6.9% [5.1%, 9.2%]	-0.3% [-1.4%, 0.8%]
Non-Hispanic White alone	1,853	6.4% [5.3%, 7.7%]	1,290	6.0% [4.7%, 7.6%]	-0.4% [-1.2%, 0.4%]	6.1% [4.8%, 7.8%]	-0.3% [-1.0%, 0.5%]
Non-Hispanic Black alone	523	4.4% [2.7%, 7.0%]	369	4.5% [2.5%, 8.0%]	0.1% [-1.1%, 1.4%]	4.7% [†] [2.6%, 8.5%]	0.3% [-1.1%, 1.8%]
Non-Hispanic other race or multiple races	416	5.3% [3.5%, 8.0%]	308	5.4% [3.4%, 8.5%]	0.1% [-1.0%, 1.2%]	5.6% [3.5%, 8.7%]	0.3% [-0.9%, 1.4%]

Table 5-9. Comparison of Wave 4 ever tobacco-use estimates between Wave 4 youth who were eligible for Wave 6 youth interview and Wave 4 youth who completed Wave 6 youth interview (Wave 4 Cohort) (continued)*

Wave 4 characteristic ^a	Wave 4 youth who were eligible for Wave 6 youth interview		Wave 4 youth who completed Wave 6 youth interview				
	Sample size	A: Weighted estimate, using Wave 4 cross-sectional weight [95% confidence interval]	Sample size	Before Wave 6 weighting adjustment		After Wave 6 weighting adjustment	
				B: Weighted estimate, using Wave 4 cross-sectional weight [95% confidence interval]	Difference in weighted estimates [B – A] [95% confidence interval]	C: Weighted estimate, using Wave 6 single-wave weight [95% confidence interval]	Difference in weighted estimates [C – A] [95% confidence interval]
Census region							
Northeast	608	5.8% [4.2%, 7.9%]	422	5.6% [3.7%, 8.2%]	-0.2% [-1.5%, 1.1%]	5.7% [3.7%, 8.7%]	-0.1% [-1.8%, 1.6%]
Midwest	965	6.1% [4.7%, 7.9%]	673	4.9% [3.5%, 6.8%]	-1.2% [-2.6%, 0.1%]	5.2% [3.6%, 7.4%]	-0.9% [-2.4%, 0.5%]
South	1,577	6.9% [5.4%, 8.7%]	1,091	6.6% [5.1%, 8.5%]	-0.3% [-1.0%, 0.4%]	6.6% [5.1%, 8.6%]	-0.3% [-1.0%, 0.5%]
West	1,111	5.4% [4.0%, 7.2%]	843	5.5% [4.0%, 7.7%]	0.1% [-0.5%, 0.7%]	5.7% [4.1%, 8.0%]	0.3% [-0.4%, 1.0%]

* An 'ever user' of a tobacco product is someone who has ever used the product, even one or two times. The products covered by the Wave 4 youth interview are cigarettes, traditional cigars, cigarillos, filtered cigars, pipes, smokeless tobacco, snus, hookah, dissolvable tobacco, electronic nicotine delivery systems or ENDS (such as e-cigarettes, vape pens, personal vaporizers and mods, e-cigars, e-pipes, e-hookahs, and hookah pens), bidis, and kreteks.

^a For each characteristic, the sum of the counts across the categories may not be equal to the count in the overall row due to missing values.

^b Table 5-9 includes a subset of the youth in Table 5-8 because there are no Wave 4 interview data for the Wave 6 youth who were shadow youth at Wave 4.

^c Hispanic refers to Hispanic or Latino; Black refers to Black or African-American. Alone means that the participant reported only one race.

[†] Estimate should be interpreted with caution because it has low precision. It is based on a denominator sample size of less than 50, or the coefficient of variation of the estimate or its complement is larger than 30 percent.

Table 5-9 shows the comparison of Wave 4 “ever tobacco-use” estimates between the Wave 4 youth respondents eligible for the Wave 6 youth interview and the Wave 4 youth respondents who completed the Wave 6 youth interview (among the Wave 4 Cohort). Table 5-9 includes a subset of the youth in Table 5-8 because there are no Wave 4 interview data for the Wave 6 youth who were shadow youth at Wave 4. The estimated 95 percent confidence intervals for both the “before Wave 6 weighting adjustment” differences and the “after Wave 6 weighting adjustment” differences included zero for all of the Wave 4 demographics examined.

These results reflect positively on the use of both demographic and tobacco-use variables from Wave 4 for calibrating the Wave 6 single-wave weight for the Wave 4 Cohort. Assuming that the Wave 4 demographic and tobacco-use characteristics in Tables 5-8 and 5-9 are correlated with key tobacco- and health-related outcome measures at Wave 6 of the PATH Study, the results do not raise concern about bias in the youth interview estimates for the Wave 4 Cohort due to nonresponse at Wave 6.

5.2.3 Comparison of Cigarette-Smoking Estimates for Youth Ages 14 to 17 to Other National Studies

This section compares estimates for two common measures of youth cigarette smoking based on the Wave 4 Cohort respondents at Wave 6 of the PATH Study to similar estimates based on data from NYRBS 2019, NHANES 2017-2020, NSDUH 2020, MTF 2021, and NYTS 2021. The primary measure of cigarette smoking among youth is whether the youth has ever tried smoking a cigarette, even one or two puffs. A second measure is having smoked cigarettes at all in the past 30 days. (See Appendix A for details.)

Tables 5-10 and 5-11 present estimates of the prevalence of cigarette smoking for youth ages 14 to 17, and for subgroups among these youth. In Table 5-10, comparisons of PATH Study estimates to NHANES and NSDUH are provided by age (14-15 and 16-17) and comparisons to MTF are provided by grade (10 and 12). In Table 5-11, estimates from the PATH Study are compared to NYRBS and NYTS estimates among high school students only.²⁸ As mentioned in Section 3.2.4, the

²⁸ To be precise, the estimates were computed for youth in grades 9 to 12 (who were ages 14 to 17).

data from all five external sources were restricted to ages 14 to 17 to match the age range for PATH Study youth at Wave 6.

The point estimates from the PATH Study were calculated using the Wave 6 single-wave weight for the Wave 4 Cohort. The corresponding replicate weights were used to calculate variances and 95 percent confidence intervals. Point estimates and confidence intervals are reported for the other national studies as well.

Table 5-10. Comparison of cigarette-smoking estimates by age and grade from Wave 6 youth interview (Wave 4 Cohort) and other national studies, for youth ages 14 to 17

Cigarette use by age					Cigarette use by grade			
Cigarette-smoking measure and subgroup	PATH Study Wave 6 (2021) sample size	Percentage from PATH Study Wave 6 (2021) [95% confidence interval]	Percentage from NHANES 2017-2020 [95% confidence interval]	Percentage from NSDUH 2020 [95% confidence interval]	Cigarette-smoking measure and subgroup	PATH Study Wave 6 (2021) sample size	Percentage from PATH Study Wave 6 (2021) [95% confidence interval]	Percentage from MTF 2021 [95% confidence interval]
Ever tried cigarette smoking, even one or two puffs					Ever tried cigarette smoking, even one or two puffs			
Age 14-15, overall	2,449	3.1% [2.5%, 3.9%]	11.0% [6.6%, 17.8%]	7.3% [5.7%, 9.3%]	10 th grade	1,124	5.1% [4.0%, 6.6%]	9.5% [8.1%, 11.0%]
Age 14-15, male	1,290	3.4% [2.5%, 4.5%]	13.6% [7.5%, 23.3%]	7.6% [5.0%, 11.4%]	10 th grade, male	589	6.1% [4.2%, 8.8%]	9.6% [8.0%, 11.4%]
Age 14-15, female	1,151	2.9% [2.1%, 4.0%]	8.2% [4.3%, 15.2%]	6.9% [4.8%, 9.8%]	10 th grade, female	531	4.2% [2.8%, 6.2%]	8.4% [7.0%, 10.0%]
Age 16-17, overall	3,113	9.0% [7.9%, 10.3%]	20.3% [14.7%, 27.3%]	11.8% [9.4%, 14.6%]	12 th grade	924	9.0% [7.3%, 11.0%]	15.9% [12.6%, 19.9%]
Age 16-17, male	1,627	9.7% [8.1%, 11.6%]	23.5% [15.0%, 34.8%]	14.3% [11.2%, 18.0%]	12 th grade, male	460	9.0% [6.6%, 12.2%]	15.8% [12.6%, 19.6%]
Age 16-17, female	1,475	8.4% [7.1%, 9.9%]	17.0% [11.0%, 25.2%]	9.5% [6.8%, 13.1%]	12 th grade, female	462	9.0% [6.7%, 12.0%]	15.6% [11.2%, 21.4%]
Has smoked cigarettes in past 30 days, age 14-17	5,577	1.1% [0.8%, 1.4%]	3.3% [1.5%, 7.0%]	1.9% [1.4%, 2.4%]	Has smoked cigarettes in past 30 days, 12 th grade	926	2.1% [1.3%, 3.5%]	3.2% [1.6%, 6.4%]

Table 5-11. Comparison of cigarette-smoking estimates from Wave 6 youth interview (Wave 4 Cohort) and other national studies, for high school students ages 14 to 17

Cigarette-smoking measure and subgroup^a	PATH Study Wave 6 (2021) sample size	Percentage from PATH Study Wave 6 (2021) [95% confidence interval]	Percentage from NYRBS 2019 [95% confidence interval]	Percentage from NYTS 2021 [95% confidence interval]
Ever tried cigarette smoking, even one or two puffs				
Overall	4,512	5.7% [5.1%, 6.4%]	22.0% [19.5%, 24.6%]	9.9% [8.7%, 11.1%]
Male	2,348	6.1% [5.2%, 7.1%]	22.7% [19.9%, 25.9%]	9.3% [8.1%, 10.8%]
Female	2,149	5.3% [4.5%, 6.4%]	21.3% [18.9%, 24.0%]	10.4% [8.9%, 12.2%]
Hispanic	1,273	5.5% [4.3%, 7.0%]	24.5% [20.1%, 29.4%]	7.8% [6.2%, 9.8%]
Non-Hispanic White alone	2,036	7.0% [5.9%, 8.2%]	23.0% [20.0%, 26.4%]	11.4% [9.8%, 13.2%]
Non-Hispanic Black alone	493	3.7% [2.4%, 5.8%]	13.1% [10.2%, 16.6%]	5.6% [4.0%, 7.6%]
Non-Hispanic other race or multiple races	464	3.5% [2.2%, 5.6%]	18.6% [13.9%, 24.4%]	11.1% [7.9%, 15.3%]
Has smoked cigarettes in past 30 days	4,525	0.8% [0.6%, 1.2%]	5.0% [4.1%, 6.1%]	1.5% [1.2%, 2.0%]

^a For the race/ethnicity subgroups, Hispanic refers to Hispanic or Latino, Black refers to Black or African American, and alone means the respondent reported only one race.

When comparing by age, the estimates of ever cigarette smoking from Wave 6 of the PATH Study were uniformly much lower than the estimates from NHANES 2017-2020 with non-overlapping 95 percent confidence intervals (see Table 5-10). For youth ages 14 and 15, the Wave 6 estimates were lower than the estimates from NSDUH 2020 with non-overlapping 95 percent confidence intervals. For youth ages 16 and 17, the PATH Study and NSDUH estimates were comparable, and both were lower than the estimates from NHANES. For past 30-day cigarette smoking among 14- to 17-year-olds, the Wave 6 estimates were lower than the estimates from NHANES and NSDUH with non-overlapping confidence intervals. Most of the confidence intervals did not overlap between Wave 6 of the PATH Study and MTF 2021 for the grade-level estimates of ever cigarette smoking, with the MTF estimates being consistently higher. However, the confidence intervals for cigarette smoking in the past 30 days among youth in 12th grade did overlap.

Similarly, the PATH Study estimates of cigarette smoking among high school students ages 14 to 17 were well below those from the comparison studies in Table 5-11. Only two of the NYTS 2021 estimates had 95 percent confidence intervals that overlapped with those from Wave 6 of the PATH

Study—the estimates of ever cigarette smoking for Hispanic and non-Hispanic Black alone youth. None of the NYRBS 2019 confidence intervals overlapped with those from Wave 6.

It is unclear whether the PATH Study estimates of cigarette smoking among Wave 6 youth ages 14 to 17 are generally lower due to nonresponse bias in the estimates from one or more of the studies compared, or for other reasons. However, a similar pattern has been observed in previous comparisons between estimates from the PATH Study and other national studies. The differences can be explained by a number of potential reasons, similar to those discussed in Section 5.1.3. In addition to the varying degrees of sampling and measurement errors, the surveys differ in terms of survey mode, context, question order and wording, and timing of data collection. The NYRBS, MTF, and NYTS are all self-administered school-based surveys. In 2019, the NYRBS was a pencil-and-paper survey that was completed by students in the classroom. In 2021, the MTF and NYTS both transitioned to fully online Web surveys that were administered in the classroom for schools conducting in-person classes and at home for schools conducting classes virtually. The NYTS is administered to students in grades 6 through 12, the MTF is administered to 8th, 10th, and 12th graders, and the NYRBS is administered to students in grades 9 to through 12. The NHANES and NSDUH use ACASI for the in-person questions about tobacco use by youth, and administer the survey individually in a mobile examination center or household setting, respectively. Due to the COVID-19 pandemic, Wave 6 of the PATH Study was administered either over the telephone without ACASI or in person using ACASI (see Section 2.8). Currivan, Nyman, Turner, and Biener (2004) found that even when telephone ACASI was used, estimates of youth smoking prevalence were lower for a telephone survey of youth smoking than for a school-based survey of the same population (see also Fowler & Stringfellow, 2001, for a discussion of higher smoking rates in school-based surveys). In addition, higher estimates of cigarette use in school-based studies as compared to home-based studies do not necessarily mean the school-based estimates are more accurate. As noted in Boyd, Veliz, Evans-Polce, Eisman, and McCabe (2020), school-based studies are often implemented in a group setting and the fact that peers could observe their answers may make respondents feel it more socially desirable to over-report risky behaviors.

As discussed in Section 5.1.3, FDA tobacco regulations went into effect before the beginning of data collection for Wave 6 of the PATH Study, NSDUH 2020, MTF 2021, and NYTS 2021, and near or after the close of data collection for NYRBS 2019 and NHANES 2017-2020. In addition, the Wave 6, NSDUH 2020, MTF 2021, and NYTS 2021 responses were gathered during the COVID-

19 pandemic, whereas data collection for NYRBS 2019 and NHANES 2017-2020 occurred prior to the pandemic.²⁹ The differences between cigarette-smoking estimates from the PATH Study and NYRBS and NHANES could be in part due to a material change in behavior in the population caused by the new FDA regulations.

The contexts and purposes of these surveys also differ: NYRBS and NHANES are general health surveys; whereas MTF and NSDUH are cross-sectional surveys on substance use (including tobacco use) and health, and NYTS is a cross-sectional survey focusing on tobacco use. Unlike the cross-sectional prevalence surveys, the PATH Study uses a longitudinal cohort design to assess within-person changes and between-person differences in tobacco-use behaviors and health over time. Differences among the questions used in the instruments of these different studies are outlined in Appendix A.

5.2.4 Comparison of ENDS-Use Estimates for Youth Ages 14 to 17 to Other National Studies

This section compares estimates for youth ENDS use based on the Wave 4 Cohort respondents at Wave 6 of the PATH Study to similar estimates based on data from NYRBS 2019, NSDUH 2020, MTF 2021, and NYTS 2021. The primary measure of ENDS use among youth is whether the youth has ever tried using ENDS. A second measure is having used ENDS at all in the past 30 days. (See Appendix B for details.) NYRBS 2019 asks about use of “electronic vapor products,” NSDUH 2020 asks whether respondents have ever “vaped nicotine or tobacco,” MTF 2021 asks respondents how often they have “vaped nicotine” in their lifetime, and NYTS 2021 asks about “e-cigarette” use. However, the definitions provided to respondents immediately prior to asking these questions are virtually identical to the definition of “electronic nicotine products” provided to PATH Study respondents.

Tables 5-12 and 5-13 present estimates of the prevalence of ENDS use for youth ages 14 to 17, and for subgroups among these youth. Table 5-12 provides Wave 6 estimates by age (14-15 and 16-17) to compare to NSDUH 2020, and by grade (10 and 12) to compare to MTF 2021. Table 5-13 shows

²⁹ NHANES 2017-2020 stopped data collection in March of 2020, due to school closures from the COVID-19 pandemic.

estimates only among high school students³⁰ to compare Wave 6 of the PATH Study to NYRBS 2019 and NYTS 2021. The data from all four external sources were restricted to ages 14 to 17 to match the age range for PATH Study youth at Wave 6.

The point estimates from the PATH Study were calculated using the Wave 6 single-wave weight for the Wave 4 Cohort. The corresponding replicate weights were used to calculate variances and 95 percent confidence intervals. Point estimates and confidence intervals are reported for the other national studies as well.

³⁰ To be precise, the estimates were computed for youth in grades 9 to 12 (who were ages 14 to 17).

Table 5-12. Comparison of ENDS-use estimates by age and grade from Wave 6 youth interview (Wave 4 Cohort) and other national studies, for youth ages 14 to 17

ENDS use by age				ENDS use by grade			
ENDS-use measure and subgroup	PATH Study Wave 6 (2021) sample size	Percentage from PATH Study Wave 6 (2021) [95% confidence interval]	Percentage from NSDUH 2020 [95% confidence interval]	ENDS-use measure and subgroup	PATH Study Wave 6 (2021) sample size	Percentage from PATH Study Wave 6 (2021) [95% confidence interval]	Percentage from MTF 2021 [95% confidence interval]
Ever tried ENDS, even one or two times				Ever tried ENDS, even one or two times			
Age 14-15, overall	2,439	11.0% [9.8%, 12.3%]	14.0% [12.4%, 15.8%]	10 th grade	1,125	14.6% [12.5%, 17.1%]	27.7% [24.7%, 30.9%]
Age 14-15, male	1,285	10.6% [9.1%, 12.4%]	11.1% [8.4%, 14.4%]	10 th grade, male	590	14.3% [11.3%, 17.9%]	24.4% [21.5%, 27.5%]
Age 14-15, female	1,146	11.4% [9.6%, 13.4%]	17.1% [13.5%, 21.4%]	10 th grade, female	532	14.9% [12.1%, 18.1%]	30.5% [27.1%, 34.1%]
Age 16-17, overall	3,108	24.9% [22.9%, 27.0%]	21.0% [18.0%, 24.4%]	12 th grade	924	27.7% [24.7%, 31.0%]	36.5% [32.8%, 40.3%]
Age 16-17, male	1,629	23.8% [21.5%, 26.2%]	19.7% [16.4%, 23.5%]	12 th grade, male	461	27.2% [23.3%, 31.5%]	34.2% [29.1%, 39.7%]
Age 16-17, female	1,470	26.1% [23.5%, 28.9%]	22.2% [17.5%, 27.8%]	12 th grade, female	461	28.3% [23.8%, 33.3%]	38.0% [33.8%, 42.5%]
Has used ENDS in past 30 days, age 14–17	5,574	5.9% [5.2%, 6.8%]	7.4% [6.3%, 8.8%]	Has used ENDS in past 30 days, 12 th grade	927	10.0% [8.1%, 12.4%]	17.5% [14.5%, 20.9%]

Table 5-13. Comparison of ENDS-use estimates from Wave 6 youth interview (Wave 4 Cohort) and other national studies, for high school students ages 14 to 17

ENDS-use measure and subgroup^a	PATH Study Wave 6 (2021) sample size	Percentage from PATH Study Wave 6 (2021) [95% confidence interval]	Percentage from NYRBS 2019 [95% confidence interval]	Percentage from NYTS 2021 [95% confidence interval]
Ever tried ENDS, even one or two times				
Overall	4,499	18.2% [16.9%, 19.5%]	48.2% [46.1%, 50.3%]	27.5% [25.1%, 30.2%]
Male	2,346	17.3% [15.8%, 18.9%]	46.9% [44.5%, 49.3%]	26.1% [23.3%, 29.0%]
Female	2,140	19.1% [17.4%, 21.0%]	49.5% [47.0%, 52.1%]	29.2% [26.3%, 32.2%]
Hispanic	1,270	17.8% [15.3%, 20.6%]	48.5% [45.8%, 51.2%]	24.2% [21.6%, 27.0%]
Non-Hispanic White alone	2,032	20.9% [19.0%, 22.8%]	52.2% [49.6%, 54.8%]	31.6% [28.4%, 35.0%]
Non-Hispanic Black alone	491	13.3% [10.3%, 16.9%]	39.9% [36.9%, 43.0%]	15.0% [12.3%, 18.3%]
Non-Hispanic other race or multiple races	466	14.8% [11.6%, 18.6%]	38.2% [31.6%, 45.2%]	29.9% [25.0%, 35.3%]
Has used ENDS in past 30 days	4,523	6.1% [5.2%, 7.0%]	31.0% [29.0%, 33.1%]	10.5% [9.0%, 12.2%]

^a For the race/ethnicity subgroups, Hispanic refers to Hispanic or Latino, Black refers to Black or African American, and alone means the respondent reported only one race.

The estimates of ever use of ENDS from Wave 6 of the PATH Study were generally lower than the estimates from NSDUH 2020 for youth ages 14 and 15, with non-overlapping 95 percent confidence intervals for the age group overall and for females (see Table 5-12). The PATH Study estimates were comparable with the NSDUH estimates for youth ages 16 and 17, with overlapping confidence intervals. The PATH Study estimate of past 30-day ENDS use among 14- to 17-year-olds was slightly lower than the NSDUH estimate, but the confidence intervals for these estimates overlapped. The estimates of ever use of ENDS from Wave 6 of the PATH Study were lower than those from MTF 2021 for all grade and sex subgroups among youth ages 14 to 17, with only the 12th-grade male subgroup having an overlap in the 95 percent confidence intervals. The PATH Study estimate of past 30-day ENDS use among 12th graders was also lower than the MTF estimate, with no overlap in confidence intervals.

As shown in Table 5-13, the estimates of ever use of ENDS from Wave 6 of the PATH Study among high school students ages 14 to 17 were uniformly much lower than those from NYRBS 2019 and NYTS 2021. None of the NYRBS 95 percent confidence intervals overlapped with those from the PATH Study, and the only overlapping confidence intervals between NYTS and the PATH Study were for non-Hispanic Black alone youth. The NYTS 2021 estimates of ever use of

ENDS were also lower than the NYRBS 2019 estimates among high school students, with only one overlap of confidence intervals for non-Hispanic youth of other or multiple races. The PATH Study estimate of ENDS use in the past 30 days among high-school students ages 14 to 17 was lower than the NYTS estimate, which in turn was lower than the NYRBS estimate—none of the corresponding confidence intervals overlapped.

As discussed in Section 5.2.3, some differences in the estimates may be attributable to differences in timing of data collection, survey mode, context, question order and wording, and measurement error. In particular, NYRBS 2019 was administered in the classroom, and MTF 2021 and NYTS 2021 were administered in the classroom for schools conducting in-person classes and at home for schools conducting classes virtually, whereas Wave 6 of the PATH Study was administered over the telephone or at home. It is uncertain which factors lead to the observed differences in ENDS-use estimates for youth ages 14 to 17 between the PATH Study and other national studies; however, the timing of data collection is likely important given changes in the prevalence of ENDS use in recent years.

6. Summary of Findings

This chapter summarizes the PATH Study's Wave 6 nonresponse bias analysis findings separately for the Wave 1 Cohort and the Wave 4 Cohort. Each of the following components of the analysis is reviewed:

- Wave 6 interview response rates.
- The national representativeness of Wave 6 interview respondents and statistical weighting adjustments to reduce potential nonresponse bias.
- The degree to which Wave 6 estimates of cigarette-smoking and ENDS use align with similar estimates from other national studies.

6.1 Wave 1 Cohort

6.1.1 Adult Interview

The Wave 1 Cohort unweighted all-waves response rate was slightly lower than the weighted all-waves response rate for the Wave 6 adult interview (conditioning on Wave 1 response) overall and for many subgroups. There were also differences in weighted all-waves response rates across subgroups.

- The weighted overall all-waves response rate was 54.4 percent.
- Wave 6 adults who were younger than age 18 and ages 18 to 24 at Wave 1 had lower weighted all-waves response rates (50.5 percent and 47.0 percent, respectively) than those in the older age groups (who had response rates ranging from 52.9 percent to 60.6 percent).
- Adults with a bachelor's degree or advanced degree at Wave 1 had a higher propensity to respond at every primary wave (61.9 percent) than adults with less education (who had response rates between 50.6 percent and 54.3 percent).
- Adults with health insurance at Wave 1 were more likely to respond at every primary wave (56.4 percent) than adults without health insurance (48.8 percent).
- The weighted all-waves response rates were 8.2 percentage points lower for Wave 1 adult current established tobacco users than for Wave 1 adults who were not current

established tobacco users, and 9.7 percentage points lower for Wave 1 youth who had ever used tobacco than for Wave 1 youth who were never users.

However, these differences in response rates do not necessarily indicate nonresponse bias in the Wave 6 adult estimates for the Wave 1 Cohort.

As a result of comparing select demographic and socio-economic characteristics between those eligible to be Wave 6 adult all-waves respondents and the Wave 6 adult all-waves respondents, the noticeable differences prior to Wave 6 weighting adjustments to address nonresponse since Wave 1 were: an underrepresentation of the male population among the all-waves respondents, and overrepresentations of adults ages 45 to 64 at Wave 1 and those with a bachelor's or advanced degree at Wave 1. When similar comparisons were made with respect to tobacco-use characteristics, the differences consistently indicated an underrepresentation of Wave 1 tobacco users among the Wave 6 adult all-waves respondents.

After Wave 6 weighting adjustments were applied, the differences between those eligible to be Wave 6 adult all-waves respondents and the Wave 6 adult all-waves respondents were greatly reduced. However, among the Wave 6 adult respondents, there were slight overrepresentations of adults 25 to 44 at Wave 1 and those with a bachelor's degree or advanced degree at Wave 1, and a slight underrepresentation of adults ages 65 and older at Wave 1. Underrepresentations of Wave 1 adults who were current established tobacco users and Wave 1 youth who were ever users of tobacco persisted overall and for a number of subgroups (see Section 4.1.2). The 95 percent confidence intervals for the estimated differences were furthest away from zero for males and adults with health insurance who were current established tobacco users at Wave 1, and for non-Hispanic Whites and youth who resided in the South census region who were ever users of tobacco at Wave 1. Despite the use of both demographic and tobacco-use variables from Wave 1 for calibrating the Wave 6 all-waves weight, which improved the representativeness of the Wave 6 adult all-waves respondents, these results demonstrate the challenges faced by panel studies as waves of nonresponse accrue.

6.1.2 Youth Interview

The Wave 1 Cohort unweighted all-waves response rate was similar to the weighted all-waves response rate for the Wave 6 youth interview (conditioning on Wave 1 response) overall and for subgroups, although there were differences in weighted all-waves response rates across subgroups.

- The weighted overall all-waves response rate was 53.7 percent.
- Moderate differences (i.e., of no more than 5 percent) were found by Wave 1 race/ethnicity and census region, with the largest difference indicating that Wave 6 youth who lived in the West census region at Wave 1 (56.2 percent) had a higher propensity to respond at every primary wave than those who lived in the Northeast census region (51.5 percent).

These differences in response rates do not necessarily indicate nonresponse bias in the Wave 6 youth estimates for the Wave 1 Cohort.

As a result of comparing select demographic characteristics between those eligible to be Wave 6 youth all-waves respondents and the Wave 6 youth all-waves respondents, there was no evidence of bias due to nonresponse since the formation of the Wave 1 Cohort, either before or after applying the Wave 6 weighting adjustments.

6.2 Wave 4 Cohort

6.2.1 Adult Interview

The Wave 4 Cohort unweighted single-wave response rate was slightly lower than the weighted single-wave response rate for the Wave 6 adult interview (conditioning on Wave 4 response) overall and for most subgroups. There were differences in weighted single-wave response rates across subgroups.

- The weighted overall single-wave response rate was 73.5 percent.
- The subgroups with the largest differences in the weighted all-waves response rates for Wave 1 Cohort adults also had large differences in the weighted single-wave response rates for Wave 4 Cohort adults.
- The weighted single-wave response rates for Wave 4 Cohort adults were lower for those younger than age 18 and ages 18 to 24 at Wave 4 (65.9 percent and 65.6 percent,

respectively) than those in the older age groups (who had response rates ranging from 72.8 percent to 78.5 percent).

- Adults with a bachelor's degree or advanced degree at Wave 4 had a higher propensity to respond (81.6 percent) than adults with less education (who had response rates ranging from 67.2 percent to 73.5 percent).
- Adults with health insurance at Wave 4 were more likely to respond at Wave 6 (75.1 percent) than adults without health insurance (67.3 percent).
- The weighted single-wave response rates were 8.4 percentage points lower for Wave 4 adult current established tobacco users than for Wave 4 adults who were not current established tobacco users, and 8.2 percentage points lower for Wave 4 youth who had ever used tobacco than for Wave 4 youth who were never users.
- The weighted single-wave response rate was much higher for Wave 6 adults who responded at Wave 5 (81.0 percent) than for those who did not respond at Wave 5 (17.4 percent).
- There was also a large difference in the weighted single-wave response rate for those recruited at Wave 4 (62.1 percent) compared to those recruited at Wave 1 (75.4 percent). The Wave 1 sample members of the Wave 4 Cohort have persisted with the PATH Study over a number of years and all completed an interview at Wave 4.

However, these differences in response rates do not necessarily indicate nonresponse bias in the Wave 6 adult estimates for the Wave 4 Cohort.

As a result of comparing select demographic and socio-economic characteristics between those eligible for the Wave 6 adult interview and the Wave 6 adult respondents, prior to Wave 6 weighting adjustments, the most noticeable underrepresentations due to nonresponse at Wave 6 among the Wave 4 Cohort were for males, adults ages 18 to 24 at Wave 4, and Wave 4 adults with at most a high school diploma. There were corresponding overrepresentations of adults ages 45 to 64 at Wave 4 and Wave 4 adults with a bachelor's degree or advanced degree. (See Section 5.1.2 for a full description of findings.) When similar comparisons were made with respect to tobacco-use characteristics, the differences consistently indicated underrepresentations of Wave 4 adult current established tobacco users and Wave 4 youth ever users of tobacco among the Wave 6 adult respondents.

After Wave 6 weighting adjustments were applied, the differences between those eligible for the Wave 6 adult interview and the Wave 6 adult respondents from the Wave 4 Cohort were mostly negligible with a few exceptions. Among the Wave 6 adult respondents there was a slight

underrepresentation of adults ages 65 and older at Wave 4, and a slight overrepresentation of those with a bachelor's degree or advanced degree at Wave 4. Although the Wave 6 weighting adjustments successfully addressed the underrepresentation of Wave 4 adult current established tobacco users, some underrepresentation of Wave 4 youth ever users of tobacco persisted overall, for males, and for Wave 6 adults who resided in the West census region at Wave 4.

There is no evidence of nonresponse bias in the Wave 4 Cohort with respect to current cigarette-smoking behavior among adults, in the sense that the PATH Study's estimates were all within the range of estimates from other national studies. In general, the Wave 4 Cohort estimates of adult current cigarette-smoking at Wave 6 were similar to the estimates from NHANES 2017-2020 and NSDUH 2020, and higher than the estimates from TUS-CPS 2018-2019 and NHIS 2021.

The Wave 4 Cohort estimates of adult current ENDS use at Wave 6 were higher than the estimates from TUS-CPS 2018-2019 but generally similar to the estimates from NHIS 2021. However, the current ENDS-use estimate for adults ages 18 to 24 was higher for Wave 6 of the PATH Study (14.5 percent) than both TUS-CPS 2018-2019 (5.3 percent) and NHIS 2021 (11.0 percent). The 95 percent confidence intervals for the PATH Study estimates overlapped with those from NHIS 2021 for almost all of the other estimates, but they only overlapped for one estimate from TUS-CPS 2018-2019.

When interpreting the results of these cross-study comparisons, caution should be taken due to differences in the timing of data collection, especially with Wave 6 of the PATH Study, NSDUH 2020, and NHIS 2021 collecting responses during the pandemic, whereas TUS-CPS 2018-2019 and NHANES 2017-2020 were completed prior to the pandemic. Other possible reasons for differences in the estimates between studies include varying degrees of sampling and measurement errors as well as differences in survey mode, question order, and context.

6.2.2 Youth Interview

The Wave 4 Cohort unweighted single-wave response rate was slightly higher than the weighted single-wave response rate for the Wave 6 youth interview (conditioning on Wave 4 response) overall and for most subgroups, although there were differences in weighted response rates across subgroups.

- The weighted overall single-wave response rate was 63.6 percent.
- The weighted single-wave response rates varied by 14.4 percentage points for Wave 4 age and 14.7 percentage points for recruitment wave. Response rates were lower for participants younger than age 12 at Wave 4 (56.7 percent) and those recruited at Wave 4 (58.5 percent)—these are highly correlated characteristics because all Wave 4 shadow youth in the Wave 4 Cohort are from the Wave 4 replenishment sample. Shadow youth were not required to complete a PATH Study interview at the time of their enrollment in order to become a study participant.
- The weighted single-wave response rate was higher for Wave 6 youth who responded at Wave 5 (77.1 percent) than for those who did not respond at Wave 5 (8.2 percent).

These differences in response rates do not necessarily indicate nonresponse bias in the Wave 6 youth estimates for the Wave 4 Cohort.

As a result of comparing select demographic and tobacco-use characteristics between those eligible for the Wave 6 youth interview and the Wave 6 youth respondents, the noticeable underrepresentation prior to Wave 6 weighting adjustments to address nonresponse among the Wave 4 Cohort was for participants younger than age 12 at Wave 4, i.e., Wave 4 shadow youth. There was also a slight underrepresentation of participants residing in the Midwest at Wave 4 and an overrepresentation of those residing in the West among the Wave 6 youth respondents.

After Wave 6 weighting adjustments, the underrepresentation of Wave 4 shadow youth was eliminated, as were the differences by Wave 4 census region. Wave 4 youth who were ever users of tobacco were appropriately represented among the Wave 6 youth respondents, both before and after applying the Wave 6 weighting adjustments.

When comparing by age and grade, the PATH Study's youth cigarette-smoking estimates based on the Wave 4 Cohort at Wave 6 were generally lower than the estimates from NHANES 2017-2020, NSDUH 2020, and MTF 2021; although the Wave 6 estimates of ever cigarette smoking among youth ages 16 and 17 were similar to those from NSDUH 2020. For high school students ages 14 to 17, the Wave 6 estimates of cigarette smoking were lower than the estimates from NYRBS 2019 and NYTS 2021, with 95 percent confidence intervals overlapping for only two subgroup estimates between the PATH Study and NYTS.

The PATH Study's Wave 4 Cohort estimates of ever ENDS use at Wave 6 were generally lower than the estimates from NSDUH 2020 for youth ages 14 and 15 with mostly non-overlapping

confidence intervals, whereas the PATH Study estimates were generally comparable to the NSDUH estimates for youth ages 16 and 17. However, the Wave 6 ever ENDS-use estimates by grade and school level for youth ages 14 to 17 were noticeably lower than the estimates from NYRBS 2019, MTF 2021, and NYTS 2021, and the differences were generally bigger than those observed for youth cigarette smoking.

It is uncertain whether the differences in estimates of youth cigarette-smoking and ENDS use are due to nonresponse bias in one or more of the studies compared or a variety of other differences between the studies, including the timing of data collection.

6.3 General Conclusions

Assuming that the demographic, socio-economic, and tobacco-use characteristics examined in this report are correlated with key tobacco- and health-related outcome measures, these results raise no serious concerns about potential nonresponse bias in youth estimates from Wave 6 of the PATH Study, for the Wave 1 Cohort or Wave 4 Cohort. However, almost all of the Wave 6 youth in the Wave 1 Cohort were shadow youth at Wave 1, and limited data are available for assessing nonresponse bias for this group. Due to its age, the Wave 1 Cohort has an increased potential for nonresponse bias, and the analyses presented in this report suggest that the attrition of tobacco users could be beginning to have a small effect on estimates for adults based on all-waves respondents. However, there is limited evidence of potential nonresponse bias in estimates based on all Wave 4 Cohort adult respondents at Wave 6. These results confirm the timeliness of the introduction of the Wave 4 replenishment sample and the PATH Study's plans to replenish the sample again at Wave 7.

There is no evidence of nonresponse bias with respect to Wave 6 estimates of current cigarette smoking among adults, in the sense that the Wave 4 Cohort estimates were within the range of estimates from other national studies. However, the Wave 6 estimates of youth cigarette smoking were generally lower than the estimates from other national studies. At Wave 6, the Wave 4 Cohort produces adult estimates of current ENDS use that are higher than those from TUS-CPS 2018-2019, but similar to those from NHIS 2021. The PATH Study's estimates of ever ENDS use among youth ages 14 to 17 are noticeably lower at Wave 6 than those from recent school-based studies. It is difficult to determine the primary reason(s) for these results due to the many differences between the PATH Study and the surveys compared.

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Appendix A

Cigarette-Smoking Questions in the PATH Study and Other Surveys

Table A-1 lists the questions used to ask about the current cigarette-smoking status of adults in the PATH Study, as well as in the surveys used for comparison, and describes the populations included in the estimates from those surveys. Table A-2 provides similar information for youth ever and past 30-day cigarette smoking.

Note that although the questions used to define current cigarette smoking are similar among the surveys, small differences could have an effect on the answers given. For example, in the PATH Study, the question used to establish whether a respondent has smoked at least 100 cigarettes in their lifetime has closed response categories:

“How many cigarettes have you smoked in your entire life? A pack usually has 20 cigarettes in it.”

For adults:

1. 1 or more puffs but never a whole cigarette;
2. 1 to 10 cigarettes (about ½ pack total);
3. 11 to 20 cigarettes (about ½ pack to 1 pack);
4. 21 to 50 cigarettes (more than 1 pack but less than 3 packs);
5. 51 to 99 (more than 2½ packs but less than 5 packs); and
6. 100 or more cigarettes (5 packs or more).

For youth:

1. 1 or more puffs but never a whole cigarette;
2. 1 cigarette;
3. 2 to 10 cigarettes (about ½ pack total);
4. 11 to 20 cigarettes (about ½ pack to 1 pack);
5. 21 to 50 cigarettes (more than 1 pack but less than 3 packs);
6. 51 to 99 (more than 2½ packs but less than 5 packs); and

7. 100 or more cigarettes (5 packs or more).

In TUS-CPS, NHANES, NSDUH, and NHIS, however, the question “Have you smoked at least 100 cigarettes in your entire life?” calls for a yes/no response.

The positioning of the questions and context also differ among the surveys. For example, the cigarette-smoking questions are near the beginning of the PATH Study’s adult and youth questionnaires, and respondents know that the questionnaires are about tobacco-use behaviors. In contrast to the PATH Study, NHIS and NHANES have cigarette-smoking questions in the middle of the survey, and respondents know that these surveys cover a range of health topics. Another example is the TUS-CPS, where the cigarette-smoking questions are near the beginning of the adult questionnaire on tobacco, but the survey is administered as part of the CPS. These differences among the surveys may be associated with differences in responses to the cigarette-smoking questions.

Table A-1. Questions and algorithms used to define adult current cigarette smoking in the PATH Study Wave 6, TUS-CPS 2018-2019, NHANES 2017-2020, NSDUH 2020, and NHIS 2021

PATH Study Wave 6	TUS-CPS 2018-2019*	NHANES 2017-2020	NSDUH 2020	NHIS 2021
Questions to define current cigarette use (answers defining current cigarette use given in parentheses)				
“How many cigarettes have you smoked in your entire life? A pack usually has 20 cigarettes in it.” (Wave 1 – Wave 6, 100 or more cigarettes (5 packs or more)) “Have you ever smoked a cigarette, even one or two puffs?” (Wave 1, Wave 4, Wave 5, Wave 5.5, Wave 6 adult, yes) “Have you ever tried cigarette smoking, even one or two puffs?” (Wave 1 – Wave 5.5 youth, yes) “In the past 30 days, have you smoked a cigarette, even one or two puffs?” (Wave 2 – Wave 6 adult, yes) “In the past 12 months, have you smoked a cigarette, even one or two puffs?” (Wave 2 – Wave 6, yes) “Do you now smoke cigarettes every day, some days, or not at all?” (Wave 6, every day, some days)	“(Have/has) (you/name) smoked at least 100 cigarettes in (your/his/her) entire life?” (yes) “(Do/does) (you/name) now smoke cigarettes every day, some days, or not at all?” (every day, some days)	“{Have you/Has SP} smoked at least 100 cigarettes in {your/his/her} entire life?” (yes) “{Do you/Does SP} now smoke cigarettes every day, some days, or not at all?” (every day, some days)	“Have you smoked at least 100 cigarettes in your entire life?” (yes) “Have you ever smoked part or all of a cigarette?” (yes) “During the past 30 days, have you smoked part or all of a cigarette?” (yes)	“Have you smoked at least 100 cigarettes in your ENTIRE LIFE?” (yes) “Do you NOW smoke cigarettes every day, some days or not at all?” (every day, some days)

Table A-1. Questions and algorithms used to define adult current cigarette smoking in the PATH Study Wave 6, TUS-CPS 2018-2019, NHANES 2017-2020, NSDUH 2020, and NHIS 2021 (continued)

PATH Study Wave 6	TUS-CPS 2018-2019*	NHANES 2017-2020	NSDUH 2020	NHIS 2021
Coding of “Yes” and “No” answers				
If R06R_A_CUR_ESTD_CIGS = 1 then “Yes” Else if R06R_A_CUR_ESTD_CIGS = 2 then “No”	If PEA1 = 1 and PEA3 in (1 2) then “Yes” Else if PEA1 = 2 or PEA3 = 3 then “No”	If SMQ020 = 1 and SMQ040 in (1 2) then “Yes” Else if SMQ020 = 2 or SMQ040 = 3 then “No”	If CIGEVER = 1 and CIGREC = 1 and CIG100LF in (1 5) then “Yes” Else if CIGEVER = 2 or CIGREC in (2 3 4 14 19 29) or CIG100LF = 2 then “No”	If SMKEV_A = 1 and SMKNOW_A in (1 2) then “Yes” Else if SMKEV_A = 2 or SMKNOW_A = 3 then “No”
Additional question to define every-day cigarette smoking (answer defining every-day cigarette smoking given in parentheses)				
“Do you now smoke cigarettes every day, some days, or not at all?” (Wave 6, every day)	“(Do/does) (you/name) now smoke cigarettes every day, some days, or not at all?” (every day)	“{Do you/Does SP} now smoke cigarettes every day, some days, or not at all?” (every day)	N/A	“Do you NOW smoke cigarettes every day, some days or not at all?” (every day)
Coding of “Yes” and “No” answers				
If R06R_A_EDY_CIGS = 1 then “Yes” Else if R06R_A_EDY_CIGS = 2 then “No”	If PEA1 = 1 and PEA3 = 1 then “Yes” Else if PEA1 = 2 or PEA3 in (2 3) then “No”	If SMQ020 = 1 and SMQ040 = 1 then “Yes” Else if SMQ020 = 2 or SMQ040 in (2 3) then “No”	N/A	If SMKEV_A = 1 and SMKNOW_A = 1 then “Yes” Else if SMKEV_A = 2 or SMKNOW_A in (2 3) then “No”
Additional question to define some-days cigarette smoking (answer defining some-days cigarette smoking given in parentheses)				
“Do you now smoke cigarettes every day, some days, or not at all?” (Wave 6, some days)	“(Do/does) (you/name) now smoke cigarettes every day, some days, or not at all?” (some days)	“{Do you/Does SP} now smoke cigarettes every day, some days, or not at all?” (some days)	N/A	“Do you NOW smoke cigarettes every day, some days or not at all?” (some days)
Coding of “Yes” and “No” answers				
If R06R_A_SDY_CIGS = 1 then “Yes” Else if R06R_A_SDY_CIGS = 2 then “No”	If PEA1 = 1 and PEA3 = 2 then “Yes” Else if PEA1 = 2 or PEA3 in (1 3) then “No”	If SMQ020 = 1 and SMQ040 = 2 then “Yes” Else if SMQ020 = 2 or SMQ040 in (1 3) then “No”	N/A	If SMKEV_A = 1 and SMKNOW_A = 2 then “Yes” Else if SMKEV_A = 2 or SMKNOW_A in (1 3) then “No”
Subsetting used for estimates				
Ages 18+	Ages 18+ Excludes proxy respondents	Ages 18+	Ages 18+	Ages 18+

Table A-1. Questions and algorithms used to define adult current cigarette smoking in the PATH Study Wave 6, TUS-CPS 2018-2019, NHANES 2017-2020, NSDUH 2020, and NHIS 2021 (continued)

PATH Study Wave 6	TUS-CPS 2018-2019*	NHANES 2017-2020	NSDUH 2020	NHIS 2021
Exclusions from population				
The Wave 6 adult target population for the Wave 4 Cohort is the resident (not incarcerated) U.S. population ages 18 and older at Wave 6 who were in the U.S. civilian, noninstitutionalized population at the time of Wave 4. Thus, it includes Wave 4 respondents who at Wave 6 were on active military duty or living in a health care institution (e.g., a mental health hospital) but not those in a correctional facility.	Includes only the U.S. civilian, noninstitutionalized population age 18 or older at the time of the interview.	Includes only the U.S. civilian, noninstitutionalized population.	Includes only the U.S. civilian, noninstitutionalized population. Excludes homeless persons who do not use shelters, military personnel on active duty, and residents of institutional group quarters, such as jails and hospitals.	Includes only the civilian, noninstitutionalized population residing in the U.S. at the time of the interview. Several segments of the population are excluded, such as: persons in long-term care institutions; persons on active duty with the Armed Forces; persons in correctional facilities; and U.S. nationals living in foreign countries.
Proxy responses allowed				
No	Yes	Yes, when there is a serious physical or mental condition, a proxy respondent may be used to conduct the interview.	No	Yes, for individuals physically or mentally incapable of responding.

Table A-1. Questions and algorithms used to define adult current cigarette smoking in the PATH Study Wave 6, TUS-CPS 2018-2019, NHANES 2017-2020, NSDUH 2020, and NHIS 2021 (continued)

PATH Study Wave 6	TUS-CPS 2018-2019*	NHANES 2017-2020	NSDUH 2020	NHIS 2021
Other comments				
		Suspended field operations in March 2020, due to the COVID-19 pandemic. Data collected from 2019 to March 2020 were combined with data from the NHANES 2017-2018 cycle to form a nationally representative sample of NHANES 2017-March 2020 pre-pandemic data.	Data collection was interrupted in March 2020, due to the COVID-19 pandemic. Consequently, the public use files contain survey data from only quarter 1 and quarter 4 of 2020, with a survey mode change from in-person to Web between the two quarters.	Due to the COVID-19 pandemic, typical data collection procedures were disrupted. During January through April 2021, contact with household members was attempted first via telephone, with subsequent personal visits allowed. Beginning in May 2021, interviewers returned to regular survey interviewing procedures, whereby first contact with household members was attempted in person, with follow-up allowed by telephone. Allowed personal visits varied by Regional Office, based on COVID-19 rates in local jurisdictions.

* A proxy response is allowed only after four failed callback attempts to the self-respondent or if the self-respondent will not return before closeout. See <https://cancercontrol.cancer.gov/sites/default/files/2020-06/cpsmay19.pdf>, p7-1.

Table A-2. Questions and algorithms used to define youth cigarette smoking in the PATH Study Wave 6, NYRBS 2019, NHANES 2017-2020, NSDUH 2020, MTF 2021, and NYTS 2021

PATH Study Wave 6	NYRBS 2019	NHANES 2017-2020	NSDUH 2020	MTF 2021	NYTS 2021
Questions to define ever tried cigarette smoking (answers defining ever tried cigarette smoking given in parentheses)					
“Have you ever tried cigarette smoking, even one or two puffs?” (Wave 1 – Wave 6 youth, yes) “In the past 12 months, have you smoked a cigarette, even one or two puffs?” (Wave 2 – Wave 6 youth, yes)	“Have you ever tried cigarette smoking, even one or two puffs?” (yes)	“About how many cigarettes have you smoked in your entire life?” (1 or more puffs but never a whole cigarette, 1 cigarette, 2 to 5 cigarettes, 6 to 15 cigarettes, 16 to 25 cigarettes, 26 to 99 cigarettes, 100 or more cigarettes)	“Have you ever smoked part or all of a cigarette?” (yes)	“Have you ever smoked cigarettes?” (yes)	“Have you ever smoked a cigarette, even one or two puffs?” (yes)
Coding of “Yes” and “No” answers					
If R06R_Y_EVR_CIGS = 1 then “Yes” Else if R06R_Y_EVR_CIGS= 2 then “No”	If QN30 = 1 then “Yes” Else if QN30 = 2 then “No”	If SMQ621 in (2 3 4 5 6 7 8) then “Yes” Else if SMQ621 = 1 then “No”	If CIGEVER = 1 then “Yes” Else if CIGEVER = 2 then “No”	For 8th and 10th grades: If V7101D = 1 then “Yes” Else if V7101D = 0 then “No” For 12th grade: If V2101D = 1 then “Yes” Else if V2101D = 0 then “No”	If QN35 = 1 then “Yes” Else if QN35 = 2 then “No”

Table A-2. Questions and algorithms used to define youth cigarette smoking in the PATH Study Wave 6, NYRBS 2019, NHANES 2017-2020, NSDUH 2020, MTF 2021, and NYTS 2021 (continued)

PATH Study Wave 6	NYRBS 2019	NHANES 2017-2020	NSDUH 2020	MTF 2021	NYTS 2021
Additional questions to define past 30-day cigarette use (answers defining past 30-day cigarette use in parentheses)*					
“When was the last time you smoked a cigarette, even one or two puffs?” (Wave 6; earlier today, not today but sometime in the past 7 days, not in the past 7 days but sometime in the past 30 days)	“During the past 30 days, on how many days did you smoke cigarettes?” (1 or 2 days, 3 to 5 days, 6 to 9 days, 10 to 19 days, 20 to 29 days, all 30 days)	“On how many of the past 30 days did {you/SP} smoke a cigarette?” (1-30)	“During the past 30 days, have you smoked part or all of a cigarette?” (yes)	“How frequently have you smoked cigarettes during the past 30 days?” (less than 1 cigarette per day, 1 to 5 cigarettes per day, about one-half pack per day, about one pack per day, about one and one-half packs per day, two packs or more per day)	“During the past 30 days, on how many days did you smoke cigarettes?” (1-30)
Coding of “Yes” and “No” answers					
If R06R_Y_CUR_CIGS = 1 then “Yes” Else if R06R_Y_CUR_CIGS = 2 then “No”	If QN32 = 1 then “Yes” Else if QN32 = 2 then “No”	If 1 SMD641 in (1 2 ... 30) then “Yes” Else if SMQ621 = 1 or SMD641 = 0 then “No”	If CIGEVER = 1 and CIGREC = 1 then “Yes” Else if CIGEVER=2 or CIGREC in (2 3 4 14 19 29) then “No”	For 8 th and 10 th grades: If V7102D = 1 then “Yes” Else if V7102D = 0 then “No” For 12 th grade: If V2102D = 1 then “Yes” Else if V2102D = 0 then “No”	If 1 QN38 in (1 2 ... 30) then “Yes” Else if QN35 = 2 or QN38 = 0 then “No”
Subsetting used for estimates					
Ages 14-17 Excludes respondents who report being “ungraded” or “home-schooled” from estimates of cigarette smoking and ENDS use by grade or for high school students.	Ages 14-17 Excludes respondents who report being “ungraded” from estimates of cigarette smoking and ENDS use for high school students.	Ages 14-17	Ages 14-17	Ages 14-17	Ages 14-17 Excludes respondents who report being “ungraded” from estimates of cigarette smoking and ENDS use for high school students.

Table A-2. Questions and algorithms used to define youth cigarette smoking in the PATH Study Wave 6, NYRBS 2019, NHANES 2017-2020, NSDUH 2020, MTF 2021, and NYTS 2021 (continued)

PATH Study Wave 6	NYRBS 2019	NHANES 2017-2020	NSDUH 2020	MTF 2021	NYTS 2021
Exclusions from population					
The Wave 6 youth target population for the Wave 4 Cohort is the resident (not incarcerated) U.S. population ages 14 to 17 at Wave 6 who were in the U.S. civilian, noninstitutionalized population at the time of Wave 4. Thus, it includes Wave 4 respondents who at Wave 6 were on active duty or living in a health care institution (e.g., a mental health hospital) but not those in a correctional facility.	Includes only students in grades 9 to 12 who are attending public and private schools in the 50 States and the District of Columbia. U.S. territories and the Virgin Islands are excluded from the sampling frame.	Includes only the U.S. civilian, noninstitutionalized population.	Includes only the U.S. civilian, noninstitutionalized population. Excludes homeless persons who do not use shelters, military personnel on active duty, and residents of institutional group quarters, such as jails and hospitals.	Includes only youth in grades 8, 10, and 12 who are attending public and private schools within the 48 contiguous states.	Includes only youth who are public and private school students enrolled in regular middle schools and high schools in grades 6 through 12 in the 50 U.S. States and the District of Columbia. Alternative schools, special education schools, Department of Defense operated schools, Bureau of Indian Affairs schools, vocational schools that serve only pull-out populations, and students enrolled in regular schools unable to complete the questionnaire without special assistance are excluded.

Table A-2. Questions and algorithms used to define youth cigarette smoking in the PATH Study Wave 6, NYRBS 2019, NHANES 2017-2020, NSDUH 2020, MTF 2021, and NYTS 2021 (continued)

PATH Study Wave 6	NYRBS 2019	NHANES 2017-2020	NSDUH 2020	MTF 2021	NYTS 2021
Other comments					
	Self-administered survey in classroom.			Separate online surveys for 8 th and 10 th grades vs. 12 th grade, self-administered in the classroom or at home.	In 2021, all surveys were administered online while students were either in the classroom or at home.

* To reduce respondent burden, the PATH Study, NHANES, NSDUH, MTF, and NYTS use computer-assisted interviews (CAIs) whereby skip patterns based on answers to questions about having ever tried cigarettes are used to determine whether to present the past 30-day cigarette-smoking question. For the PATH Study, NHANES, and NSDUH, youth who indicated that they had never tried cigarettes or who did not answer the question were not asked about cigarette smoking in the past 30 days. Thus, when creating estimates for these three studies, youth who answered that they had never tried cigarettes were logically coded a “No” value for past 30-day cigarette smoking. For MTF and NYTS, the skip pattern rules for these two questions were slightly different: youth who had never tried cigarettes were not asked the past 30-day question and were logically coded a “No” value when computing estimates of past 30-cigarette smoking; but youth who did not answer the question about having ever tried cigarettes were asked about cigarette smoking in the past 30 days. MTF and NYTS responses to the past 30-day question were used when creating estimates of past 30-day cigarette smoking, even if the youth had not responded about having ever tried cigarettes; however, their missing value was retained when computing estimates of ever cigarette smoking, regardless of the response to the past 30-day question.

The NYRBS is a paper-and-pencil survey, so youth were able to answer the question about past 30-day cigarette smoking regardless of their answer about having ever tried cigarettes. When computing cigarette-smoking estimates for NYRBS, no attempt was made to resolve inconsistencies between the responses to these two questions; the responses to the ever question were used to compute estimates of ever cigarette smoking, and the responses to the past 30-day question were used to compute estimates of past 30-day cigarette smoking.

Appendix B

ENDS-Use Questions in the PATH Study and Other Surveys

Table B-1 lists the questions used to ask about current ENDS use of adults in the PATH Study, TUS-CPS, and NHIS. The observations in Appendix A regarding the differences in positioning of the cigarette-smoking questions among the surveys also apply to the questions about use of ENDS.

Table B-2 lists the questions used to ask about ever and past 30-day ENDS use of youth in the PATH Study as well as in the surveys used for comparison. Administration of the questions differs between the PATH Study, NYRBS, NSDUH, MTF, and NYTS (see Sections 5.1.3 and 5.2.3 for details).

The information provided in Tables A-1 (for adults) and A-2 (for youth) about age restrictions, population exclusions, proxy responses, and “other comments” applies to both the cigarette-smoking and ENDS-use estimates and so is not repeated in Tables B-1 and B-2.

Table B-1. Questions and algorithms used to define adult current ENDS use in the PATH Study Wave 6, TUS-CPS 2018-2019, and NHIS 2021

PATH Study Wave 6	TUS-CPS 2018-2019	NHIS 2021
Questions to define current ENDS use (answers defining current ENDS use given in parentheses)		
“Have you ever used an e-cigarette, such as NJOY, Blu, or Smoking Everywhere, even one or two times?” (Wave 1, yes) “Have you ever used an electronic nicotine product, even one or two times?” (Wave 2, Wave 4, Wave 5, Wave 5.5, Wave 6, yes) “In the past 30 days, have you used an electronic nicotine product, even one or two times?” (Wave 3 – Wave 6 adult, yes) “In the past 12 months, have you used an electronic nicotine product, even one or two times?” (Wave 3 – Wave 6, yes) “Have you ever used e-cigarettes fairly regularly?” (Wave 1, yes) “Have you ever used electronic nicotine products fairly regularly?” (Wave 2 – Wave 6, yes)* “Do you now use electronic nicotine products every day, some days, or not at all?” (Wave 6, every day, some days)	“(Have/Has) (you/name) EVER used E-cigarettes EVEN ONE TIME?” (yes) “(Do you/Does [name]) NOW use an E-cigarette every day, some days or not at all?” (every day, some days)	“Have you ever used an e-cigarette or other electronic vaping product, even just one time, in your entire life?” (yes) “Do you NOW use e-cigarettes or other electronic vaping products every day, some days, or not at all?” (every day, some days)
Coding of “Yes” and “No” answers		
If R06R_A_CUR_ESTD_EPRODS = 1 then “Yes” Else if R06R_A_CUR_ESTD_EPRODS = 2 then “No”	If PEJ1A3_5 = 1 and PEJ2A3_5 in (1 2) then “Yes” Else if PEJ1A3_5 = 2 or PEJ2A3_5 = 3 then “No”	If ECIGEV_A = 1 and ECIGNOW_A in (1 2) then “Yes” Else if ECIGEV_A = 2 or ECIGNOW_A = 3 then “No”
Additional question to define every-day ENDS use (answer defining every-day ENDS use given in parentheses)		
“Do you now use electronic nicotine products every day, some days, or not at all?” (Wave 6, every day)	“(Do you/Does [name]) NOW use an E-cigarette every day, some days or not at all?” (every day)	“Do you NOW use e-cigarettes or other electronic vaping products every day, some days, or not at all?” (every day)

Table B-1. Questions and algorithms used to define adult current ENDS use in the PATH Study Wave 6, TUS-CPS 2018-2019, and NHIS 2021 (continued)

PATH Study Wave 6	TUS-CPS 2018-2019	NHIS 2021
Coding of “Yes” and “No” answers		
If R06R_A_EDY_EPRODS = 1 then “Yes” Else if R06R_A_EDY_EPRODS = 2 then “No”	If PEJ1A3_5 = 1 and PEJ2A3_5 = 1 then “Yes” Else if PEJ1A3_5 = 2 or PEJ2A3_5 in (2 3) then “No”	If ECIGEV_A = 1 and ECIGNOW_A = 1 then “Yes” Else if ECIGEV_A = 2 or ECIGNOW_A in (2 3) then “No”
Additional question to define some-days ENDS use (answer defining some-days ENDS use given in parentheses)		
“Do you now use electronic nicotine products every day, some days, or not at all?” (Wave 6, some days)	“(Do you/Does [name]) NOW use an E-cigarette every day, some days, or not at all?” (some days)	“Do you NOW use e-cigarettes or other electronic vaping products every day, some days, or not at all?” (some days)
Coding of “Yes” and “No” answers		
If R06R_A_SDY_EPRODS = 1 then “Yes” Else if R06R_A_SDY_EPRODS = 2 then “No”	If PEJ1A3_5 = 1 and PEJ2A3_5 = 2 then “Yes” Else if PEJ1A3_5 = 2 or PEJ2A3_5 in (1 3) then “No”	If ECIGEV_A = 1 and ECIGNOW_A = 2 then “Yes” Else if ECIGEV_A = 2 or ECIGNOW_A in (1 3) then “No”

*The exact wording of this question differed slightly across waves, and at some waves it was asked about specific electronic nicotine products individually.

Table B-2. Questions and algorithms used to define youth ENDS use in the PATH Study Wave 6, NYRBS 2019, NSDUH 2020, MTF 2021, and NYTS 2021

PATH Study Wave 6	NYRBS 2019	NSDUH 2020	MTF 2021	NYTS 2021
Questions to define ever ENDS use (answers defining ever ENDS use given in parentheses)				
“Have you ever used an e-cigarette, such as NJOY, Blu, or Smoking Everywhere, even one or two times?” (Wave 1, yes) “Have you ever used an electronic nicotine product, even one or two times?” (Wave 2 – Wave 6 youth, yes) “In the past 12 months, have you used an electronic nicotine product, even one or two times?” (Wave 3 – Wave 6 youth, yes)	“Have you ever used an electronic vapor product?” (yes)	“Have you ever, even once, vaped nicotine or tobacco with an e-cigarette or other vaping device?” (yes)	“In your LIFETIME how often have you vaped nicotine?” (once or twice, occasionally but not regularly, regularly in the past, regularly now)	“Have you ever used an e-cigarette, even once or twice?” (yes)
Coding of “Yes” and “No” answers				
If R06R_Y_EVR_EPRODS =1 then “Yes” Else if R06R_Y_EVR_EPRODS= 2 then “No”	If QN34 = 1 then “Yes” Else if QN34 = 2 then “No”	If VAPNICEVR = 1 then “Yes” Else if VAPNICEVR in (2 91) then “No”	If V7780 in (2 3 4 5) then “Yes” Else if V7780 = 1 then “No”	If QN6 = 1 then “Yes” Else if QN6 = 2 then “No”

Table B-2. Questions and algorithms used to define youth ENDS use in the PATH Study Wave 6, NYRBS 2019, NSDUH 2020, MTF 2021, and NYTS 2021 (continued)

PATH Study Wave 6	NYRBS 2019	NSDUH 2020	MTF 2021	NYTS 2021
Additional question to define past 30-day ENDS use (answers defining past 30-day ENDS use given in parentheses)				
“When was the last time you used an electronic nicotine product, even one or two times?” (Wave 6, earlier today, not today but sometime in the past 7 days, not in the past 7 days but sometime in the past 30 days)	“During the past 30 days, on how many days did you use an electronic vapor product?” (1 or 2 days, 3 to 5 days, 6 to 9 days, 10 to 19 days, 20 to 29 days, all 30 days)	“How long has it been since you last vaped nicotine or tobacco with an e-cigarette or other vaping device?” (within the past 30 days)	“On how many DAYS (if any) during the LAST 30 DAYS have you vaped nicotine?” (1-2 days, 3-5 days, 6-9 days, 10-19 days, 20-29 days, 30 days)	“During the past 30 days, on how many days did you use e-cigarettes?” (1-30)
Coding of “Yes” and “No” answers				
If R06R_Y_CUR_EPRODS = 1 then “Yes” Else if R06R_Y_CUR_EPRODS = 2 then “No”	If QN35 = 1 then “Yes” Else if QN35 = 2 then “No”	If VAPNICREC = 1 then “Yes” Else if VAPNICEVER in (2 91) or VAPNICREC in (2 3 19) then “No”	If V7782 in (2 3 4 5 6 7) then “Yes” Else if V7782 = 1 then “No”	If QN9 in (1 2 ... 30) then “Yes” Else if QN6 = 2 or QN9 = 0 then “No”